European Commission

Joint Research Centre

AI for Border Management and Customs Controls

Anthology of EU-funded projects, 2015-2024

Anderson, D., Belcineanu, A.-I., Tzvetkova, M.

2025



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Contact information
Name: Alexandru-Ioan Belcineanu
Address: Joint Research Centre, Geel
Email: Alexandru-Ioan.BELCINEANU@ec.europa.eu

The Joint Research Centre: EU Science Hub
https://joint-research-centre.ec.europa.eu

JRC143223
EUR 40536

PDF ISBN 978-92-68-33762-2 ISSN 1831-9424 doi:10.2760/1286492 KJ-01-25-575-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: Anderson, D., Belcineanu, A.-I. and Tzvetkova, M., *AI for Border Management and Customs Controls - Anthology of EU-funded projects, 2015-2024,* Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/1286492, JRC143223.

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Abstract

This report examines how artificial intelligence (AI) is being applied to border management and customs controls in EU-funded research projects. This report features a careful review of 75 projects funded between 2015 and 2024 – primarily under Horizon 2020 and Horizon Europe – representing a combined EU investment of approximately EUR 376 million. Key trends include the integration of diverse sensing technologies and AI-driven analytics for maritime surveillance, threat detection, and real-time decision-making. Projects often combine mobile platforms (e.g. drones and underwater vehicles) with fixed infrastructure and open-source intelligence to monitor borders more effectively. Unmanned vehicles are increasingly deployed for tasks ranging from port surveillance to smuggling detection. In areas like identity verification, projects are advancing contactless and on-the-move biometric systems, while also addressing privacy, ethics, and data protection concerns. Several projects explore predictive analytics for migration flows, cyber threats, and organised crime. Some projects directly address the societal and legal implications of AI, proposing safeguards and ethical frameworks for responsible deployment. Overall, EU-funded initiatives demonstrate the significant potential of AI to transform border management and customs controls. While some challenges remain, the breadth of innovation suggests strong momentum toward more intelligent, integrated, and ethically-grounded border management and customs controls solutions.

Acknowledgements

The authors gratefully acknowledge the thorough and constructive reviews of their colleagues, Giulio Mancini and Solon Mias, which significantly improved the quality of the final report.

Authors

David Anderson, Alexandru-Ioan Belcineanu, Marina Tzvetkova

1. Introduction

1.1. EU policy context

This report examines the use of AI for border management and customs controls through the lens of EU-funded research projects. In the EU context, **border management**[1] is distinct from **customs**; they are parallel but coordinated systems that both operate at or near borders. In simple terms, border management focuses on persons, customs controls focus on goods. This report features research projects from both domains, because border management and customs both deploy AI- enabled technologies at border crossing points.

Border management, or more precisely, European integrated border management (EIBM) is defined in Regulation (EU) 2019/1896 on the European Border and Coast Guard, and comprises multiple components such as border control, search and rescue, risk analyses, information exchange and cooperation (European Commission, 2019). The **border control** component is further defined by reference to Regulation (EU) 2016/399 on the Schengen Borders Code to comprise both **border checks** and **border surveillance** (European Commission, 2016a).

The Union Customs Code (UCC) is laid down in Regulation (EU) 952/2013, and focuses on the movement of goods, controlling imports, exports and transit. The aim is revenue collection (duties, VAT), ensuring product standards, and preventing illicit trade (e.g. drugs, weapons, counterfeit goods). The regulation includes a definition of **customs controls**.

Border management and customs act at the same physical borders but serve different legal mandates. They are distinct activities, as specified in Regulation (EU) 2019/1896:

> *"European integrated border management does not alter the respective competences of the Commission and the Member States in the customs area, in particular regarding controls, risk management and the exchange of information."*

However, cooperation between the two is formally recognised in Article 68 of Regulation 2019/1896:

> *"The Agency [Frontex] shall also engage in cooperation with the Commission and, where relevant, with Member States and the EEAS in activities relating to the customs area, including risk management, where those activities support each other".*

Concerning **artificial intelligence (AI)**, the EU has established a comprehensive policy and regulatory landscape for AI, with a focus on promoting innovation, trust, and protection of fundamental rights. Regulation (EU) 2024/1689 on artificial intelligence – also known as the AI Act (European Commission, 2024b) – establishes a risk-based approach, with stricter requirements for high-risk AI systems. High-risk AI systems, such as, remote biometric identification systems, AI systems intended to be used for biometric categorisation, and AI systems intended to be used for emotion recognition (see Article 6 of the AI Act) including those used in biometric identification, critical

[1] The term *border security* is sometimes used in everyday parlance. However, this term is not defined in EU legislation. In the EU policy context, the preferred term is *border management*.

infrastructure, and law enforcement, are subject to stringent requirements for transparency, data governance, human oversight, and accountability. It is important to note that the AI Act applies to AI systems when they are placed on the market, put into service, or used in the Union, but explicitly excludes scientific research and development activities (Article 2(6)). This means that AI research projects, such as those analysed in this report, are not restricted by the AI Act while they remain in the research domain. However, once project outputs move from research into operational testing or deployment, they fall within the scope of the Regulation. Additionally, the Commission has established the European AI Office to oversee the implementation of the Regulation, provide guidance, and support the development and adoption of AI in the EU (European Commission, 2024c), as well as developing guidelines on high-risk use cases. The EU's regulatory framework for AI also includes the General Data Protection Regulation (GDPR), which applies to the processing of personal data in AI systems (European Commission, 2016b). Key challenges remain such as civil liberties concerns and the challenges around building interoperable governance, certification, and accountability systems across the EU, especially for AI-powered border and law enforcement tools.

Finally, in terms of **research and innovation**, border management has been a strategic priority within the EU's Horizon Framework Programmes, Horizon 2020 (2014–2020) and Horizon Europe (2021–2027). These programmes (European Commission, 2013, 2021) fund research and innovation projects aimed at developing advanced, ethical, and interoperable technologies to strengthen the EU's external borders. Under Horizon 2020's "Secure Societies" pillar, numerous projects were launched to explore smart surveillance, risk-based border management, and biometric systems. In Horizon Europe, border security remains a key component of Cluster 3: Civil Security for Society, which emphasizes innovation in areas such as interoperable border control systems, AI and big data for threat detection, and integrated surveillance. The programme also places emphasis on ethical compliance, fundamental rights, and societal acceptance.

1.2. Related work

Several studies map how AI is used in border management. LeGallais (2024) analysed 42 Horizon 2020 projects and identified three main clusters: observing territory, controlling flows of people/goods, and protecting infrastructure. These partly overlap with the EU Civil Security Market Taxonomy (European Commission, 2022a), which also distinguishes irregular migration, travel facilitation, and flow of goods. Ontology and knowledge-representation work, such as EUCISE-OWL for maritime information sharing (Riga et al., 2020) and BUAS for border surveillance (Kontopodis et al., 2022), reflects an interest in standardising how border-related capabilities and systems are described. Readers of this report might be interested to follow the *Detection Hub* group on LinkedIn[2], which includes EU funded projects working on detection of drugs, explosives, and illicit goods.

Dumbrava (2021) reviews current and emerging AI applications for border surveillance, risk assessment, and biometric identification, mostly at the pilot or early deployment stage. Angelanna (2023) highlights the use of predictive modelling in Integrated Border Management, but also notes that training data are often synthetic or incomplete, which can affect accuracy. Industry reports (IntelexVision, 2024) and advocacy groups (EuroMed Rights, 2023) describe video analytics, anomaly detection, and biometric databases as part of the technological landscape. Dumbrava (2025) expands

[2] https://www.linkedin.com/company/detection-hub

the scope to asylum procedures, where AI supports tasks such as translation, document checks, and language analysis. Many of these applications are still experimental or used in limited contexts.

Ethical and legal issues are a central theme. Amnesty International (2024), EuroMed Rights (2023), and others warn about surveillance risks, bias, and lack of accountability. Ozkul (2023) and Ciudad Fontecha (2023) point to challenges of transparency and fairness in automated decision-making. Blasi Casagran (2021) examines fundamental rights concerns linked to interoperability of EU databases. Kloza et al. (2020) discuss data protection impact assessments as a way to operationalise safeguards. These issues are directly linked to the EU's regulatory framework: the AI Act (European Commission, 2024b), which imposes stricter obligations on high-risk AI systems, and the GDPR, which regulates personal data processing. Spartalis et al. (2024) highlight tensions between explainability, privacy, and security in sensitive use cases.

RAND Europe (2021) identifies three main barriers to the uptake of AI at borders: knowledge gaps between stakeholder groups, organisational and cultural obstacles, and limited empirical evidence of impact. Studies on migration forecasting (Angenendt et al., 2023; Boss et al., 2023) also show how difficult it is to achieve reliable predictive models. These findings suggest that more systematic evaluation of outcomes, better access to representative datasets, and follow-up studies on long-term impacts of research investments would be valuable.

In summary, the literature shows that AI offers opportunities in areas such as surveillance, risk analysis, and biometrics, but deployment remains limited and raises important governance issues. Existing studies also underline the need for more systematic evidence about effectiveness and impact. This report builds on that foundation by mapping EU-funded projects and highlighting their scientific outputs.

1.3. Scope of this report

This report is a deliverable planned in the JRC's 2025 Work Programme under the *Border Security Solutions (BOSS)* project. This report describes EU projects involving the use of AI for border management and customs controls. It does not cover operational deployments at borders, but rather focuses on research and innovation projects and their scientific outputs.

The timeframe covered in this report is projects having a starting date between 2015 and 2024 (inclusive). The methodology and data sources are described in section 3.2. This report is intended for policymakers, researchers, and practitioners, and is complementary to other reports on this subject for two main reasons:

- we identified relevant projects using a bottom-up approach, i.e. by searching for relevant search terms within a single, harmonised dataset of projects across multiple funding programmes;
- in addition to presenting the projects, we highlight and discuss the projects' results and scientific publications in the field, including over 120 references to external publications.

1.4. Disclaimer

- The authors used a semi-automated process and human judgement to identify EU-funded projects related to AI for border management and customs controls, as described in section 3.2. In general, we have selected only projects that explicitly mention both AI and border security or customs (based on Cordis data and project deliverables).

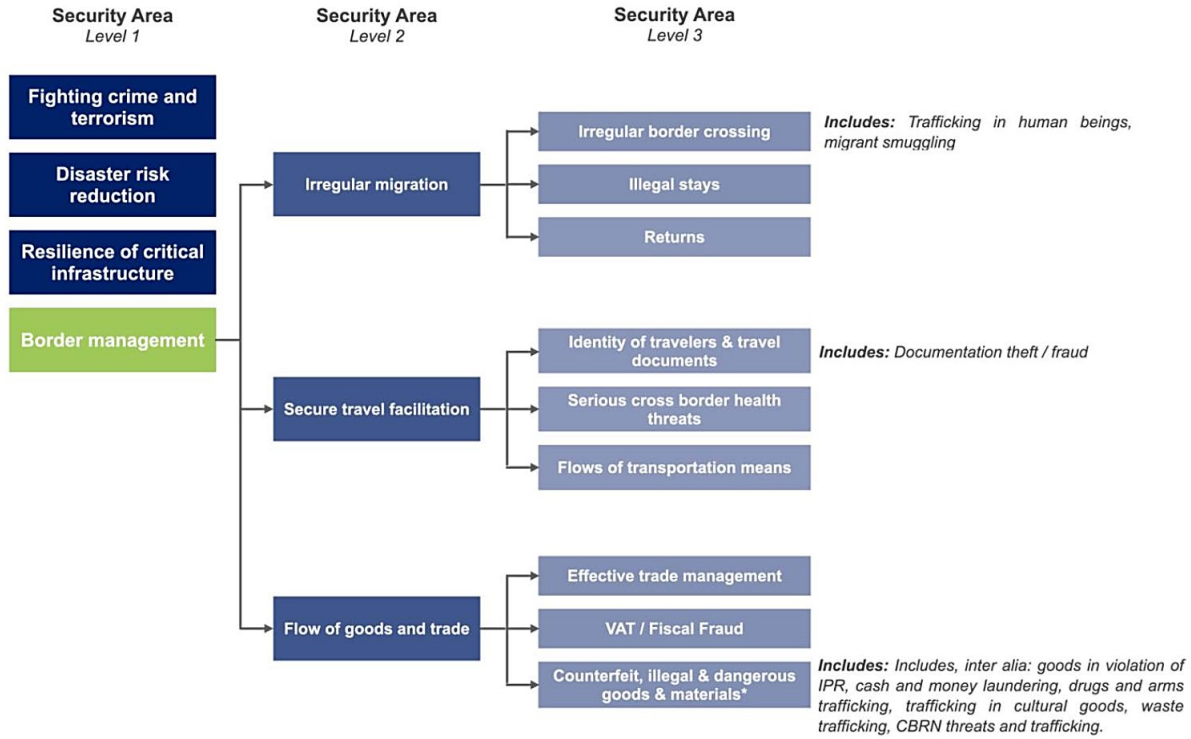
- The inclusion of a project in this report does not constitute an endorsement of the project outcomes, nor can this report be considered an exhaustive list of projects in this area.
- The project descriptions in this report include the "objective" of the project as reported on Cordis. The actual work performed might deviate from this original plan. The authors are not responsible for any discrepancies.
- Reported results and selected publications included in this report for each project are not intended as exhaustive lists. The authors selected only the results and publications they considered most relevant in the overall context of this report.
- Budget figures are for the projects as a whole; the authors did not attempt to assess the proportion of each project's budget attributable specifically to the use of AI for border management or customs controls.

2. Technology overview

There are many ways to classify the application of AI in border management and customs controls. After performing a literature search, we have summarised the classification schemes used in five related reports in Table 1.

We also highlight the EU Civil Security Taxonomy (European Commission, 2022a), which was developed in the frame of the EU Security Market Study (European Commission, 2022b). The taxonomy aims to create a common language or harmonised terminology for security products and services, and one of the top-level areas is "border management" (see Figure 1).

Figure 1. Overview of the EU Civil Security Taxonomy



Source: European Commission (2022a)

In this report, we have crafted a categorisation scheme that reflects the distribution of EU research projects featuring AI for border management and customs controls. Although some projects correspond to more than one category, we have endeavoured to classify each project according to the category where the main innovation lies. This classification scheme is also reflected in the chapter structure and colour scheme of this report, as shown in Table 2.

Table 1. Various classifications schemes for AI for border management used in other related reports.

AI in Border Security Control and Surveillance (EuroMed Rights, 2023)	**Artificial intelligence at EU borders** (Dumbrava, 2021)	**The Digital Border – Migration, Technology & Inequality** (Amnesty International, 2024)	**Enhancing Border Security through AI** (IntelexVision, 2024)	**How AI can support the European Border and Coast Guard** (RAND Europe, 2021)
- AI at borders • chatbots • risk assessment tools • computer vision - surveillance technologies • high altitude pseudo satellites • IoT • intelligent video surveillance • Radar • underwater sensors • video synopsis • drones & micro-drones • algorithmic surveillance - biometrics unveiled • contactless friction ridge recognition • face recognition • iris recognition - privacy and rights considerations • privacy • freedom • asylum • human rights	- biometric systems • Fingerprint identification • Face recognition - emotion detection - algorithmic profiling - fundamental rights • bias and discrimination • data protection and privacy • the risk of unlawful profiling	- before border • Predictive Forecasting Tools • Border Deterrence - Technology - at the border • Immigration Surveillance • Biometrics • Algorithmic Decision Making - after the border • technology-enabled detention and alternatives to detention • economic, social and cultural rights • social security and welfare • work, migration, and technology	- automated entry and exit checks • Facial recognition • Verify identities and travel • Documents - anomaly detection in crowd behaviours - predictive threat analysis - surveillance of remote and rugged terrains	- automated border control - maritime domain awareness - machine learning optimisation - surveillance towers - heterogeneous robotic systems - small UAS - predictive asset maintenance - object recognition - geospatial data analytics

Table 2. Categorisation scheme used in this report to best reflect the distribution of EU research projects identified as featuring AI for border management and customs controls, with mapping to the EU Civil Security Taxonomy.

Chapter	EU Civil Security Taxonomy	Examples
maritime and border surveillance	• irregular migration • flow of goods and trade	• sensor fusion for maritime monitoring, • anomaly detection, • object detection at sea, • critical infrastructure mapping.
goods inspection and detection technologies	• flow of goods and trade	• automated inspection of cargo, containers, post, etc., • safeguarding cultural heritage and artifacts, • predictive maintenance for screening processes.
risk analysis and predictive intelligence	• irregular migration • flow of goods and trade	• predictive risk profiling, • real-time database cross-checking, • false positive reduction, • dynamic risk scoring.
identity, biometrics and document security	• irregular migration • secure travel facilitation	• facial recognition under challenging conditions, • multimodal biometric fusion, • spoof and liveness detection, • real-time eGate processing.
unmanned vehicles and autonomous systems	• irregular migration • flow of goods and trade	• autonomous patrols, • onboard AI processing, • swarm coordination.
social and ethical dimensions	• secure travel facilitation	• privacy-preserving techniques, • transparency and explainability, • bias detection in AI models.

Source: JRC

3. EU funding and projects

3.1. Funding context

Horizon 2020 (H2020) was the EU's research and innovation funding programme for the period 2014-2020, with a budget of nearly EUR 80 billion (European Commission, 2013). Horizon 2020 was structured around three main pillars:

- Excellent Science: Focused on strengthening the EU's scientific base through grants from the European Research Council (ERC), Marie Skłodowska-Curie Actions (MSCA) for researcher mobility and training, and support for Future and Emerging Technologies (FET) and research infrastructures.
- Industrial Leadership: Aimed at boosting innovation by supporting small and medium-sized enterprises (SMEs) and large companies. It included funding for Leadership in Enabling and Industrial Technologies (e.g., ICT, nanotech, space), Access to Risk Finance, and dedicated SME instruments.
- Societal Challenges: Addressed key global challenges such as health, energy, transport, and climate change by funding multi-disciplinary research and innovation projects.

The main sub-programme of relevance for AI for border management was "*3.7 Secure societies - Protecting freedom and security of Europe and its citizens*".

Horizon 2020 was succeeded by **Horizon Europe (HE)**, which is the EU's key funding programme for research and innovation for the period 2021-2027, with an indicative budget of EUR 93.5 billion (European Commission, 2021). Horizon Europe contains the following new elements:

- European Innovation Council: Support for innovations with potential breakthrough and disruptive nature with scale-up potential that may be too risky for private investors. This is 70% of the budget earmarked for SMEs.
- Missions: Sets of measures to achieve bold, inspirational and measurable goals within a set timeframe. There are five main mission areas in Horizon Europe.
- Open science policy: Mandatory open access to publications and open science principles are applied throughout the programme
- New approach to partnerships: Objective-driven and more ambitious partnerships with industry in support of EU policy objectives

Other funding programmes of partial relevance for this study are:

- **Customs Control Equipment Instrument (CCEI)**: the CCEI provides financial support to the customs authorities of European Union Member States for the purchase, maintenance and upgrade of customs control equipment, including detection technology.
- **Internal Security Fund (ISF)**: the ISF comprises the Internal Security Fund Police (ISFP) and Internal Security Fund Borders and Visa (ISFB), and aim at a high level of security in the Union.
- **Erasmus+**: the EU's programme to support education, training, and youth in Europe.

In parallel to EU-funded research projects, Frontex runs its own research and innovation programme (Frontex, 2025a). The budget assigned to activities launched and ongoing in 2024 amounts to over EUR 5 million. We did not consider this programme in the scope of this report. Interested readers are directed to a recent progress report (Frontex, 2025b) that describes – among others – work on training technologies, the reference architecture for the European Border and Coast Guard surveillance system, and studies on the impact of the "polycrisis" on policing and border security.

3.2. Methodology

For this study, project meta data was downloaded for both the **Horizon 2020 and Horizon Europe** framework programmes from the official portal for EU data. For each programme, a ZIP file containing several Excel files (e.g. project.xlsx, organisation.xlsx, euroSciVoc.xlsx, topic.xlsx) was downloaded [3, 4]. As of January 2025, a total of 35 386 projects were downloaded for Horizon 2020, and a total of 14 970 projects for Horizon Europe.

Meta data for projects funded under **non-Horizon programmes** were retrieved from the EU Funding and Tenders Portal. The section of the website containing the project data is also known as the Single Electronic Data Interchange Area (SEDIA) [5]. The website offers an application programme interface (API) that allows data to be downloaded programmatically. An Excel workbook was created to download the data from SEDIA using the API. The result is returned in JSON format, which is then parsed and outputted to a new worksheet in Excel. As of January 2025, 21 704 projects were downloaded for the period 2014-2027. The projects related to AI and border management and customs controls were provisionally identified by searching in three fields (*title*, *objective*, *freeKeywords*) for any of the terms indicated in Table 3.

Table 3. Search terms used to provisionally identify projects on AI for border management and customs controls.

Concept	**Search terms***
border management	border security, border guard, border surveillance, border control, border check, border patrol, border management, border protection, border crossing, border risk, border authorit, Schengen, smart border, freedom of movement, illegal migra, maritime security, maritime surveillance, maritime border, sea border, FRONTEX, EUROSUR, border crime, Common Information Sharing Environment
artificial intelligence	artificial intelligence, \cAI, machine learning, deep learning, neural network, \cCNN, big data, algorithm, predictive model, data mining
customs controls	\wcustoms, customs union, \excise

[*] Where \c = case sensitivity, \w = whole words, < = start of word, > = end of word.

[3] https://data.europa.eu/data/datasets/cordish2020projects

[4] https://data.europa.eu/data/datasets/cordis-eu-research-projects-under-horizon-europe-2021-2027

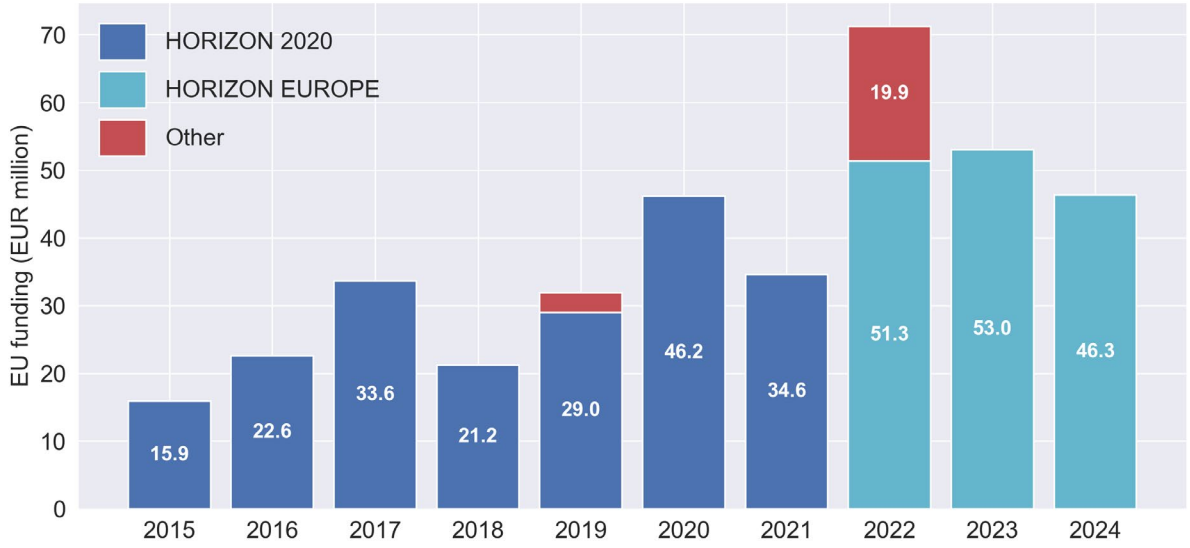
[5] https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-results

A manual review was subsequently performed in which the authors read the meta data of each project – in particular the objective – and decided if the project was relevant or not. In the end, we identified **75 projects related to AI for border management and customs controls**. Selected results and selected publications were identified by the authors based on the Cordis project information pages of each project. It should be noted that this analysis was based exclusively on publicly available project information (e.g. CORDIS summaries and open deliverables), and that classified or commercially sensitive reports were excluded.

3.3. Featured projects

Following the semi-automated process and manual review described above, a total of 75 projects were identified as being relevant for AI for border management & customs controls. Most of the projects and funding (93 %) were awarded under the two Horizon programmes (Horizon 2020 and Horizon Europe). The remaining projects were funded under the Customs Control Equipment Instrument (CCEI), Erasmus+ (ERASMUS+), Internal Security Fund Borders and Visa (ISFB), and Internal Security Fund Police (ISFP). The EU funding allocated per year for AI for border management and customs controls projects is illustrated in Figure 2 for the 10-year time frame covered by this report (2015-2024). The data comprises 75 projects with a combined EU funding of EUR 376 million.

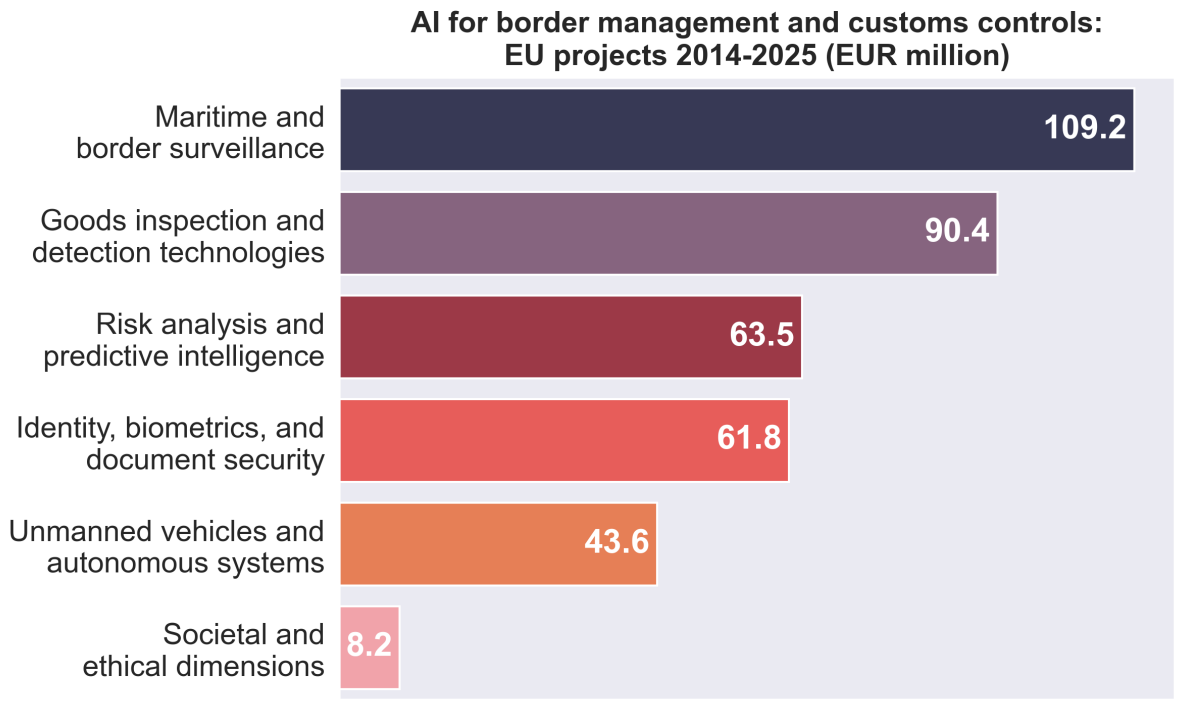
Figure 2. EU funding allocated for AI for border management and customs controls projects, 2015-2024 (based on project start date).



Source: JRC

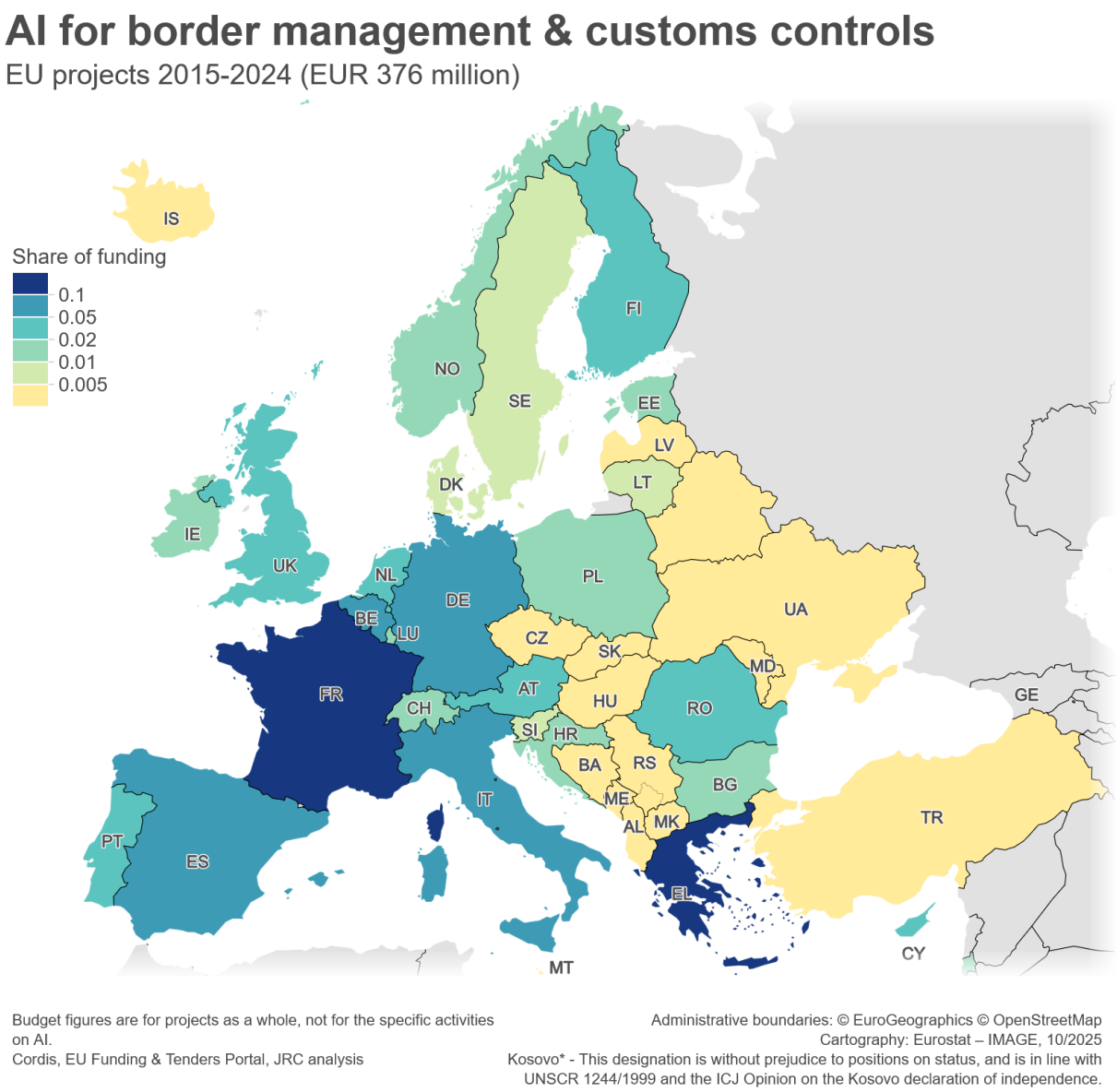
The topics covered are shown in Figure 3. The corresponding geographical distribution of the EU funding for projects involving AI for border management and customs controls is indicated in Figure 4. The top five recipient countries were France, Greece, Belgium, Italy, and Germany, which reflects the established research and innovation ecosystems in the border management domain.

Figure 3. EU projects on AI for border management and customs controls by topic (2015-2024).



Source: JRC

Figure 4. Distribution of EU-funding 2015-2024 on projects involving AI for border management and customs controls.



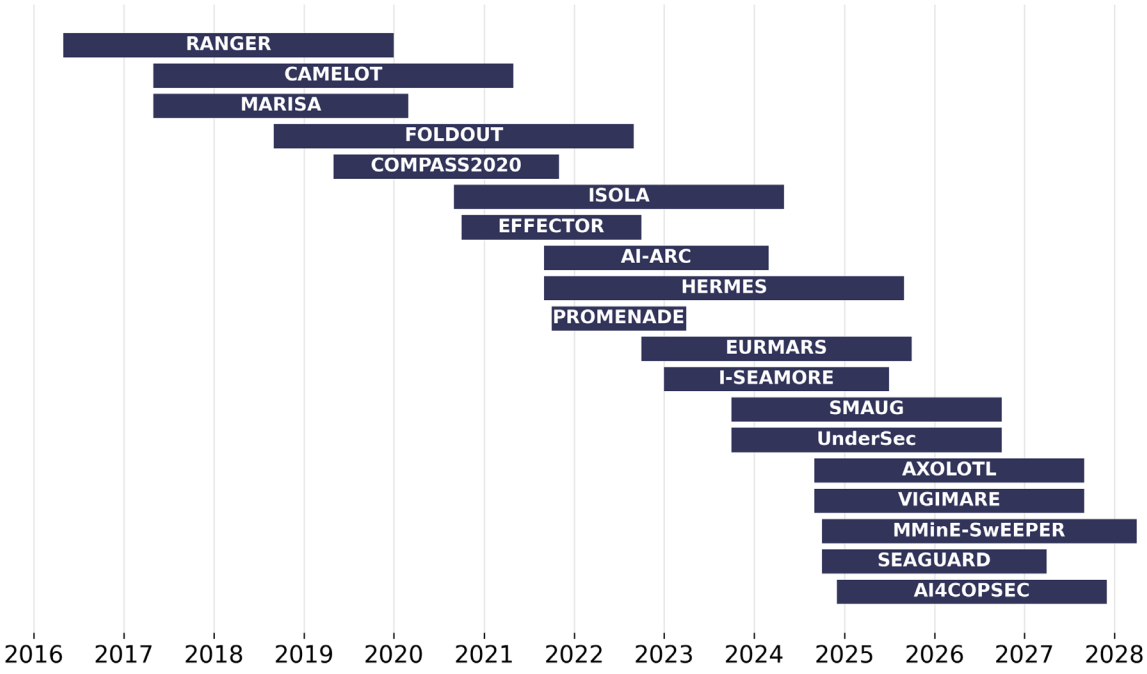
Source: Cordis, EU Funding & Tenders Portal, JRC

4. Maritime and border surveillance

4.1. Introduction

We identified 19 EU-funded projects that are strongly focussed on maritime surveillance and border surveillance. A timeline of these projects is showing in Figure 5.

Figure 5. EU projects featuring AI for maritime and border surveillance.



Source: JRC

A common theme across many of these projects is **data fusion and interoperability**, combining multiple sensing techniques, both mobile and fixed. Mobile sensing often refers to unmanned aerial vehicles (UAVs)[6], unmanned surface vehicles (USVs) and unmanned underwater vehicles (UUVs) - known collectively as UxVs. Fixed sensing includes radar installations, underwater cables, and fixed buoys. The *CAMELOT* project developed and demonstrated different advanced command and control service modules for multiple platform domains. The *MARISA* project combined surveillance platforms with open-source intelligence, including Internet and social networks, with the aim to support decision-making and reaction capabilities. The *COMPASS2020* project addressed operational challenges in wide-area surveillance by demonstrating coordinated use of manned and unmanned assets, supported by an interoperable mission system that enables real-time data fusion and threat analysis. These projects were complemented by later projects such as *EFFECTOR*, *EURMARS*, and *I-SEAMORE*.

Common Information Sharing Environment (CISE) is an EU initiative to improve cooperation and information exchange across the many authorities involved in maritime surveillance (EMSA, 2025).

[6] The preferred term used in EU legislation (i.e. Regulation EU 2019/945) is *unmanned aircraft* (UA).

Each of these actors traditionally uses separate IT systems, which makes it difficult to achieve a common, real-time picture of maritime activity. CISE addresses this challenge by creating a framework for interoperability. It allows systems operated by different authorities and countries to securely share data on vessel movements, incidents, environmental hazards, or criminal activities at sea. Projects like *EFFECTOR*, *EURMARS*, and *I-SEAMORE*, were integrated with CISE to enable consolidation of heterogeneous data sources, such as AIS signals, radar, satellite imagery, etc., into a single situational awareness picture. For land-based border surveillance, the *FOLDOUT* project developed a system that combines various sensors and technologies and intelligently fuses these into an effective and robust intelligent detection platform, to detect people through dense foliage in extreme climates.

In maritime awareness, a common application of AI is **vessel tracking and anomaly detection.** For example, the *AI4COPSEC* project addressed anomaly detection and ship recognition in maritime traffic using high-resolution optical images and advanced segmentation algorithms to enhance the detection of small vessels independently of the type of material. The project aimed at providing e intelligence for enhancing existing Copernicus products through the usage of deep learning, geomatics and social media data The *SEAGUARD* project featured a cross-border surveillance system comprising multiple sensing technologies integrated in both fixed and mobile platforms. The system was designed to detect various types of threats, including illegal border crossings, smuggling of contraband, and illegal fisheries.

The EU is paying increasing attention to the security and resilience of **underwater and cable infrastructure** (European Commission, 2024a, 2025a). The *UnderSec* project aimed at developing a modular and holistic system featuring multimodal sensors and robotic assets. The system provides underwater security, enhance situational awareness, and offer decision-making support, including response capabilities for ships, ports, and maritime infrastructures. The specific risk to underwater critical infrastructure is addressed by the *VIGIMARE* and *SMAUG* projects. Activities include mapping submarine systems, cables and landing stations, assessing vulnerabilities in Europe, and developing automated threat detection. *SMAUG* focuses on underwater threat detection in ports and their access routes by integrating acoustic sensing, sonar scanning, and swarms of autonomous underwater vehicles. The *AI-ARC* project featured a highly innovative and AI-based platform known as the Virtual Control Room (VCR), which has the potential to greatly improve maritime situational awareness and decision-making in the Arctic Sea.

Surveillance platforms can also be used to support **search and rescue** operations (*RANGER*, *I-SEAMORE*, *AI4COPSEC*), and also **environmental applications** such as response to oil spills (*AI4COPSEC*) and biodiversity assessment (*AXOLOTL*). The *MMinE-SwEEPER* project aims at advance knowledge and capability in Europe for dealing with marine munition in the non-military aspect of clearance of unexploded ordnance (UXO). The project addresses automated munition detection, environmental monitoring of chemical contaminants, predictions of UXO corrosion and spread of contaminants.

Some projects focus on the development of **innovative sensing technology**. For example, the *RANGER* project aims at combining innovative Radar technologies with novel technological solutions for enhanced capability to detect, identify and track suspicious vessels. The *HERMES* project proposes the fusion of AI and deep sub-micron CMOS technology to open a new generation of wireless transceivers.

4.2. Data fusion and interoperability

4.2.1. MARISA

740698

Maritime Integrated Surveillance Awareness

LEONARDO - SOCIETA PER AZIONI (IT)

€7,997,492

01/05/2017 – 29/02/2020

https://doi.org/10.3030/740698

Programme
HORIZON 2020

Topic title
SEC-19-BES-2016 – Data fusion for maritime security applications

https://www.youtube.com/@MARISAHProject

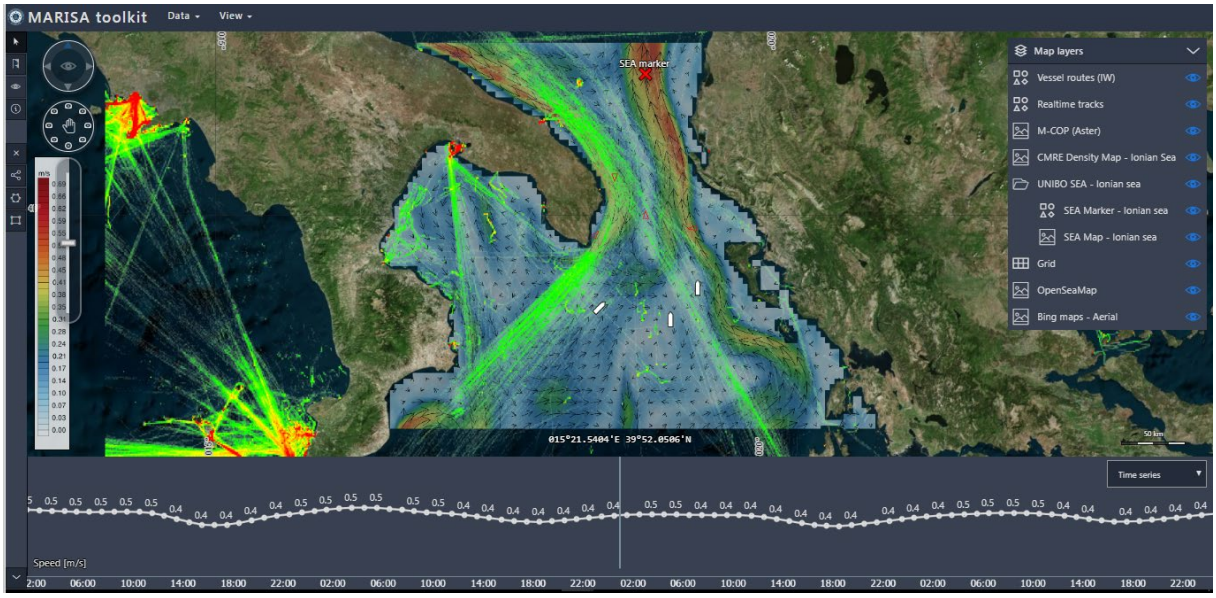
Objective

Combating irregular migration, human smuggling, terrorism at sea, piracy, as well as arms and drug trafficking has become a high priority on Europe's security agenda. Securing the sea requires a day-to-day collaboration activities among European actors of maritime surveillance, Member States' administrations and European agencies principally, and a significant number of initiatives are being taken at EU level to address this challenge. The large amount of 'raw data' available today are not usable by systems supporting maritime security since they are not accessible at the same time and, often, they are not interoperable.

Therefore, the overarching goal of MARISA project is to provide the security communities operating at sea with a data fusion toolkit, which makes available a suite of methods, techniques and modules to correlate and fuse various heterogeneous and homogeneous data and information from different sources, including Internet and social networks, with the aim to improve information exchange, situational awareness, decision-making and reaction capabilities.

The proposed solution will provide mechanisms to get insights from any big data source, perform analysis of a variety of data based on geographical and spatial representation, use techniques to search for typical and new patterns that identify possible connections between events, explore predictive analysis models to represent the effect of relationships of observed object at sea. Enterprise and ad-hoc reporting and services, within the CISE context, will be provided to support users and operational systems in their daily activities, as well as presentation tools for navigating and visualizing results of data fusion processing. The involvement of 5 practitioners as full partners will allow on the one hand to align innovation to user needs, on the other hand to validate the toolkit through a number of trials addressing cross country/cross domain applications.

Figure 6. MARISA console for maritime situational awareness.



Source: Cordis

Selected publications

The "MARISA Game", an innovative analytical tool designed to support knowledge acquisition for intelligent maritime behavioural analysis systems, was presented in de Rosa and De Gloria (2020). By engaging domain experts in structured gameplay, the method facilitates the elicitation of conditional probabilities for dynamic Bayesian networks, enhancing the realism and accuracy of behaviour models.

Pirinen (2017) extended the 'Learning by Research and Development' model into the domain of security-related innovation. By integrating interdisciplinary perspectives ranging from service science and pedagogy to engineering and management, the study illustrates how co-creative environments foster the development of both human and technological capabilities.

Ruoslahti (2020) investigated the inherent complexity of EU-funded innovation projects, particularly how it shapes the co-creation of knowledge across institutional and national boundaries. Using narratives from project experts, the study identified key complexity elements such as self-organization, interdependence, and co-evolution as central to successful innovation. Understanding the complexity of collaboration for innovation in relation to the input-throughput-output model of organisational communication is a contribution to theory that may help future projects achieve faster innovation.

4.2.2. CAMELOT

740736

C2 Advanced Multi-domain Environment and Live Observation Technologies

EXODUS ANONYMOS ETAIREIA PLIROFORIKIS (EL)

€8,020,921

01/05/2017 – 30/04/2021

https://doi.org/10.3030/740736

Programme
HORIZON 2020

Topic title
SEC-20-BES-2016 - Border Security: autonomous systems and control systems

border surveillance, Command and control, unmanned platforms, MSMDC2

Objective

The creation of the Schengen area has been one of the major achievements of the EU. However, this agreement requires countries to cooperate tightly in order to keep a high level of security at their internal borders, as well as to share the responsibility of managing external borders. Such a variety of borders (land, sea and air) and current challenges requires a consistent approach to border surveillance, based on a plethora of heterogeneous assets. These can be manned or unmanned, ranging from sensors (optical, radar, IR) to unmanned platforms (UAV, UGV, USV or UUV), and need to be combined to offer an integrated situational picture of the area under surveillance and of their location. In order to effectively control their operation and manage the large amounts of data collected by them, new approaches for command and control need to be considered, allowing efficient interaction between the operator and the different assets in the field. CAMELOT proposes to develop and demonstrate different advanced command and control service modules for multiple platform domains, based on a SOA architecture that specifies internal and external interfaces, allowing the development of a modular and scalable command and control station, customisable to the user needs. This architecture can be based on results of previous studies and work or open architectures that may prove more suitable and the interfaces can take advantage of the standardisation work that has been done already. After the definition, CAMELOT partners will prototype service modules according to their expertise, background individual technologies and practitioner needs. These will be integrated progressively in specific testing along the project. This prototype development approach will culminate in 2 final demonstrations involving end users and relevant stakeholders, to achieve a maturity of TRL6 (for most individual technologies supporting the functionalities for border surveillance) and an IRL of 7 for CAMELOT.

4.2.3. FOLDOUT

787021

Through-foliage detection, including in the outermost regions of the EU

AIT AUSTRIAN INSTITUTE OF TECHNOLOGY GMBH (AT)

€8,199,388

01/09/2018 – 31/08/2022

https://doi.org/10.3030/787021

https://foldout.eu/

Programme
HORIZON 2020

Topic title
SEC-16-BES-2017 – Through-foliage detection, including in the outermost regions of the EU

border surveillance, harsh environment

Objective

In the last years irregular migration has dramatically increased and is no longer manageable with existing systems. Improved methods for border surveillance are necessary to ensure an effective and efficient EU border management. FOLDOUT focus is on through foliage detection in the inner and outermost regions of the EU. Foliage penetration is an unsolved important part of border surveillance. By solving the problem of unreliable detections in such harsh environments border guards' workloads are reduced, costs are reduced and, last but not least, lives can be saved. Detecting people through dense foliage in extreme climates with only a penetration technology is prone to high fault rates.

FOLDOUT will build a system that combines various sensors and technologies and intelligently fuses these into an effective and robust intelligent detection platform. Fusing several sensor signals increases the effectiveness of detection. Further, sensors will be influenced (i.e. detection parameters adapted) by events detected by other sensors in the vicinity. By integrating data, such as vehicle traffic, from outside the immediate border area pre-events can be detected and learned. The events will be analysed with machine learning tools to continuously increase the systems detection and tracking capability. FOLDOUT will make the tasks of Border Guards simpler and faster by combining events from various sensors to give a complete situation threat assessment combined with suggested reaction scenarios.

A two-year pilot in Bulgaria and demonstrators in Greece, Finland and French Guiana FOLDOUT will provide fundamental enhancements in the domain of border surveillance and improved search & rescue scenarios.

Reported results

- The FOLDOUT system has been tested and evaluated in all demonstration sites (**Bulgaria, Greece, Finland, French Guiana**) under the most diverse environmental conditions. Results of the KPI evaluation are reported in each of the test reports D2.3 to D2.6.
- The SAR demonstrator airborne campaign was successfully accomplished and took place in November 2021 at the Bulgarian test site.

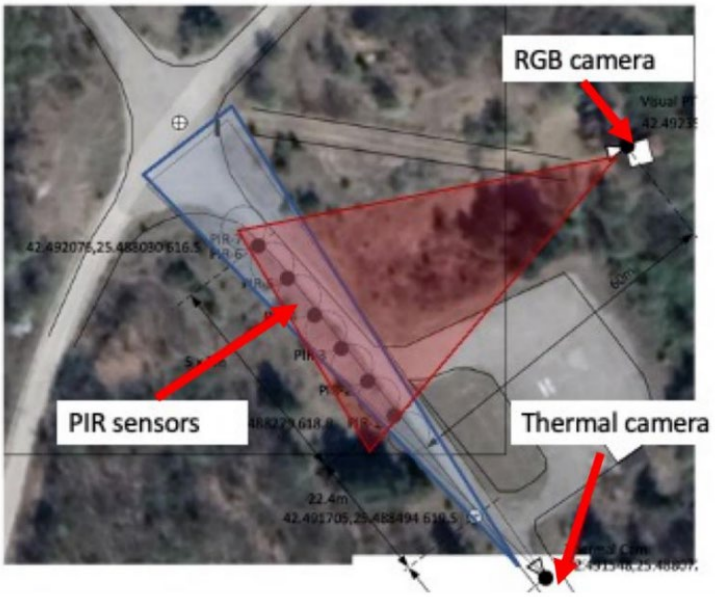
- Following sensors and detection technologies are part of the FOLDOUT system: **AIT Thermal camera, BHE EO camera, AIT smart camera platform, ITTI acoustic-seismic sensors, ONERA terrestrial LiDAR, ONERA Multispectral IR camera, ONERA LiDAR on un-manned aircraft, BHE RF detector, VTT active hyperspectral sensor, VTT radar fence**. Data collected by the airborne sensors are processed offline and integrated into the FOLDOUT framework.
- Novel algorithms based on machine learning techniques were explored to provide **automatic event detection**.
- Test studies and preparation of satellite and stratosphere missions are finalised.
- Five software components are fully developed and tested to improve situational awareness through fusion: **Fusion and Reasoning Platform, Registration Module, Fusion Module, Local Dynamic Maps and Situational Awareness**.
- All data information is standardised into a **Common Alerting Protocol** based data model and documented.
- A Mobile Command Centre has been developed as a base station for deploying the FOLDOUT System in remote areas.
- A GIS-based Command and Control platform has been developed that provides a map-based and layered approach, to ensure that all different pieces of information are all placed in one single interface, for advancing the operators' experience and efficiency when dealing with a complex situation.
- A public annotated dataset for visual tracking and surveillance has been made available to the scientific community and the analytics challenged launched at the PETS workshop 2021, organized by FOLDOUT within the AVSS 2021 conference.

Selected publications

Patino et al. (2022a) tackles the challenge of automatically detecting activities in forbidden areas, like forested land borders, which are critical for border control. The authors propose a **multi-sensor fusion** and tracking system to reduce false detections common with single sensors like radar. By integrating passive infrared detectors with automated person detection from **thermal and visual video cameras**, the system uses weighted maps and a rule engine to process data. Tested on real data from the EU FOLDOUT project at representative forested EU borders, the approach improves detection accuracy by up to 50%, effectively minimizing false positives from individual sensors.

Patino et al. (2022b) presents a **fusion and tracking system** developed under the FOLDOUT project, designed to assist border guards by combining sensor data and automatically tracking objects in surveillance areas. By integrating multiple sensor signals, the system enhances detection effectiveness, particularly in forested areas. The approach uses weighted maps, or heatmaps, to merge multi-sensor information, creating and updating tracks based on the cost of associating fused detections over time. The study compares tracking outcomes between individual sensors and fused data in a simulated EU border scenario in Bulgaria. Results indicate improved tracking accuracy when using fused data compared to individual sensor information.

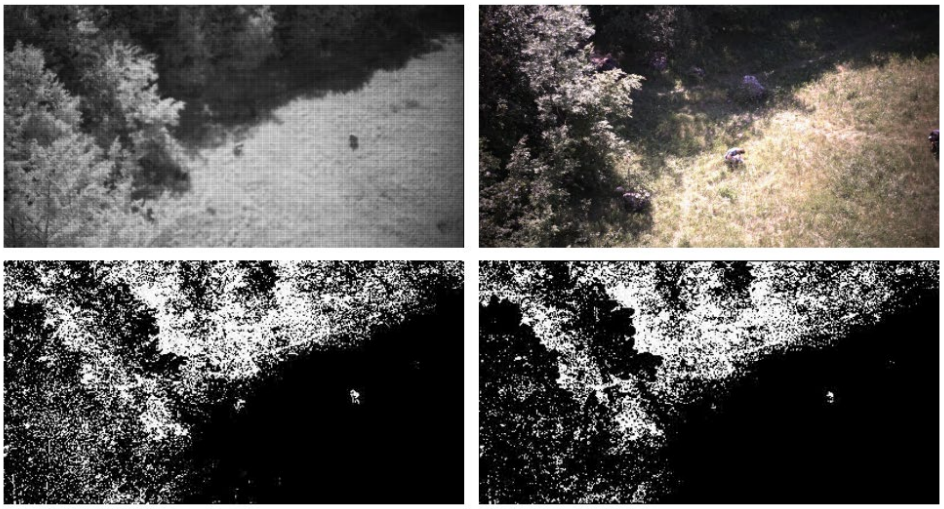
Figure 7. Simulated border and the sensors.



Source: Patino et al. (2022b)

Papp et al. (2020) addresses the challenge of **detecting people and cars in dense vegetation**, where traditional image and video processing faces difficulties due to factors like landscape and illumination changes. The authors propose using a **hyperspectral camera**, **potentially mounted on drones**, to detect objects based on spectral signatures. They introduce an automatic method for annotating these signatures using advanced deep learning techniques. By collecting millions of samples, they trained a classifier achieving a Matthews correlation coefficient (MCC) of 0.83 for detection accuracy. The study assesses this method across varying vegetation conditions and discusses the technological limitations, highlighting that while high accuracy is possible in uncontrolled outdoor settings, challenges remain due to high dynamic range illumination and low image resolution in hyperspectral imaging technology.

Figure 8. Matthews correlation coefficient for varying classification thresholds.



Source: Papp et al. (2020)

4.2.4. COMPASS2020

833650

Coordination Of Maritime assets for Persistent And Systematic Surveillance

DIRECCAO GERAL DA AUTORIDADE MARITIMA (PT)

€4,838,490

01/05/2019 – 31/10/2021

https://doi.org/10.3030/833650

Programme
HORIZON 2020

Topic title
SU-BES03-2018-2019-2020 – Demonstration of applied solutions to enhance border and external security

https://www.youtube.com/@COMPASSEUProject

Objective

The external borders of the EU have historically been under great pressure, subject to a variety of threats, which include irregular migration and trafficking of narcotics. Within this context, authorities in charge of border and maritime patrol are faced with different challenges that include the heterogeneity of the traffic that undertakes illegal activities in European waters, limitations in the ability to collect and share timely available data among institutional organizations, as well as a lack of assets by the relevant authorities to cover the wide maritime areas under their mandate. Although there has been an expressive investment done in the domain of surveillance technologies and tools, the intake by the competent authorities has been slow, due to lack of uniformity in the integration of such systems with existing surveillance infrastructures.

In order to address these challenges, project COMPASS2020 aims to demonstrate the combined use and seamless coordination of manned and unmanned assets to achieve greater coverage, better quality of information and shorter response times in maritime surveillance operations. The proposed solution will be based on an innovative CONOPS that makes use of multiple aerial and underwater unmanned vehicles with improved capabilities, deployed from OPVs or from land, and will be supported by a central, multi-domain and interoperable Mission System (MS) that enables the operation of these platforms from both locations. UxVs may act as deported ship sensors, providing critical mission data to the MS that can then be exploited through dedicated services to be developed in the scope of the project (e.g. Data Fusion and Threat Risk Analysis).

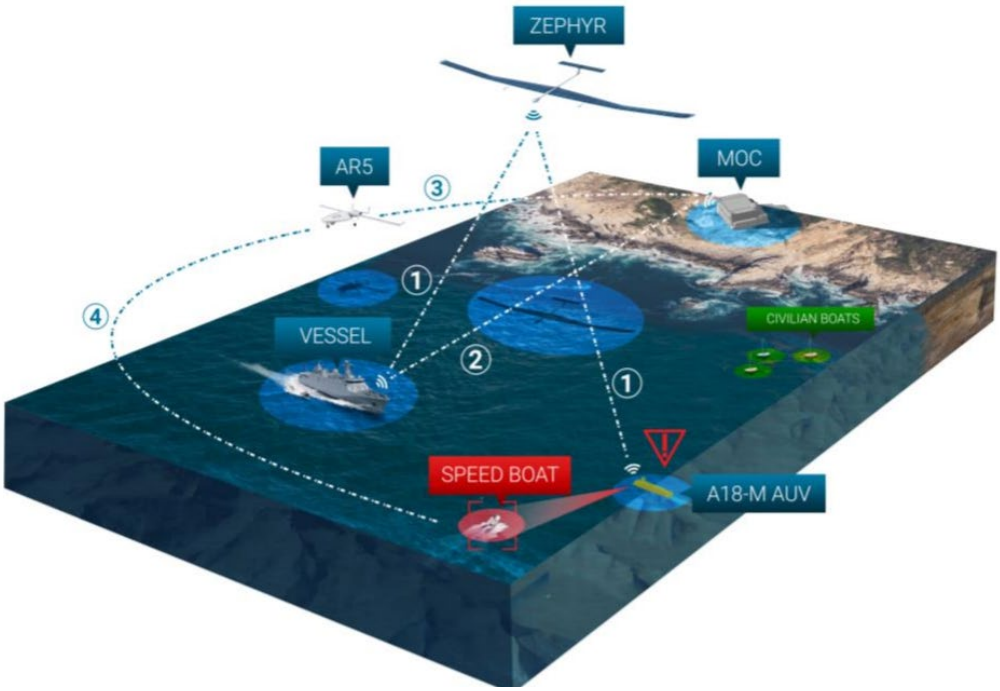
The major goal of COMPASS2020 is to demonstrate an operational solution to ensure long range and persistent surveillance, increasing the situational awareness of coast guards and maritime authorities, and, thus, increasing the cost-effectiveness, availability and reliability of the operations.

Reported results

COMPASS2020 laid the foundation for a 'system of systems' by demonstrating the ability of existing offshore patrol vessels (OPV) to autonomously launch and operate unmanned aircraft and autonomous underwater vehicles (AUVs). The unmanned aircraft and AUVs significantly enhance OPVs' surveillance range and the types of data available to them. The OPVs can also receive data from long-endurance unmanned aircraft deployed from shore. The project integrated many systems, including an AIRBUS system that enables automated take-off and landing of unmanned aircraft from

a moving offshore patrol vessel. COMPASS2020 demonstrated that data from different available aerial and underwater systems can be integrated to form a wide-area, persistent maritime surveillance network.

Figure 9. COMPASS2020 concept of operations.



Source: Cordis

Selected publications

Bauk et al. (2020a) presented the key features of unmanned aircraft systems (UAS) deployed in the context of maritime safety, search and rescue at sea, controlling irregular migration and securing European maritime borders from narcotics smuggling. This paper considers only the aerial segment, i.e., the Zephyr pseudo-satellite, as well as the AR3 Net Ray and the AR5 Life Ray Evolution crafts. Comparative advantages and disadvantages of these systems are presented.

Bauk et al. (2020b) gives an overview of autonomous underwater vehicles (AUVs) for marine surveillance and reconnaissance missions in European seas. The AUVs navigate autonomously due to the dedicated navigation algorithms and collect surrounding information. Basic data on the AUVs are given, along with some general directions for further research in the filed with the aim of achieving more efficient and environmentally friendly underwater missions in the future.

4.2.5. EFFECTOR

883374

An End to end Interoperability Framework For MaritimE Situational Awareness at StrategiC and TacTical OpeRations

Secrétariat général de la mer (FR)

€4,999,529

01/10/2020 – 30/09/2022

https://doi.org/10.3030/883374

https://www.effector-project.eu/

Programme
HORIZON 2020

Topic title
SU-BES03-2018-2019-2020 - Demonstration of applied solutions to enhance border and external security

Interoperability, Information systems, Maritime Surveillance Systems, Assets, CISE, EUROSUR, Situational Awareness, National, Regional, Local Coordination Centers

https://youtu.be/3-PI4Vms0fs?si=Qmfw8x3jjm3PWhpT

Objective

EFFECTOR aims to enhance maritime surveillance, improve decisions support, and foster collaboration of maritime stakeholders by implementing an Interoperability Framework and associated Data Fusion and Analytics services for Maritime Surveillance and Border Security that will allow faster detection of new events, better informed decision making, achievement of a joint understanding and undertaking of a situation across borders, allowing seamless cooperation between operating authorities and on-site intervention forces ensuring that all existing privacy and data protection rules are fully respected. Specifically, EFFECTOR will unlock the full capabilities of maritime surveillance systems and data sharing at tactical and strategic level by introducing applied solutions for enhanced border and external security, including the implementation of a multilayered data lake platform for end-to-end interoperability and data exploitation, the exchange of enhanced situational awareness pictures at different level with CISE and EUROSUR, the adoption of interoperability standards for exploiting data sources and systems currently underutilized in maritime environment and the demonstration of new concepts and tools for knowledge extraction, semantic representation, data fusion, analytics, and federated querying that can scale from local to regional and up to national and transnational level.

The EFFECTOR solution will be tested, validated and demonstrated in real operational scenarios together with maritime authorities, End Users and practitioners in France, Portugal and Greece.

The project will leverage on the developments, results and experience from current and previous research projects (EUCISE2020, MARISA, RANGER, PERSEUS, BLUEMASSMED), from National Procurement projects of CISE Nodes and Adaptors and on the CISE infrastructure of the End Users.

Reported results

EFFECTOR developed and demonstrated a multi-layer data lake architecture that enables the integration and processing of heterogeneous maritime data in real time. The system incorporates tools for knowledge extraction, semantic representation, AI-supported analytics, and federated querying, allowing information to be shared and exploited at local, national, and transnational levels. A key achievement was the development of a CISE–JORA adaptor, bridging EFFECTOR's framework with the European Common Information Sharing Environment (CISE) and Frontex's Joint Operation Reporting Application (JORA). This ensured seamless interoperability with EUROSUR and enhanced anomaly detection services.

The system significantly extended maritime domain awareness by fusing traditional vessel tracking and radar data with satellite communications, Copernicus Earth observation services, national vessel traffic systems, and even geo-localised images shared by individuals. Field trials in Greece and France demonstrated its ability to support rapid detection, cross-border collaboration, and operational response. In one case, authorities intercepted a migrant smuggling operation within hours, triggered by a fisherman's photo upload. EFFECTOR thus proved its value in accelerating detection, improving decision-making, and enabling coordinated responses across agencies, while fully respecting data protection requirements.

Selected publications

Concerning **interoperability** and data management centres on extending the Common Information Sharing Environment (CISE) with big data infrastructures. Mihailovic et al. (2022) describe trial scenarios for shared data retrieval across borders using a "data lake" concept. Paladin et al. (2022a) present related contributions on integrating a big data framework within CISE, applying multilayered and semantically enabled data lakes for maritime interoperability (Paladin, 2022b). A case study of data lake deployment in operational trials is reported in Paladin, 2022c.

A second group of articles examines **unmanned aircraft platforms**. Douklias et al. (2022) outline the design of a UAV system capable of running computer vision and machine learning applications onboard. Vasilopoulos et al. (2022a, 2022b) evaluate and implement object detection algorithms for autonomous identification and tracking of maritime objects, with applications in search and rescue and safety operations.

Finally. Paladin et al. (2022d) review the **use of AI** modules for vessel behaviour monitoring, anomaly detection, and collision risk assessment, integrated into the EFFECTOR architecture. These include trajectory fusion tools and early warning mechanisms designed to detect anomalies and provide better decision support and proper information to be distributed among CISE maritime safety stakeholders.

4.2.6. EURMARS

101073985

An advanced surveillance platform to improve the EURopean Multi Authority BordeR Security efficiency and cooperation

EUROPEAN DYNAMICS LUXEMBOURG SA (LU)

€5,884,214

01/10/2022 – 30/09/2025

https://doi.org/10.3030/101073985

https://eurmars-project.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-01 – Enhanced security and management of borders, maritime environment, activities and transport, by increased surveillance capability, including high altitude, long endurance aerial support

High Altitude Platforms, Border Risk Management, Multi-Authority Cooperation

Objective

Security risks and threats in the maritime domain are becoming increasingly more complex day by day. Within the EU there have been significant increases in irregular migration flows and human trafficking and smuggling but also of other illegal activities, such as drugs and arms trafficking and illegal, unreported and unregulated fishing, and –according to the FRONTEX "Risk Analysis for 2021" report- the challenge becomes even bigger. The European Union Maritime Security Strategy, and other EU initiatives such as EUROSUR and CISE, highlight that closer collaboration between authorities at national, regional and EU levels can constitute the key action to increase both situational awareness and operational efficiency. In particular, there is increased interest in assessing the viability of multi-authority cooperation in (especially high altitude) sensing and capability advancement for enhanced wide area surveillance of land, air and sea borders.

EURMARS's ground-breaking vision is to expand the common risk assessment practices currently deployed by authorities to enable the development, deployment and evaluation of a secure multitasking surveillance platform that improves sensing capabilities for a wide range of security risks and threats in wider border areas by clustering high altitude platforms technology, satellite imagery, UxVs and ground-based sensors into a novel joint surveillance capability.

As part of the EURMARS framework, the various existing and future systems for maritime surveillance will be integrated, to allow for the collaborative operation and the provision of the sensing results to related authorities. The open architecture will build on the lessons learnt of previous initiatives, assimilate the knowledge of the stakeholders and their practice on CISE and other relevant systems, exploit the latest AI, risk assessment and visualization innovations, and undergo extensive technical and user acceptance tests and ethical and legal impact assessments.

4.2.7. I-SEAMORE

101073911

Integrated Surveillance Ecosystem for European Authorities Responsible for Maritime Operations Leveraged by Reliable and Enhanced Aerial Support

ATOS IT SOLUTIONS AND SERVICES IBERIA SL (ES)

€6,481,677

01/01/2023 – 30/06/2025

https://doi.org/10.3030/101073911

https://iseamore-project.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-01 – Enhanced security and management of borders, maritime environment, activities and transport, by increased surveillance capability, including high altitude, long endurance aerial support

https://www.youtube.com/@i-seamore

Persistent maritime surveillance, Aerial platforms, Interoperability, Detection, identification and tracking, multi-authority collaboration, multi-tasking capabilities, situational awareness

Objective

I-SEAMORE is an Ecosystem composed of an advanced platform solution to host and manage the operation of several innovative assets, services and systems that aim to provide European Authorities with increased situational awareness and operational capabilities for Maritime surveillance operations resorting to aerial and water surface support. The core platform (infrastructure and software layers) is conceptualized to be deployed and operated at Maritime Operation Centres (MOCs) with interfaces to other systems including the UxVs Ground Control Stations (GCSs), as well as external systems. It thus provides end-users with a holistic platform capable of handling several multipurpose tasks including, e.g., wide maritime border and coastal areas monitoring, analysis of potential threats, support to search and rescue operations, detection of illegal activities, among others.

Such tasks will be possible since I-SEAMORE platform provides a complete set of functionalities and capabilities to mission commanders, focusing on 4 main pillars: 1) employment and indirect tasking of multiple types of long-endurance Unmanned Assets (aerial and water surface), 2) exploitation of heterogeneous data sources e.g. payload data and open data sources including Copernicus Services, 3) provision of a common operational picture empowered by a novel and comprehensive suite of data fusion services based on Artificial Intelligence (AI) and Big Data Analysis, for optimal decision making and successful mission execution of the desired missions, and 4) interoperability within the Ecosystem and its interface with key existing external systems.

Moreover, the project will also generate additional knowledge to support the uptake of the solution at EU level, as well as multi-country, multi-authority collaboration, including novel concepts of operation, standard operating procedures for joint operations, and new methodologies for co-creation and validation of maritime security solutions by end-users.

Reported results

I-SEAMORE has reported that improvements are being implementing for the unmanned assets, starting with the integration of additional payloads, such as software defined radio (SDR)-based sensor for augmented SIGINT capabilities (UA) and IR camera for enabling nocturnal operations (USVs). Other innovations include developing appropriate algorithms for on-board processing of data from the new payloads, improving the navigational performance of unmanned aircraft by using EGNOS, as well as augmenting the endurance and decreasing the acoustic signature of part of the unmanned aircraft by modernizing their current power systems for stealthier operations. Interoperability is achieved with the development of a CISE adaptor for further integration with external compliant systems.

Selected deliverables

The following project deliverables are available to download from the project's Cordis webpage[7]:

- Assessment of I-SEAMORE UxVs and Payloads Capabilities: Catalogue of all the UxVs and payloads brought by I-SEAMORE partners, including their operational capabilities.
- Analysis of citizens' awareness and acceptance of EU maritime security system: Results of the activities performed regarding citizens' awareness and acceptance.
- Monitoring and Evaluation/Validation Framework: This report defines the overall methodology for the implementation of monitoring and evaluation/validation activities.
- Model based State-of-the-art: Model based SotA resulting from the activities performed regarding concepts and capabilities for Maritime Surveillance
- Preliminary Data Sets from UxVs & Copernicus: Preliminary experimental data sets from UxV payloads and Copernicus services.

Reported results

Hjelmervik et al. (2024) described the use of both conventional methods and deep learning (DL) algorithms for detecting the presence of targets close to underwater acoustic sensors. The DL leverages convolutional neural networks (CNNs) to recognise patterns and anomalies in complex sensory data like sonar signals. By automatically extracting features from raw data, CNNs can identify subtle acoustic signatures of different types of objects, distinguishing between benign and targets. An augmentation technique was used to generate training data by mixing hydrophone recordings of ships with ambient noise. The ShipsEAR dataset[8] was used to demonstrate the overall method.

[7] https://doi.org/10.3030/101073911

[8] http://atlanttic.uvigo.es/underwaternoise

4.3. Vessel tracking and anomaly detection

4.3.1. ISOLA

883302

Innovative & Integrated Security System on Board Covering the Life Cycle of a Passenger Ships Voyage

AIRBUS DEFENCE AND SPACE SAS (FR)

€ 7,006,081

01/09/2020 – 30/04/2024

https://doi.org/10.3030/883302

https://isola-project.eu

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 - Technologies to enhance border and external security

Maritime, Cruise ships, Security, Monitoring, Detection, Threat Recognition

https://www.youtube.com/@ISOLAProject-ud2sv

Objective
ISOLA project combines state-of-the-art technologies blended with novel solutions to enhance the Situational Awareness and support the assigned Ship Security Officer & Crew to the execution of their duties especially to the Ship Security Plan. The suggested system is contextually aligned and harmonised with the existing regulations. Constraints to integration will also be looked at with attention given to key architecture framework criteria: standardisation, interoperability, openness, flexibility, adaptability, ability to evolve, scalability, modularity, simplicity, usability, robustness, information management, decision support and mission effectiveness, and legacy integration. ISOLA develops, integrates, tests, deploys, demonstrates and validates a systematic and fully automated security approach by incorporating innovative technologies for sensing, monitoring, data fusion, alarming and reporting real-time during illegal incidents. This will ensure high level of security among all the passengers of the ship and augmentation of the Ship Security Plan. The main objectives of ISOLA are: (i) to create strategies and methods in order to a ship to easily integrated solutions regarding passengers and crew safety in the existing ship systems; (ii) to propose innovative sensor and visual technologies to support security safety; (iii) to create a complex collaborative system for monitoring and detecting security incidents and events; (iv) to create early warning methods for the ship security crew to prevent security issues; (v) to collect incident evidences by exploiting the Augmented Reality; (vi) to allow easy engagement of different authorities in a ship related crisis; (vii) to model, classify and easily report a security event.

Reported results

The project identified five use cases: detection of intoxicated persons, detection of theft, piracy attack, illegal boarding/stowaway, and cyberattack. The reported tools include (i) visual analysis for threat detection, (ii) crowd monitoring, (iii) passenger and crew identification and tracking system, (iv) hazard dispersion prediction and reporting, (vi) swarm of UA used for monitoring.

4.3.2. AI-ARC

101021271

Artificial Intelligence based Virtual Control Room for the Arctic (AI-ARC)

LAUREA-AMMATTIKORKEAKOULU OY (FI)

€6,889,792

01/09/2021 – 29/02/2024

https://doi.org/10.3030/101021271

https://ai-arc.eu/

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 - Technologies to enhance border and external security

Search and Rescue, AI, EU CISE, Security, Safety, Arctic, Sea, Cooperation, Anomaly Detection, Surveillance, Communication, Risk Assessment, Information, Virtual Reality, Information, Interoperability

https://www.youtube.com/@ai-arc5269

Objective

The AI-ARC proposal presents a highly innovative and user-friendly artificial intelligence (AI) based platform known as the Virtual Control Room (VCR). Due to the vast amounts of information collected the potential for information overload is real. This reality can complicate the operational picture; reduce situational awareness and often results in delayed and impaired decision-making. On the other hand, areas such as the Arctic Sea suffer from a lack of communication, surveillance data and rescue assets and without action taken to address these vulnerabilities, the consequences are potentially dramatic in terms of accidents, pollution, border infringements and criminal activities.

The AI-ARC VCR supports all these challenges by applying AI, machine-learning and virtual reality (VR) technologies to filter numerous validated and statistical data streams and databases to a user-friendly interface. The VCR improves situational awareness by assisting end users to customize a “smart” operational picture. The VCR will permit users to specify their preferences in terms of threat levels, abnormal behaviour, interoperability and risk management by flagging detected anomalies with confidence and providing threat or risk levels according to a predefined model based on user preferences. This means that users can create awareness for their own purposes that reflects their needs without increasing their workload.

AI-ARC‘s principal objectives align fully with the H2020 BES-SU-open, and are of crucial relevance to it. The Virtual Control Room (VCR) has the power to greatly improve maritime situational awareness, decision-making, communication, available rescue resources, and thus the safety of all maritime actors, particularly in the Arctic Sea. Furthermore, the enhanced communication and collaboration provided by AI ARC’s innovative technology encourages and enables further development of symbiotic services and fosters much needed Arctic cooperation.

Selected publications

Sheikh et al. (2023a) developed a two-stream Convolutional Neural Network (CNN) to estimate sea ice concentration in the Baltic Sea by fusing Sentinel-1 radar data (high spatial resolution) with AMSR2 radiometer data (wind-independent). The model used focal loss and skip connections to improve performance, achieving an R^2 score of 90.6 %, demonstrating the potential of AI-based sensor fusion for operational ice monitoring. Similarly, Sheikh et al. (2023b) proposed a CNN fusion model for ice-water detection using Sentinel-1 SAR and AMSR2 data. By combining complementary imaging modalities, the model produced pixel-wise classification maps of sea ice, improving accuracy and edge sharpness over single-modality approaches. The fusion approach achieved 94.6 % pixel accuracy and a 94.99 % F1-score, supporting automated sea ice chart generation.

Farahnakian et al. (2022) reviewed 45 studies on machine learning methods for detecting anomalous vessel behaviours in maritime environments. Techniques were categorised by method (e.g., supervised, unsupervised learning) and data type (e.g., AIS, radar, satellite). The review outlined research challenges and highlighted the role of AI in developing automated frameworks for identifying illegal or unsafe maritime activities. Farahnakian et al. (2023) used AIS data from the Baltic Sea to evaluate clustering methods – K-means, DBSCAN, Affinity Propagation, and Gaussian Mixtures – for detecting abnormal vessel behaviours. The focus was on “dark ships” (AIS switched off) and spiral movement patterns linked to illicit activities. Results showed that K-means clustering effectively identified these behaviours, demonstrating how unsupervised AI techniques can support maritime anomaly detection.

Figure 10. The AI-ARC project aimed to develop the Virtual Control Room (VCR) based on artificial intelligence to provide situational awareness in the Arctic Ocean region.



Source: Cordis (© wifesun/stock.adobe.com)

4.3.3. PROMENADE

101021673

imPROved Maritime awarENess by means of AI and BD mEthods

MINISTRY OF MARITIME AFFAIRS AND INSULAR POLICY (EL)

€4,997,556

01/10/2021 – 31/03/2023

https://doi.org/10.3030/101021673

https://www.promenade-project.eu

Programme
HORIZON 2020

Topic title
SU-BES03-2018-2019-2020 – Demonstration of applied solutions to enhance border and external security

https://www.youtube.com/@Promenade-EUHResearchInnovatio

Maritime Domain Awareness, vessel, ship, Artificial Intelligence, Big Data, High Performance Computing, classification, behaviour analysis, pattern detection, anomaly detection, risk assessment

Objective

Maritime Domain Awareness is the combination of activities, events and threats in the maritime environment that could have impact on marine activities and EU territory. During the past decades, advances in Information and Communication Technologies have provided better means to monitor and analyse vessel activity. Today private and public source of data such the Automatic Identification System or space related data can be combined with Vessel Traffic Services, Vessel Traffic Management Systems and Vessel Traffic Monitoring & Information Systems data enabling the development of value added information resulted by the combination of such data.

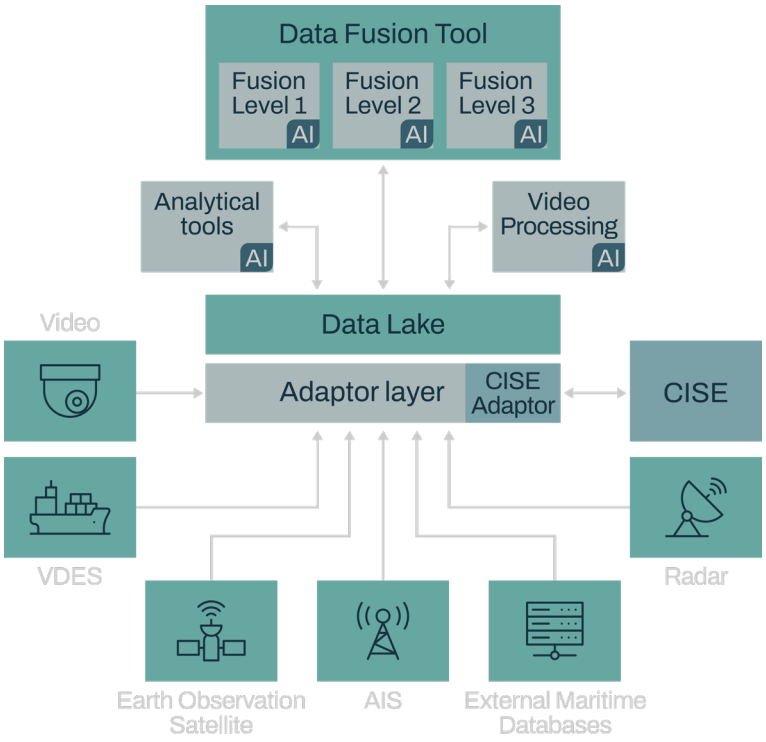
European waters are navigated daily by some 12,000 vessels, which share their positions to avoid collisions, generating a huge number of positional messages every month. It is important that this overabundance of information will not overwhelm the marine operator in charge for decision-making. The challenge is twofold: on one side encourage the exchange of heterogeneous data among administration valorising the CISE network currently in place, on the other exploit at the best these datasets by means of automated processing in a way to minimise false alert that might results by an incorrect processing or interpretation of the results.

PROMENADE will improve solutions for the vessel tracking, behaviour analysis and automatic anomaly detection by means of the application Artificial Intelligence (AI) and Big Data (BD) technologies, and to promote collaborative exchange of information between maritime surveillance authorities, shortening the time to market and assuring the compliance with legal and ethical regulations. An open, service-based toolkit implementing “state of art” AI / BD techniques also benefiting of HPC (High Performance Computing) platform is the core activity of the project. The project’s developments will be demonstrated and evaluated in 3 operational scenarios and 1 simulated defined by Border Guards Authorities.

Reported results

The results of the PROMENADE have been reported in the literature, and the following description of results is adapted from Astyakopoulos et al. (2025). In PROMENADE, various AI-based services for maritime awareness were developed and demonstrated in operational trials. These services use machine learning and deep learning models that have been trained on historic maritime data sets, including Radar, automatic identification system (AIS), images, videos, maritime databases, and OSINT data sources. Large datasets representative of real-world scenarios were used in most cases. Leonardo S.p.A. made available its own high powered computing facility, DaVinci-1, during the data and computing intensive phases of AI model development and training. PROMENADE developed a data lake infrastructure, leveraging open-source big data technologies and acting as the backbone of the system for data ingestion, data storage, data analytics, data, and streaming to integrate diverse data formats.

Figure 11. PROMENADE high-level architecture.



Source: Cordis

Overall, the PROMENADE toolkit includes 22 AI-based and CISE compliant services, for example, vessel detection, route classification, and trajectory prediction. Trials took place in Greece in February 2023. The overall outcome of the AI services tested with real vessels during the Hellenic trial was very positive. The main lesson learnt was related to the dynamic adjustment of the system alert parameters to depict the 100% operational challenges that the end users have to face in the field. Participant feedback confirmed the use of AI and big data provides a significant advantage in upgrading deprecated systems and achieving early detection of targets by detecting abnormal behaviour.

4.3.4. SEAGUARD

101168489

Sea Environmental Awareness and Guard enhanced with Unmanned AI Robotic Detection

INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA (PT)

€4,463,268

01/10/2024 – 31/03/2027

https://doi.org/10.3030/101168489

https://theseaguardproject.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-BM-01-01 – Capabilities for border surveillance and situational awareness

Objective

Illegal activities across external borders (land and maritime) are a threat to the security of the European Union. Better border surveillance capabilities are required to deter and monitor irregular border crossings. These should be interoperable across member states, ranging from the Mediterranean to the Nordic Countries, which have different border surveillance challenges. These should also be scalable, flexible and relocatable, since the needs along borders shift rapidly in time, with changing and improving threats.

SEAGUARD project aims to address these needs. Situational awareness technologies will play a crucial role in the future of maritime border security by providing real-time information about the maritime environment, including vessel movements, cargo, and potential threats. The SEAGUARD project aims at developing an innovative and integrated approach to maritime cross-border surveillance that will enhance border security and management in Europe. The scope of this project includes the design, development, testing, and deployment of a cross-border surveillance system that will incorporate multiple sensing technologies integrated in both fixed and mobile platforms. This project will combine unmanned vehicles (Unmanned Aerial, Surface and Underwater Vehicles - UxVs), fixed buoys, (SMART) submarine cables, with advanced analytics and decision support. The system will be able to monitor and detect various types of threats, including illegal border crossings, smuggling of contraband, illegal fisheries and terrorist activities.

The project will leverage new technologies and data analytics, enhanced by multi-domain autonomous cooperative platforms and multimodal communications, to provide real-time information and situational awareness to multiple border guards and other authorities, enabling them to detect and respond to potential threats coming from the air, surface or underwater, more effectively.

4.3.5. AI4COPSEC

101190021

Security enhancement through heterogeneous data fusion and improved AI/ML-powered Copernicus maritime and border surveillance services

SIMULA RESEARCH LABORATORY AS (NO)

€4,044,013

01/12/2024 – 30/11/2027

https://doi.org/10.3030/101190021

https://ai4copsec.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL4-2024-SPACE-01-36 – Copernicus for Security

ML, GeoAI, AIS, EO and non-EO data, OSINT data, Ocean conditions, SaR, Oil spill response, irregular migration, Illegal fishing, AI, data fusion, non-space data, maritime surveillance

Objective

The Copernicus Security Service (CSS) is instrumental to support the European Union's security, safety, law enforcement, and international commitments with the European Commission advocating for its advancement. The AI4COPSEC project aims to demonstrate the potential of advanced ML and GeoAI models to provide relevant intelligence for enhancing existing Copernicus products and services such as oil spill response, illegal fishing alert and create new services intended to support Search-and–Rescue operations and irregular migration detection.

The project will enhance CSS services through the usage of self-supervised deep learning models, geomatics and social media data (OSINT) extracted from heterogenous multi-sources data. AI4COPSEC is expected to have a transformative effect on the operational capabilities of EU security, surveillance and safety by leveraging advances to enhance the timeliness, accuracy, and relevance of information derived from EO and non-EO data sources, including the innovative use of thermal imagery for ship detection, the combination of satellite automatic identification system (AIS) data with structured data extracted from social media sources and the usage of in-situ environmental measurement derived from AIS data and IoT devices. This integration will lead to more effective and efficient security operations, environmental monitoring, and disaster response.

AI4COPSEC is also set to detect anomalies and threat in the maritime traffic and advance ship detection capabilities, utilising high-resolution optical images and advanced segmentation algorithms to enhance the detection of small vessels independently of the type of material (metallic structures, wood, rubber...), a critical factor for a comprehensive maritime surveillance in line with the CSS and the EU's maritime security strategy (EUMSS).

4.4. Underwater and infrastructure protection

4.4.1. UnderSec

Underwater Security | **101121288**

FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV (DE)

€ 5,986,393

1/10/2023 – 30/9/2026

https://doi.org/10.3030/101121288

https://undersec-project.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-01 – Improved underwater detection and control capabilities to protect maritime areas and sea harbours

Objective
Ensuring maritime security is of paramount importance to guarantee the safe transportation and continuous operation of coastal, river, and maritime infrastructure. Implementing round-the-clock underwater security surveillance to deter intruders, conducting on-demand hull inspections, and inspecting ports underwater are all essential measures. The EU-funded UnderSec project aims to develop a modular and holistic prototype system featuring multimodal sensors and robotic assets. This system will provide underwater security, enhance situational awareness, and offer decision-making support, including response capabilities for ships, ports, and maritime infrastructures. The project will conduct testing and comparative analysis of the technology in a controlled environment and evaluate the system through real-life demonstrations, leading to the generation of new knowledge and innovative solutions.

Reported results

- Design, develop, and deploy the first layer of protection technologies for continuous underwater surveillance, threat detection, verification, and deterrence actions. It focuses on integrating sensing technologies with the UnderSec platform to process real-time raw data. Designing and developing the second and third layers of protection robotic technologies for autonomous and remote scanning, automatic detection, classification.
- Advances underwater acoustics through controlled experiments, creating a comprehensive database of ship hull surfaces and anomaly positions. This data will drive **machine learning algorithms and automatic detection systems**. It also includes pre-evaluation testing and comparative analysis of underwater sensing technologies to optimize integration, refine pilot validation scenarios, and streamline sea trials.
- Integrating the UnderSec system's hardware and software modules in both modular and holistic setups, ensuring real-time data exchange and seamless operation. The package aims to design a data management framework encompassing acquisition, harmonization, collection, fusion, processing, analysis, exchange, and storage.

– Stress-tests the UnderSec system through a pre-evaluation phase and two field demonstration phases. It validates the technical and operational relevance of four PUC scenarios, designs an evaluation framework, and conducts demonstrations across these real-life scenarios. Additionally, the package delivers a comprehensive training program for efficient system utilization and integration into existing CONOPS.

4.4.2. SMAUG

101121129

Smart Maritime and Underwater Guardian

INDRA SISTEMAS SA (ES)

€5,998,864

1/10/2023 – 30/9/2026

https://doi.org/10.3030/101121129

https://smaug-horizon.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-01 – Improved underwater detection and control capabilities to protect maritime areas and sea harbours

https://www.youtube.com/@SMAUG_EU

Objective
The primary goal of SMAUG is to improve the underwater detection of threats in ports and their entrance routes, by means of a integrated system capable of providing data concerning threat analysis between 3 main elements: ports security infrastructure, advanced underwater detection systems and surveillance vessels. Underwater detection and location will be performed by four primary methods: i) acoustic detection, where a series of hydrophones will listen for sounds emitted by small underwater vehicles and will be processed by artificial intelligence methods, ii) rapid sonar hull scan, used to scan ships hulls and perform harbour floor scanning, iii) high resolution sonar inspection, to inspect objects in water with poor visibility and iv) collective autonomous location, where a swarm of autonomous underwater vehicles will act cooperatively. This will provide information to Artificial Intelligence modules which will improve the way detecting illicit and dangerous goods and/or of threats hidden below the water surface is currently done, taking into account sources such as Unmanned Surface Vehicle Systems, (USV), underwater remote operation vehicle (ROV), UAV (unmanned aerial vehicle) and Port current information sources. The combination of these tools will allow SMAUG to prompt solutions capable of detecting possible threats to infrastructure or vessels, as well as identify vessels with concealed goods.

4.4.3. VIGIMARE

101168016

Vigilant Maritime Surveillance of Critical Submarine Infrastructure

LAUREA-AMMATTIKORKEAKOULU OY (FI)

€4,749,863

01/09/2024 – 31/08/2027

https://doi.org/10.3030/101168016

https://vigimare.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-INFRA-01-02 – Supporting operators against cyber and non-cyber threats to reinforce the resilience of critical infrastructures

Critical Infrastructure, Maritime Security, Energy, Anomaly Detection, Risk Management, Vulnerability assessment, Cyber threats, Hybrid threats, CISE, AI, Virtual Control Room

Objective

The European submarine networks are vital critical infrastructure for the EU member states. This submarine cable and pipeline network is crucial for the European society and any failure or damage to it could potentially have an enormous effect on the societies. Due to the rapidly evolving threat and geopolitical landscape, especially with incidents like the sabotage of the Nord Stream 1 and 2 gas infrastructure, organizations overseeing critical infrastructure face significant challenges. They must handle intentional threats while also grappling with growing problems caused by accidental and natural factors as well as climate change. These issues can worsen the effects of deliberate actions. To address these challenges effectively, it is crucial for these entities to implement resilience-enhancing measures.

The aim of VIGIMARE is to strengthen the resilience of Critical Infrastructure operators against threats to the European submarine critical infrastructure by developing, testing and evaluating an innovative solution to enhance security and reduce risks of physical and cyber-attacks on submarine cables and pipelines. This will be accomplished by 1) mapping submarine systems, surveying cables and landing stations, and assessing vulnerabilities in the European submarine critical infrastructure network, 2) develop an automated threat detection approach, creating a real-time shared awareness of the surface above the submarine infrastructure to promptly indicate incidents to both critical infrastructure owners and member states authorities. By integrating a comprehensive resilience ecosystem with real incidents at sea, the project seeks to 3) identify early warning signals, support analysis, and pinpoint potential response activities to address physical, cyber and hybrid threats. The result will be enhanced resilience for submarine critical infrastructure owners, supporting authorities to enforce the CER and NIS2 directives implementation in the EU member states.

4.5. Environmental aspects

4.5.1. AXOLOTL

101158669

From sky to seafloor observation: Achieving eXcellence in Oceanic surveiLlance and cOnservation Through deep Learning

CMMI CYPRUS MARINE AND MARITIME INSTITUTE (CY)

€1,496,433	Programme HORIZON EUROPE
01/09/2024 – 31/08/2027	Topic title HORIZON-WIDERA-2023-ACCESS-02-01 – Twinning Bottom-Up
https://doi.org/10.3030/101158669	

Artificial Intelligence, Automatic Identification Systems, Autonomous reef monitoring structures, Computer vision, Deep learning, Earth observation, Marine conservation, Maritime surveillance

Objective

Technological areas such as Artificial Intelligence (AI) and ecosystems such as shipping, maritime and space have been strategically prioritised by Cyprus to improve its Research and Innovation (R&I) performance. At their intersection lies the need to enhance marine conservation efforts and maritime surveillance by leveraging deep learning (DL) methodologies built on standardized and robust data sets. Indeed, in-situ sampling stands as a cornerstone in marine conservation, offering a direct approach to monitoring marine biodiversity; while remote sensing stands as a pivotal addition to maritime surveillance, expanding the scope beyond traditional Automatic Identification System capabilities. DL, as a cutting-edge AI tool, holds immense potential to enhance the analysis of in-situ samples and remotely sensed data.

AXOLOTL is proposed as a transformational international endeavour capable of enhancing the R&I capacity of CMMI, Cyprus and Europe in the interdisciplinary fields of DL-enhanced in-situ biodiversity assessment and DL-enhanced remote sensing for maritime surveillance. The project will contribute to closing the gap between a H2020 Teaming Centre of excellence and 2 strong innovators from France and Belgium through capacity-building, knowledge transfer, networking, and outreach activities at regional and international levels. Activities will go beyond the strictly scientific scope and support the mutual development, consolidation, and reinforcement of administrative, dissemination and entrepreneurial competencies, access to networks of excellence and the sustainable linkage between partners. The project's R&I component will develop new strategies for improving data quality, standardization, and synchronization issues, devise novel interdisciplinary methodologies, develop robust DL models from state-of-the-art computer vision methods, and validate its proposed solutions in 2 relevant real-world contexts (biodiversity assessment and maritime surveillance).

Reported results

Hadjipieris et al. (2025) describe the use of AI to identify and map port berths. Information about these berths is often missing or inaccurate. They propose an unsupervised spatial modelling method

that leverages automatic identification system (AIS) data clustering and hyperparameter optimization to identify berthing sites. Their models significantly outperform competing methods, revealing more precise berth boundaries and improved spatial resolution across diverse port environments.

4.5.2. MMinE-SwEEPER

101167839

Marine Munition in Europe - Solutions with Economic and Ecological Profits for Efficient Remediation

HELMHOLTZ-ZENTRUM FUR OZEANFORSCHUNG KIEL (GEOMAR) (DE)

€5,916,096

01/10/2024 – 31/03/2028

https://doi.org/10.3030/101167839

https://mminesweeper-munition.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-BM-01-02 – Identify, inspect, neutralise Unexploded Ordnance (UXO) at sea

hydroacoustic seafloor mapping, optical object detection, AI-supported data analyses, smart robots, secure data exchange, knowledge compilation, training and capacity building, stakeholder co-design

Objective

The MMinE-SwEEPER project will advance knowledge, capability and capacity in Europe for dealing with marine munition in the non-military aspect of UXO-clearance. 21 partners from 7 EU and 2 associated countries bring experience and capabilities from the civil science community (9), military research (3), coast/border guards or EOD-services (4) as well as industry (3) and intergovernmental organizations (2). Jointly they will work in 8 technical work packages to a) advance automated munition detection, identification and data analyzing technologies and software, b) environmental monitoring of chemical contaminants, c) predictions of UXO-burial, contaminants spread and the state-of-corrosion, and will d) enable secure exchange of sensitive data (64% of person months).

Three additional WPs will compile existing knowledge, learn about the legal responsibilities and technical approaches, assess remediation and mitigation measures and create training material for building European capacity (14% PM). Two WPs will engage in an intense stakeholder dialog and dissemination/outreach activities by utilizing capacities already existing at HELCOM and JPIO (14% PM).

Outcomes with respect to specific technical advancements will include a) AI-supported detection of munition in hydroacoustic spatial mapping data (MBES, SSS; SAS), b) detection of buried munition objects (magnetic, SBP, LF-SAS), c) AI-supported object identification in optical and acoustic cameras and d) implementing trained AI-models into Smart-AUVs and -USVs for adaptive and cooperative mission execution. All this will utilize secure data exchange possibilities through a demonstrator data platform that will be refined during the project. Industry partners see a great benefit in developing technologies that later can be advanced further to become commercial products. At the end of the project most TRLs will be at 5 to 6, meaning they have been applied under real conditions and proved their applicability.

4.6. Sensing innovations

4.6.1. RANGER

700478

RANGER: RAdars for loNG distance maritime surveillancE and SaR opeRations

EXUS SOFTWARE LTD (UK)

€7,992,312

01/05/2016 – 31/12/2019

https://doi.org/10.3030/700478

https://ranger-project.eu/

Programme
HORIZON 2020

Topic title
BES-01-2015 – Maritime Border Security topic 1: radar systems for the surveillance of coastal and pre-frontier areas and in support of search and rescue operations

https://www.youtube.com/@RANGERH-xm7yy

Objective

EU trade, transport, tourism and economic development are directly dependent on open and safe seas and oceans. EU's maritime borders are widely spread including various topologies from open sea to semi-enclosed cabins with islands and islets. This poses great challenges that affect securing maritime border areas. Failing to protect against a wide array of maritime threats and risks may result in these areas becoming arenas for international conflicts, terrorism or organized crime, where smuggling, irregular immigration and drug trafficking are the most common ones.

RANGER aims at re-enforcing EU by combining innovative Radar technologies with novel technological solutions for early warning, in view of delivering a surveillance platform offering detection, recognition, identification and tracking of suspicious vessels, capabilities exceeding current radar systems. It will be a platform, consisting of 2 radar technologies, a novel Over-The-Horizon Radar combined with a Multiple Input Multiple Output one implemented exploiting the latest photonics advancements, and an Early Warning System exploiting deep and adaptable machine learning schemes able to Automatically detect radar Targets. It safeguards seamless fitting and interoperability with CISE (enhanced maritime surveillance and cross border SaR operations), through the development of a CISE translation Gateway, exporting on-demand CISE services directly to end-users, by strengthening the information exchange between national authorities and the European Agency.

RANGER leverages the experience of its consortium, a balanced blend of technology providers, domain experts and end-users, delivering a cost efficient, environmental friendly solution, abiding to regulations and legislation for the protection of human lives. Two pilot exercises are foreseen to thoroughly assess RANGER's ability to deliver on its promises, enhancing its potential to become a flagship platform for the European Maritime Surveillance industry.

Reported results

The RANGER project created an innovative surveillance platform by combining ground-breaking radar technologies with state-of-the-art early warning solutions in order to detect, track, recognize and identify vessels far beyond existing legacy systems. The RANGER system was tested in four pilot demonstrations in two areas in the Mediterranean Sea (in France and Greece).

Some of the most challenging aspects in maritime surveillance have been addressed by the project; continuous long-range detection and tracking, short range detection and tracking of small vessels, prediction of vessel route, improved situational awareness and early warnings provision through novel data fusion and machine learning techniques, interoperability and seamless integration with legacy systems.

RANGER was the first EU funded project that exchanged operational data through the CISE (Common Information Sharing Environment) network with the Greek CISE node.

Selected publications

In terms of hardware, the project published on the topic of wide-band signal generation and design of high-performance radar for surveillance applications (Drechsel et al., 2019a; Drechsel et al., 2019b; Sakalas et al., 2018; Sakalas et al., 2017a; Sakalas et al., 2017b; Sakalas et al., 2019). There was also a paper on multiple-input multiple-output radar with iterative adaptive beamforming (Eckhardt et al., 2018).

The project also led to several publications describing algorithms for combining radar and other sensor data to enhance situational awareness, using multisensor data fusion and target tracking (Gaglione et al., 2018; Soldi & Braca, 2018; Soldi et al., 2019a; Soldi et al., 2019b).

Detection of outliers in radar signals is a considerable challenge in maritime surveillance applications. Protopapadakis et al. (2017) explored the applicability of deep learning techniques for detecting deviations from the norm in behavioural patterns of vessels (outliers) as they are tracked using radar. A comparative experimental evaluation of the approach showed promising results in terms of the proposed methodology's performance.

Finally, in terms of visualization interfaces for surveillance systems, Ouzounoglou et al. (2019) published a paper describing a multisource communication gateway and an advanced visualization interface for maritime surveillance systems based on the inter-VTS Exchange Format Service.

4.6.2. HERMES

964246

High Efficiency wiReless CMOS transceiver boosted by artificial intelligencE for 6G bandS and beyond

UNIVERSITE DE BORDEAUX (FR)

€3,271,458

01/09/2021 – 31/08/2025

https://doi.org/10.3030/964246

Programme
HORIZON 2020

Topic title
FETOPEN-01-2018-2019-2020 – FET-Open Challenging Current Thinking

sub-THz communications, artificial intelligence, deep submicron CMOS technology, Walsh signal processing, RFIC, wideband, 6G, smart robots

Objective
The future of wireless communications will go beyond connecting people or things to connecting smart robots or unmanned vehicles without human intervention. Only sub-THz frequencies can offer the spectrum to achieve unprecedented communication throughput. Yet current technologies suffer from barriers that prevent mass-market operations, such as high cost, limited bandwidth or power-hungry hardware not compatible with autonomous systems.

HERMES proposes the fusion of Artificial Intelligence (AI) and deep sub-micron CMOS technology to open a new generation of wireless transceivers. The project will focus on sub-THz frequencies between 140 and 160 GHz and will break new ground in the conversion of information from digital to THz using European CMOS technology to develop a highly integrated transceiver. For the first time, HERMES will deliver to the telecommunications industry a disruptive way of designing transceivers, with impact on production of billions of units that can be implemented in any autonomous system to communicate.

This new wireless link will be a springboard for an innovation leap in the robotics and the security industry. Our ambitious and risky approach goes significantly beyond the SoA: we will demonstrate that the power of AI developed by computer sciences research and associated with an original electronics signal processing technique can push CMOS technology to release outstanding performances. The project will produce a chipset of a low-cost radio that exchanges tens of Gbps and will test it in use cases of unmanned vehicles applied to maritime border surveillance operated by civil authorities. The HERMES consortium brings together 6 partners from 5 European countries bridging the full value chain of academic research, technology transfer and industry.

Selected results and publications

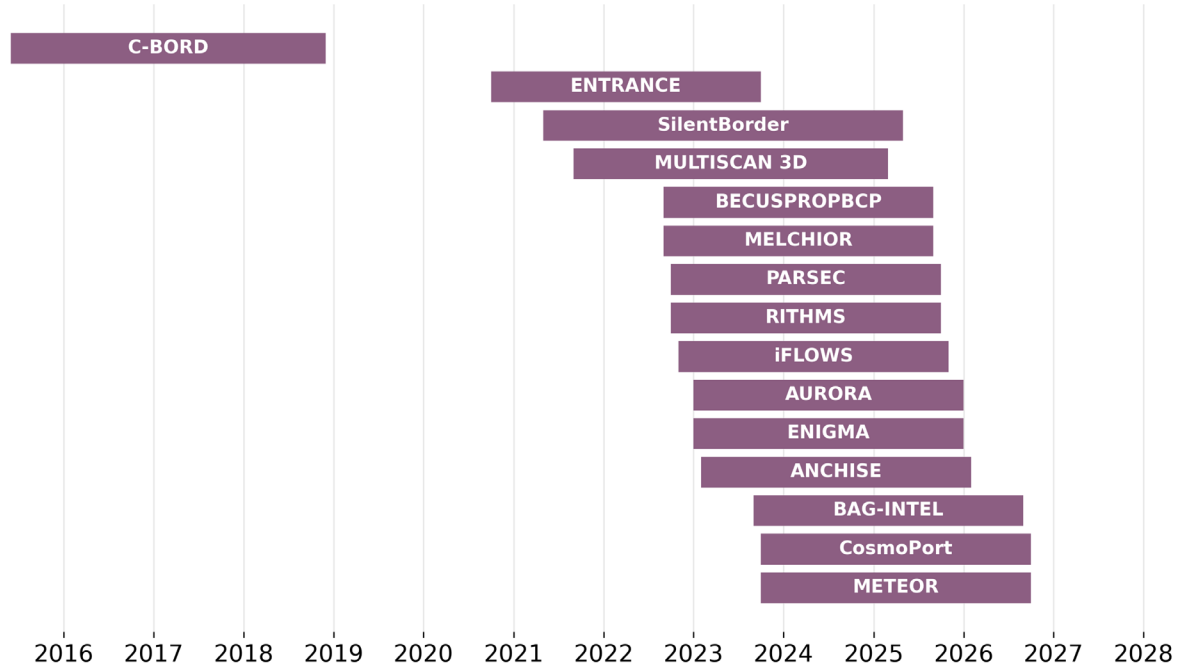
The project demonstrated how combining neural networks with Walsh-domain signal processing can significantly enhance wireless system efficiency and adaptability. Autoencoders for end-to-end deep learning communication modelling were used to predicted both symbols and the Walsh coefficients, taking into account the transmitter, channel, and receiver as a unified system. The signal processing pre-distortion techniques exhibited faster convergence and lower power consumption (Fellmann, 2024; Thys et al., 2024).

5. Goods inspection and detection technologies

5.1. Introduction

We identified 15 EU-funded projects that address AI-enabled inspection and detection technologies for goods, cargo, luggage, parcels and cultural artefacts. These range from physics-based scanning methods to advanced AI algorithms for image analysis. A timeline of these projects is showing in Figure 12.

Figure 12. EU projects featuring AI for goods inspection and detection technologies.



Source: JRC

These projects involved more than 10 different techniques to detect objects from cargo and/or luggage, including **X-ray imaging**, **Raman spectroscopy** (shining a laser on a sample, causing an energy shift that produces a spectrum of peaks unique to the sample's molecular structure), **muon tomography** (high-energy subatomic particles that penetrate materials differently based on density to create detailed images of the internal structure of large and dense objects), **target neutron interrogation**, **ion mobility spectrometry**, **photofission**, **acoustic and infrasound**. All these physics/chemistry-based techniques generate an image or graph that can easily be integrated into an AI tool to detect objects based on patterns. In addition, **electro-optical cameras** (EO cameras) and **satellite imagery** are also used in three projects, along with other sensors such as IR, RF, radar etc.

Content sniffing is used in one project. This technique usually uses scent-detection technologies or trained dogs for this specific task. A recent study, Lim et al. (2025), examines the future of scent detection, by creating an AI electronic nose. This device utilises t-distributed stochastic neighbour embedding (t-SNE) and support vector machine (SVM)-based machine learning to mimic the human sense of smell. It achieved 95% accuracy in distinguishing nine fragrances commonly found in cosmetics and perfumes. Table 4 reveals a summary of the projects and the technology involved in each of them.

Table 4. Summary of techniques used in EU projects with AI for goods inspection and detection technologies.

Category	Project	Border type: Land (L) Air (A) Sea (S)	II [1]	NII (Non-intrusive inspection techniques)													Tested / deployed
			Bioaerosol sampler	X-ray imaging	Raman spectroscopy	Ion mobility spectrometry	Gamma Ray Spectroscopy	Muon tomography	Target neutron interrogation	Spectral fluorescence signature	Photofission	Acoustic and Infrasound	Unmanned Aircraft	Content sniffing	EO camera	Satellite imagery	Social networks / websites analysis	
Screening of cargo and containers	**C-BORD**	L A S		X					X		X			X				X (Rotterdam port, Gdańsk port, Hungarian land border)
	ENTRANCE	L S		X					X		X				X	X		X (Rijeka, Croatia)
	SilentBorder	L S						X										X (Tartu, Estonia)
	MULTISCAN 3D	L S		X			X		X		X							
	BECUSPROPBCP	A S			X													X (Ports of Antwerp and Zeebrugge; Airports of Zaventem and Liège)
	CosmoPort	L A S						X										
	METEOR	L		X		X												X (Netherlands; Belgium; Spain; Ireland)
Screening of parcels and luggage	**PARSEC**	–		X			X											
	iFLOWS	–		X	X													

	BAG-INTEL	A		X											X			X (plan - 3 European airports)
Person inspection	**MELCHIOR**	L A S	X									X						X (Spain; Greece; Romania; Finland)
Cultural heritage	**RITHMS**	–															X	
	ENIGMA	–														X	X	
	AURORA	–		X														X
	ANCHISE	–								X			X			X		

[1] Intrusive inspection techniques.

Source: JRC

Regarding the **AI techniques** used for object detection in images (for persons, luggage, and vehicles), projects analysed algorithms such as **YOLO** (You Only Look Once – real-time deep learning model that simultaneously detects and classifies multiple objects within an image by dividing it into a grid and predicting bounding boxes and class probabilities), and **Transformers** (particularly as seen in architectures like DETR *DEtection TRansformers*, transformers uses self-attention mechanisms to model relationships between different parts of an image; this approach allows the model to consider the entire image context when detecting objects, unlike traditional convolutional methods that primarily focus on local features; transformers process the image as a sequence of patches, enabling them to capture global dependencies and interactions, leading to accurate localization and classification of objects without the need for predefined anchors or complex post-processing steps).

Even though, as mentioned above, some AI techniques are mentioned and revealed in some articles published as part of the projects, most of the projects doesn't describe the technical details of the AI usage in the projects. This can lead to a certain confusion regarding the degree of AI techniques used in the projects.

Table 5. Computer vision and object detection projects summary on the application areas. ([1])

Category	Project / EC contribution	Smuggling					Irregular migration
		Counterfeit goods	Illicit substances / materials ([2])	Tobacco	Firearms	Cultural heritage	
Screening of cargo and containers	**C-BORD** €11,826,452	X	X	X			X
	ENTRANCE €6,398,686		X				
	SilentBorder €7,471,578	X	X	X			
	MULTISCAN 3D €6,989,608	X	X	X			
	BECUSPROPBCP €19,879,200		X	X			
	CosmoPort €2,480,335	X	X	X			
	METEOR €2,745,160		X				
Screening of parcels and luggage	**PARSEC** €3,185,442		X		X		
	iFLOWS €3,998,125		X		X		
	BAG-INTEL ([3]) €3,942,406		X	X	X		
Person inspection	**MELCHIOR** €4,999,238	X	X		X		
Cultural heritage	**RITHMS** €4,996,972					X	
	ENIGMA €3,993,500					X	
	AURORA €3,466,354					X	
	ANCHISE €4,037,259					X	

1 Some projects do not specify the category of smuggling they are targeting. In this case, based on the description and deliverables documents provided, the most appropriate categories of smuggling were chosen.

2 E.g.: drugs, explosives, nuclear material, chemical and biological warfare.

3 Application areas not specified.

Source: JRC

5.2. Screening of cargo and containers

5.2.1. C-BORD

effective Container inspection at BORDer control points 653323

COMMISSARIAT A L ENERGIE ATOMIQUE ET AUX ENERGIES ALTERNATIVES (FR)

€11,826,452

01/06/2015 – 30/11/2018

https://doi.org/10.3030/653323

https://www.cbord-h2020.eu

Programme
HORIZON 2020

Topic title
BES-09-2014 – Supply Chain Security topic 2: Technologies for inspections of large volume freight

https://www.youtube.com/watch?v=HgyjSjaTWj0

Objective
Efficient NII (non-intrusive inspection) of containerised freight is critical to trade and society. Freight containers are potential means for smuggling (e.g. tobacco), illegal immigration, trafficking of drugs, mis-declared goods and dangerous illicit substances, including explosives, nuclear material, chemical and biological warfare agents and radioactively contaminated goods. One inspection NII technology cannot cope with all these targets.

The C-BORD Toolbox and Framework will address all these targets and enable customs to deploy comprehensive cost-effective container NII solutions to potentially protect all EU sea- and land-borders, satisfying a large range of container NII needs. The C-BORD Toolbox will include 5 complementary innovative detection technologies: delivering improved X-rays, Target Neutron Interrogation, Photofission, Sniffing and Passive Detection. User interfaces and data will be integrated to optimise effectiveness and efficiency of end-users and systems. The C-BORD Framework will help customs analyse their needs, design integrated solutions, and optimise the container inspection chain; it will address detection levels, false alarm levels, throughput, health & safety, logistics and cost & benefits.

C-BORD will increase the probability of finding illicit or dangerous content with at least equal throughput of containers per time unit, reduce the need for costly, time-consuming and dangerous manual container inspections by customs officials, and in case a container is opened, increase the probability of finding illicit materials. C-BORD involves stakeholders from 8 EU countries, as partners (5) and advisory group members (3). On 3 custom sites integrated solutions will be trialled, respectively addressing the needs of big seaports, small seaports and mobile land-borders. To optimise sustainable impact, C-BORD will actively engage with a large community, will support policy implementation, evolution and start early exploitation planning.

Reported results

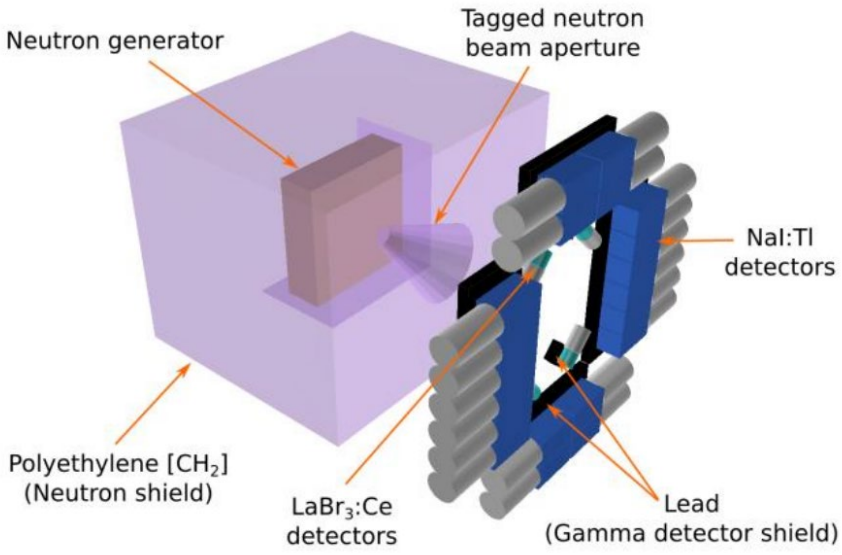
The C-BORD live Demonstration of the **X-ray Users Interface** showed higher resolution, higher performance on un-even ground, and improved scatter compensation. The use **Nuclear Detection** within a common User Interface Integration Platform convinced the customs of its potential to

contribute to the decision making. In C-BORD in this second line of detection, an anomaly in first line is further probed by tagged neutron inspection or photofission depending on the nature (Drugs, explosives or SNM) of the suspected target. Without opening, the container, within a short time frame, additional information in produced either by identifying the threat as such or by abrogating the anomaly because it happens to be benign. Both the successful **detection of targeted drugs and explosive within containers** (RRTNIS) and the **deep penetration of SNM verification within containers** (Photofission), were done in real condition, with real targets. This represents a great European success for research and customs.

Selected publications

Sardet et al. (2018) discusses the development of a Rapidly Relocatable Tagged Neutron Inspection System (RRTNIS) designed for the non-intrusive detection of explosives, chemical threats, and other illicit goods in cargo containers. The system uses gamma spectroscopy for material identification, employing twenty NaI detectors and four $LaBr_3$ detectors to analyse the specific gamma signatures generated by fast neutron interactions. An unfolding algorithm is utilized to break down the energy spectrum of suspicious items, previously selected using X-ray radiography, by comparing it to a database of gamma signatures from pure elements. The paper presents simulated signatures for these detectors, created using the MCNP6 code, and includes initial experimental spectra for several elements of interest.

Figure 13. Schematic view of RRTNIS of the C-BORD project.



Source: Sardet et al. (2018)

Most of the research articles published during the C-BORD project approach the topic of nuclear instruments and methods in physics.

5.2.2. ENTRANCE

883424

EfficieNT Risk-bAsed iNspection of freight Crossing bordErs without disrupting business

COMMISSARIAT A L ENERGIE ATOMIQUE ET AUX ENERGIES ALTERNATIVES (FR)

€6,398,686

01/10/2020 – 30/09/2023

https://doi.org/10.3030/883424

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 – Technologies to enhance border and external security

non-intrusive inspection, NII, automated threat detection, ATR, risk-based selective control, customs risk management

Objective

The mission of ENTRANCE is to develop and validate a comprehensive user-based ENTRANCE Toolbox for risk-based non-intrusive inspection (NII) of cross-border freight movements, with particular focus on the EU Customs Union border. The aim of the ENTRANCE Toolbox is to enhance border security practitioners' capabilities to protect society against a wide range of dangerous and illicit materials with minimum disruption of cross-border flow of goods.

ENTRANCE will deliver five main results: 1) Automated Risk Assessment, Threat Recognition and Information Sharing Platform (ENARTIS), 2) suite of Non-Intrusive Inspection (NII) technologies for detecting contraband hidden in high-density cargo, 3) Enhanced relocatable unit for non-intrusive detection of wide number of threats including explosives, illicit drugs, chemical warfare agents, nuclear and radioactive materials and special nuclear materials (SNM) such as enriched uranium and plutonium, 4) Trans-European RPM network for passive detection of illicit nuclear and radioactive (NR) materials combining detection facilities of different types and technologies, and 5) Novel high-speed RPM detection technology for passive detection of NR with minimal disturbance of flow.

The ENTRANCE Toolbox will be validated at the EU Customs Union border by five practitioner-led field trials chosen for their relevance, strategic position and feasibility.

Reported results

In the ENTRANCE project, deep learning algorithms enhanced X-ray imaging to detect contraband in dense cargo without disrupting freight flow. Real-time video feeds, sensor data, and satellite imagery were used to identify suspicious activity or anomalies at borders using AI. The ENTRANCE consortium assessed developed technologies through field tests conducted over three weeks in May 2023 at the seaport of Rijeka, Croatia, with operational and logistical support from RBI, MOF, and MOI. The Non-Intrusive Inspection (NII) technologies were tested on large container trucks to detect dangerous and illicit materials with minimal disruption to cross-border trade. Real targets — guns, explosives, drugs, contraband, nuclear material, and live radioactive sources — were hidden in commercial containers to evaluate the tools under realistic conditions.

These tests showcased improvements in X-ray imaging, deep learning, and material discrimination (organic/intermediate Z/metal), validating the tools' "container wall removal" functionality. Combined

with depth-function analysis, this enabled more accurate interpretation of container contents. The ability to differentiate individual objects (beyond 2D overlays) enhanced material discrimination for customs use.

The Rapidly Relocatable Tagged Neutron Inspection System (TNIS) was used to detect explosives, illicit drugs, toxic chemicals, and contraband. The photofission inspection system identified Special Nuclear Material (SNM), using a 7 MeV mobile accelerator — replacing the 9 MeV fixed installation from the previous C-BORD project. Trans-border control also tested a new fast RPM system with a passive detection portal at speeds up to 30km/h. A novel method for inspecting high-density iron ore cargo combined three technologies: 1) real-time Declared vs. Mass Measurement Comparison (DMMC), 2) real-time Scattered Radiation Imaging, and 3) second-line Muon Scattering Imaging.

Figure 14. Neutron-based inspection of a truck (top), and composition vs. distance (bottom).



Source: ENTRANCE project

Selected publications

de Sainte Foy et al. (2022) developed a new and upgraded database of pure element gamma signatures and performed laboratory tests placing simulants of explosives, illicit drugs and chemical elements inside wood and iron matrices.

5.2.3. SilentBorder

101021812

Cosmic Ray Tomograph for Identification of Hazardous and Illegal Goods hidden in Trucks and Sea Containers

TARTU ULIKOOL (EE)

€7,471,578

01/05/2021 – 30/04/2025

https://doi.org/10.3030/101021812

https://silentborder.eu/

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 – Technologies to enhance border and external security

Scanning technology, passive scanning technology

Objective

Everyday our customs workers need to tackle counterfeit goods and piracy to protect the health and safety of our citizens, yet it is estimated that only a small fraction of cargo is inspected, and even smaller fraction of illegal goods are detected. Today, the most widely used technology for scanning vehicles, ranging from vans and trucks to railcars, is gamma-ray and X-ray radiography. But new technologies are required for overcoming current technological shortcomings like inability to detect the materials, usage of radioactive and harmful source, low throughput to name some.

Cosmic-ray tomography (CRT) is considered as beyond the state-of-the-art technology in cargo screening. Cosmic-ray muons are highly penetrating, their average energy is about 10,000 times the energy of a typical X-ray and they are practically non-absorbable. They are suitable to identify materials hidden inside of shielded material, too thick or deep for other imaging methods. The CRT is completely passive, exploiting naturally occurring secondary cosmic radiation. Contrary to conventional X-ray or gamma-based imaging techniques it allows to distinguish between different materials and localizes it inside the cargo or vehicle by providing visualised 3D image.

We will bridge the major security gap for fast and safe inspection of large number of cargos by developing the Multi-Functional Passive Detection System. The detection capability is based on using high accuracy sensors for particle tracking in combination with beyond state-of-the art tomographic reconstruction and material classification algorithms.

The main objective of SilentBorder is to develop and validate a new high-technology CRT scanner for border guard, customs and LEAs that enables safe and fast screening, detection and identification of hazardous and illegal goods (e.g. SNM), contraband (e.g. tobacco or explosive) as well as hidden persons in up to 20' iso containers.

Reported results

In the first work package, the focus was on defining technological specifications and the design of the hodoscope. The mechanical design and dimensions of a module hodoscope have been determined and tested. The production technology has been developed and QA techniques in place.

In the second work package, the requirements for DAQ systems were defined. DAQ design for the prototype has been developed, testing and verification done. Design for the integration of **detection special nuclear materials** has been developed.

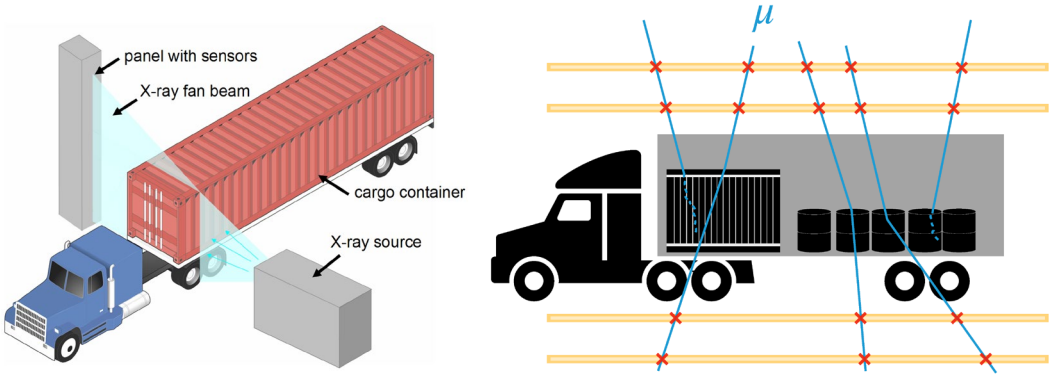
In the third work package, different programs for modelling cosmic particle sources were identified, analysed, and compared to each other. A simulation framework of full-scale tomography system for the sea container with a cargo was also developed using the Geant4 code. Reconstruction algorithms were tested and compared for different **muon energies** and detector spatial resolutions. Low and high muon momentums were compared. The parameters for optimisations have been discussed and agreed. Simulations have been started to address questions on the optimum values of such parameters. Development of tools for an optimization of future commercial tomography system has started.

The fourth work package is to develop a **tomographic reconstruction** and material classification method based on **multi-parameter reconstruction implemented by machine learning** including utilization and sharing prior data central image store of images. Validation of secondary particles as a complimentary information source for reconstruction under simulated conditions has been done. First bulk of simulated data has been reviewed and inspected with first visualizations of data using the developed code for reconstructing of simulated objects in cargo containers. Pipeline for large scale machine learning dataset generation produced.

Selected publications

Barnes et al. (2023) discusses the challenges customs workers face in intercepting hazardous, illegal, and counterfeit goods due to limited inspection capabilities. Current scanning technologies, like γ-ray and X-ray radiography, have limitations, including the inability to identify material composition, reliance on harmful ionizing radiation, and low throughput. The article highlights **cosmic ray tomography (CRT)** as a promising alternative for cargo screening. CRT uses cosmic ray muons,, which can penetrate large and dense materials and provide 3D images for material differentiation and anomaly localization. Unlike current methods, CRT is safe for humans and animals as it is passive and uses natural cosmic radiation. The article reviews the current state of CRT technology, evaluates its strengths and weaknesses, and suggests directions for future development.

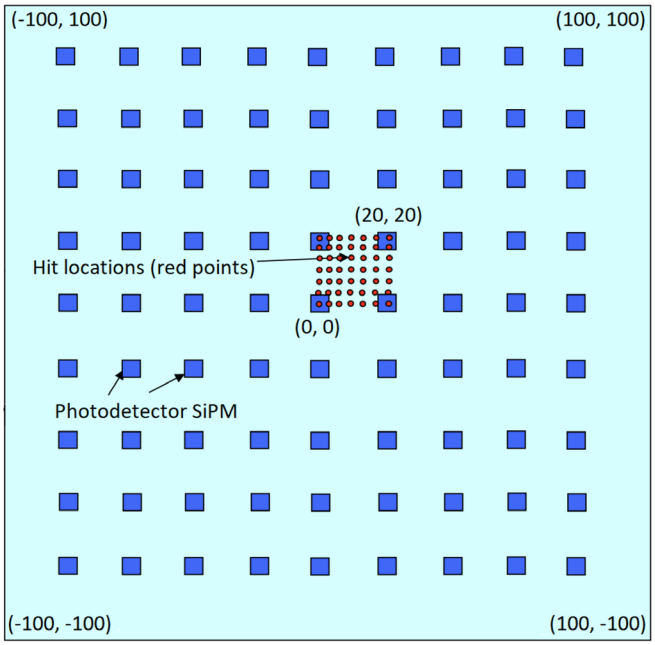
Figure 15. Conventional high energy X-ray cargo scanning (left) vs concept of a muon screening portal for cargo scanning (right).



Source: Barnes et al. (2023)

Aktas et al. (2022) examines the performance of cosmic-ray tomography systems, which depends heavily on their tracking accuracy. Traditional scintillation detector technology can achieve good precision by having a small pitch between detector elements, but enhancing resolution involves increasing the number of read-out channels, leading to higher complexity and cost. As an alternative, the paper suggests using a scintillation plate detector with multiple silicon photomultipliers, which offers a simpler technical solution. The study compares two **deep-learning**-based methods with a conventional **centre of gravity algorithm** to calculate cosmic-ray muon hit positions using photomultiplier signals (see Figure 16). Using the GEANT4 Monte Carlo simulation toolkit, a dataset of muon hits on a detector plate was generated for the analysis. The deep-learning methods significantly outperformed the CoG algorithm. A proposed method, the **fully connected network**, achieved a 0.72 mm average error in predicting hit coordinates, a noticeable improvement over CoG's 1.41 mm error. The article also explores how different sensor configurations impact performance.

Figure 16. Simulated muon hit locations on the simulated plate, in mm.



Source: Aktas et al. (2022)

5.2.4. MULTISCAN 3D

101020100

Laser-plasma based source 3D Tomography for cargo inspection

COMMISSARIAT A L ENERGIE ATOMIQUE ET AUX ENERGIES ALTERNATIVES (FR)

€6,989,608

01/09/2021 – 31/08/2025

https://doi.org/10.3030/101020100

https://multiscan3d-h2020.eu/

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 - Technologies to enhance border and external security

Tomography 3D, Laser-plasma based source

Objective

Within the field of security, Customs and Border inspection have not had breakthrough technological developments in the last 20 years, since the introduction of X-ray screening. The limits of these current technologies are accentuated by the increasing diversity and novelty in trafficking materials, tools and methods. These limitations combined with the growing needs of inspection and control call for a disruptive innovative solution.

Wanting to move a step up from the existing planar scanning methods with limited material identification results, several studies have identified potential solutions focused on: high energy 3D X-ray tomography, neutron interrogation/photofission, nuclear resonance fluorescence (NRR).

While these show good results and performances, they also have several important drawbacks, which limits their possible uses. Moreover, these solutions do not have common technological bricks meaning they can only lead to separate disposals.

The proposed MULTISCAN3D investigates a new all-in-one system whose purpose is to become simultaneously a user-friendly, flexible, relocatable solution offering high-quality information for:

- Fast high energy 3D X-rays tomography (as first line)

- Neutron interrogation/photofission (as second line)

- Narrow gamma ray beam based NRR (as second line)

MULTISCAN3D will start by investigating and defining needs and requirements, in a technologically-neutral way, with Europe's most prominent Customs Authorities which will be translated to technical specifications. The main body of the research will be focused on three parts, following which, lab validations and real-environment demonstration will be carried out. These three work areas are (i) laser-plasma based accelerators as X-ray sources, (ii) 3D reconstruction for multi-view configurations and data processing, (iii) detectors and source monitoring. At the same time complementary techniques with chemical and SNM identification capabilities will be investigated.

Reported results

Research projects have demonstrated that the additional information produced from computed tomography enables an increase in inspection performance when compared to the use of single and **dual view X-ray scanners**. The cooperation between end-users and technology partners in the process of setting requirements, developing specifications, and designing technology is a unique opportunity for all involved and crucial stage in delivering an innovative solution.

This project will develop a single system that delivers novel **3D transmission imaging for first line inspection** with additional capabilities to discern between materials on the atomic level at second line. Thus, this project has the potential of providing to end-user's technology that is revolutionary different to current capability and able to deliver technology that better fits the logistical flow, makes inspections more effective and efficient by resolving more and reducing costs associated with delays and physical inspection processes.

The project proposes to go beyond the state of the art on the following challenges described below:

- Develop a new NII Inspection technique
- Optimise X-ray and electron sources and their transportation
- Assess and develop a 3-dimensional reconstruction for multi-view configurations
- Ensure the detection of the multiple X-ray sources with a single ring
- Ensure monitoring the dose
- Ensure the second line identification
- Ensure the real environment demonstration

Selected publications

Garrido et al. (2023) discusses the use of imaging systems with embedded X-ray accelerators for security applications, such as in rail, seaports, and border controls. These systems face challenges with the wide energy spectrum of X-ray beams produced, particularly with determining the endpoint energy, which is crucial for applications like material discrimination. Fluctuations in beam stability, especially with new laser-plasma accelerators, complicate image analysis. The study explores a method for **direct beam measurement using a scintillator**, aiming for minimal footprint and quick response to provide real-time data. It examines different scintillator materials (PVT, CsI, BGO) to find the optimal choice, using simulations to analyse the feasibility of reconstructing the X-ray spectrum with various algorithms. Results suggest that the **Sequential Least SQuares Programming (SLSQP) algorithm** is effective, and CsI scintillators offer better spectral information due to their intermediate dose profile.

5.2.5. BECUSPROPBCP

101079010

2022- BELGIAN CUSTOMS – PROPOSAL FOR BCP

SERVICE PUBLIC FEDERAL FINANCES (BE)

€19,879,200

01/09/2022 – 31/08/2025

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43251534/101079010/CCEI

Programme
Customs Control Equipment Instrument (CCEI)

Topic title
CCEI-2021-EQUIP-IBA-BCROSS –

increasing flow of goods; illicit trade; E-commerce; state of the art technology; 100% scanning; X-ray; Automatic analysis of images; reinforced control in ports and airports

Objective

The objective of the proposal for Border Crossing Points of Belgian Customs (BECUS) is a reinforced control policy in 4 strategic points of entry. In order to step up its efforts Belgian Customs established a strategy (100% scanning) with some important guidelines that are identical to the CCEI's set-up. The key points of this reinforced control policy are the following: the first is maximum use of state-of-the-art technology to improve and strengthen non-intrusive inspection and control of traffic. Secondly acquired technology will be integrated in the logistics processes of the economic operators. Thirdly BECUS will establish cooperation structures with other authorities and enforcement bodies. Last but not least will the risk management techniques be updated in the light of the tightened control policy and the new technological approach, in particular by using artificial intelligence.

In the context of CCEI, BECUS has decided to focus on 4 BCPs: the Ports of Antwerp and Zeebrugge and the Airports of Zaventem and Liege. These BCP's have to process an ever-growing volume of traffic and are confronted with a number of specific threats: importation of illegal drugs, illicit trade of tobacco, enormous increase in E-commerce activities and diverse fiscal infringements. The opportunity offered by the European Commission and DG TAXUD will be seized to better equip this 4 BCP's. On the one hand, BECUS will replace, extend and upgrade technological resources that are already in use (upgrade of fixed scanner and radiation portal monitors, extension of RAMAN analysers, ...) On the other hand, BECUS also wants to focus on more innovative technology to tackle specific problems (mobile scanners with drive through mode, 3D scanner for parcels). For each BCP, the required additional technology has been determined, and the tenders are being prepared. Training, maintenance and technical support will be included in every public tender.

5.2.6. CosmoPort

101121379

Using cosmic rays for better, more portable and efficient analysis and detection for customs

GSCAN OU (EE)

€2,480,335

01/10/2023 – 30/09/2026

https://doi.org/10.3030/101121379

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-03 - Better, more portable and quicker analysis and detection for customs

atmospheric ray tomography, drug detection, mobile scanners, drug analysis, postal services

Objective

The number of goods and parcels shipped is growing with unstoppable speed. With more individual shipments the means of logistics have changed from central locations to small logistics companies, postal agencies, airports, ports, private vehicles etc. However, the analysis and detection capabilities of such locations are very low, making it an easy target for transporting illicit goods.

Currently, there are no effective technologies available that would be able to cope with this growing problem. Despite the widespread use of X-ray technologies, they are still inefficient when it comes to mobile scenarios. In addition, their detection capability is based solely on visual imaging with no way to examine the exact material content of small shipments.

We are proposing a completely novel, yet highly effective solution for this. The CosmoPort project will develop the next generation of scanner systems using Atmospheric Ray Tomography (ART). Such systems are equipped with advanced Machine Learning-based risk assessment tool. As muon tomography itself is a technique already in use, then we will develop the first mobile solution combined with AI/ML tools to enhance material and object classification for the first time ever.

Reported results

The CosmoPort project has developed key components for a mobile cosmic ray **muon tomography system** designed for non-intrusive customs inspections. Notable achievements include the production and testing of custom **Silicon Photomultipliers (SiPMs)** optimized for light detection, the creation of an efficient Data Acquisition (DAQ) system with reduced power consumption, and the prototyping of a novel fiber mat design to increase production speed. Additionally, algorithms for material classification were verified, and a beta version of a graphical user interface (GUI) for 3D data visualization was developed. **A mobile trailer platform** to house the system was also designed and contracted for construction. These advancements aim to enhance security by improving illicit material detection, increase efficiency in customs operations, and reduce costs. To ensure further success, the project needs continued research and development, successful piloting with customs agencies, a clear commercialization strategy, intellectual property protection, addressing regulatory hurdles, international expansion, and comprehensive end-user training.

5.2.7. METEOR

101121149

Rapid, portable and reliable cargo screener - New concept of vapour screening technology - Ion Mobility Chemical Fingerprint Detector

VLAAMSE INSTELLING VOOR TECHNOLOGISCH ONDERZOEK N.V. (BE)

€2,745,160

01/10/2023 – 30/09/2026

https://doi.org/10.3030/101121149

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-03 – Better, more portable and quicker analysis and detection for customs

Detection, cocaine, drugs, explosives, fast screening, container, customs

Objective

Safe and legal global transport is essential to human welfare, but it demands that customs authorities implement security regulations with a special focus on illicit goods. To enable enforcement of regulations tackling illicit goods trafficking, customs authorities have expressed their need for portable, affordable, non-intrusive, reliable screening technologies aiming to facilitate on-site and rapid inspections. The current technologies are mainly based on X-ray screening, but have its limitations in detecting illicit goods, are very expensive and the process is time consuming. This makes that currently less than 5% of the containers may be inspected, which is undesirable for customs authorities and society.

In the METEOR project, the consortium will develop a prototype of a portable and versatile air sampling-based screening system. This will enable customs administrations worldwide to rapidly and reliably inspect for the presence of illicit goods. The METEOR technology will provide a new concept of cargo screening detector with a highly efficient **air sampling technique** and the **ion mobility multidetector sensing technology**. The METEOR analyser relies on a very innovative concept of a multi-detector differential mobility analyser (DMA), that classifies molecular ions based upon their electrical mobility. This generates a chemical fingerprint of the samples, and after the processing with non-targeted screening techniques, it is possible to accurately classify and detect the threats. The focus will be on the detection of illicit drugs, but also explosives and other substances.

The goal is to develop the METEOR technology up to TRL7, and validate it in the operational environment. This is done by the four customs administrations (The Netherlands, Belgium, Spain and Ireland) involved in the project, that cover some of the main ports in Europe.

5.3. Screening of parcels and luggage

5.3.1. PARSEC

101073963

Parcel and Letter Security for Postal and Express Courier Flows

CBRA SERVICES (BE)

€3,185,442

01/10/2022 – 28/02/2026

https://doi.org/10.3030/101073963

https://www.parsec-project.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-04 - Advanced detection of threats and illicit goods in postal and express courier flows

terrorism, photons, spectroscopy

https://youtu.be/CO1ouzYe7tg?si=sd2qDzWQ5tdv9Nrh

Objective

PARSEC is a project about parcel and letter security in the context of postal and express courier services. The project delivers an ambitious set of solutions by developing, configuring, customising, and piloting innovative tools, services and security management views to fight the abuse of postal and express courier flows for criminal and terrorist purposes. The four PARSEC innovation areas and three use cases strengthen risk analysis and redefine threat detection and resilience capabilities of parcel service providers, customs authorities, police agencies, and other relevant stakeholders. PARSEC develops and tests three next-generation non-intrusive detection technologies (multi-energy photon counting detector, neutron-induced gamma-ray spectroscopy, and X-ray diffraction) and combine them into a detection architecture (= system-of-system) for optimal detection accuracy, speed and reliability. With PARSEC solutions postal and express operators, customs, and police authorities will be more capable to fight crime and terrorism, put in place a stronger deterrent, and to ensure safe and undisrupted postal and express services.

Reported results

The technologies being examined and tested in PARSEC have the capability to be more specific about material discrimination, than conventional devices and in some cases, technology can identify a substance.

It is accepted that the vast majority of consignments are regular, legitimate traffic. Therefore, the goal is to be able to only select risky consignments without interfering with the bulk of them. In order to facilitate the flows, as well as ensuring that customs physically inspect the highest risk items, the tools that PARSEC is developing are intelligence-focussed. Data-driven risk management tools will select what should be selected and the consignment will be directed to the first line technology. If this technology determines that there is no risk, then the consignment will be released back to the flows. However, if it either determines there is a known threat, or it identified there is still

something it is not happy with, then the consignment will be referred to the next technology, which has the ability to spectrally discriminate and identify compounds. If that device agrees there is an illicit material, or it believes it cannot clear the risk, then the consignment is sent for physical inspection. If the second device determines that the risk can be resolved and is not of interest to customs, then the consignment will be returned to the commercial flows.

PARSEC is making use of **multi-energy x-ray and diffraction x-ray (XRD)**. XRD is usually seen in labs in a bench-top device or is starting to be seen supporting aviation security applications for alarm resolution. It is rather unique to use this combination for customs applications. During some of our data capture work, we have evidenced that some of the drug's profiles created from pharmaceutical grade examples, worked when applied to illicit samples and an explosive was successfully identified during an in-house demonstration.

The indicators show that that technology can play a role and deliver to customs technology that is beyond-state-of-the-art. Although reporting period 1 only has brief glimpses of a successful future, the work taking place next will link those strands together.

Although the technologies are being geared towards the postal and courier express domains, the individual devices, or system, will have application in other customs areas too, such as screening of bags. What this development could deliver is an increased assurance in how the border is protected without impacting vast number of the goods and travellers.

5.3.2. iFLOWS

101073990

Advanced technologies for scanning and detection of illicit material for postal services and express courier flows (iFlows Toolkit)

RESEARCH CENTER FOR NON DESTRUCTIVE TESTING GMBH (AT)

€3,998,125

01/11/2022 – 31/01/2026

https://doi.org/10.3030/101073990

https://www.iflows-project.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-04 - Advanced detection of threats and illicit goods in postal and express courier flows

QCL based THz Scanner, plasmonic structure detector, AI-enabled Risk, Threat Intelligence, C4I, UWB Scanner, Computer vision, Illicit goods & dangerous substances detection, Counterfeit ID, Cash detection

https://www.youtube.com/@iFLOWSproject

Objective

Preventing and fighting illicit material and dangerous substances postal and courier services flows in EU areas is of top priority. However, ensuring efficiency, speed and continuity in screening and detection tasks remains yet a challenge. Currently, custom, police and courier authorities involve a range of screening systems in place but there is still much to do in terms of detection efficacy and creation of intelligence towards risk and threat identification. On top of that, criminals or smugglers exploit the multitude of flows and easily spot security gaps that allow parcels or letters to be transported within or across EU with relative ease. To overcome these constraints, iFLOWS proposes a unique approach to assist customs, police and postal/courier services in preventing and detecting such crimes by improving operational readiness, enhancing situation awareness, reducing detection time, providing operational efficiency and achieving superior mission performance. iFLOWS will design, develop, integrate, test and validate a novel multi-tier synergistic Toolkit in enhancing intelligence extraction, screening and detection operations of illicit material and hazardous substances within courier and postal flows moving across and within EU borders. iFLOWS works upon these 3 building blocks to deliver a) a tailored to such operations C4I that boosts preparedness and cooperation be-tween police, customs and courier/postal services, b) enhanced Intelligence in terms of threats, risks and suspicious flows and c) novel detection technologies (i.e. THz Scanning system and UWB Scanner) featuring enhanced detection capabilities over a large span of illicit material and substances, which, coupled with advanced image-based feature extraction in combination to legacy systems (e.g. X-Ray and Raman) optimise the detection process without interrupting the postal/courier flows.

5.3.3. BAG-INTEL

101121309

Intelligent system for improved efficiency and effectiveness of the customs control of passenger baggage from international flight arrivals

LEGIND TECHNOLOGIES AS (DK)

€3,942,406

01/09/2023 – 31/08/2026

https://doi.org/10.3030/101121309

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-04 – OPEN TOPIC

Customs automation, baggage customs control, baggage customs risk assessment, baggage customs scanning, automatic detection, luggage reidentification, customs decision support, digital twins

Objective

The BAG-INTEL project will provide robust AI based information utilization and decision support tools, within the context of advanced detection systems to support customs for increased effectiveness and efficiency of the customs control of air traveller baggage in inland border airports, while minimizing the human customs resources needed. This aim addresses the challenge of maintaining effective and efficient customs control of passenger baggage in the situation of the substantial growth of the volume of air travellers arriving in inland border airports with the limited human customs resources available.

For this aim, the project will develop an integrated system solution comprising: (1) new AI powered functionality for enhanced detection of contraband in x-ray scanning of luggage, (2) AI camera based end-to-end reidentification of luggage, (3) digital twin for system visualisation and performance optimization for the operational context of an airport, (4) use case for test demonstration and evaluation in 3 European airports, a small, a medium sized, and a big airport, and (5) wide dissemination and elaboration of easy-to-use training material for end users.

For the customs, BAG-INTEL solution aims to: increase the successful detection of contraband in luggage by at least 20%; demonstrate the possibility and utility in automatically to derive risk indicators from external data such as the Advanced Passenger Information; demonstrate the effectivity of AI camera based reidentification of luggage, when the traveller carries it into the customs space at the exit of the carousel area; increase the fluidity of passenger flow and control by at least 20%; decrease the customs personal resources mobilisation by at least 20%; derive data useful in flights risk assessment; derive data useful in flights risk assessment; demonstrate the autolearning capacity of this smart risk engine.

5.4. Person inspection

5.4.1. MELCHIOR

101073899

MECHANICAL IMPEDANCE and MULTIPHYSICS CONCEALED AND HIDDEN OBJECTS INTERROGATION

UNIVERSIDAD DE ALCALA (ES)

€4,999,238

01/09/2022 – 31/08/2025

https://doi.org/10.3030/101073899

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-05 – Improved detection of concealed objects on, and within the body of, persons

Objective

This project aims to improve substantially a novel technology for fast detection of drugs, explosives, weapons and illicit goods concealed on individuals and in critical cavities of the human body based on infrasound mechanical impedance interrogation, optionally complemented with other harmless and non-contact technologies. Mechanical Impedance interrogation was validated in lab -TRL4- in the multiply awarded H2020 MESMERISE Project, detecting items concealed on trunk under clothes -including critical cases like warm molded materials that remain undetected to other technologies.

The present Project aims to develop up to TRL7-8 blast-proof prototypes with detection capability extended to limbs and body cavities and improved sensitivity. Infrasound computer aided modelling improved Artificial Intelligence, altogether with improved new infrasound generators and sensors will enable an improvement in reliability and sensitivity. Combination with other non-contact technologies like low-cost mm waves and THz point echo sensors with no image formation will be explored.

Additionally, compatibility and complementary combination of specific bioaerosols and chemical vapour detection novel technologies will be explored in the prototypes as well. Slight variations of the prototypes will be demonstrated in operational environments (TRL7) of customs and border authorities in EU external air, land and sea borders crossing points in Spain, Greece, Romania and Finland. Demonstrations in other operational environments like an arena or prisons will increase and broaden the impact on security of society. Additionally, CE mark according to EU safety directives will be applied.

Safety of the border staff will be drastically increased - not needing to touch inspected individuals. Simultaneously, the traveller experience is totally improved. Respect to privacy and intimacy is absolute, as no contact is needed, and no image is formed at any step and individuals do not have to undress at all

5.5. Cultural heritage

5.5.1. RITHMS

101073932

Research, Intelligence and Technology for Heritage and Market Security

FONDAZIONE ISTITUTO ITALIANO DI TECNOLOGIA (IT)

€4,996,972

01/10/2022 – 30/09/2025

https://doi.org/10.3030/101073932

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-FCT-01-08 – Fight against trafficking in cultural goods

Social Network Analysis, AI-based evidence, Knowledge graph, Money laundering, Legal guidelines, Public engagement

Objective
RITHMS project intends to boost the operational capacity of Police and Customs/Border Authorities in addressing the increasingly organised and poly-criminal nature of trafficking in cultural goods through research, technological innovation, outreach and training. Illicit trade in cultural heritage has progressed from a local phenomenon limited to the initiative of a few individuals to a highly remunerative source of income for criminal organisations and terrorist groups that have taken advantage of the opportunities offered by the web and social media to further expand an already flourishing market. Against this background, RITHMS proposes an interdisciplinary approach made necessary by the transnational character of this crime and its links to other criminal networks. The project will define a replicable strategy to counter the challenges in addressing the illicit trafficking of stolen/looted cultural goods and to investigate the mechanisms underpinning it, including its connection with organised crime. RITHMS will foster cross-cutting research bringing together all the domains relevant to expand the understanding of this type of crime (art market, criminology, law studies, forensic science, etc.). Inputs from these disciplines will inform the theoretical framework underlying an interoperable AI-based Platform able to identify criminal organised networks and to provide investigators with valuable intelligence on the activities and evolution of such networks. The Platform will be leveraging the methods of Social Network Analysis (SNA), a methodology that gives a way to better understand human behaviour through people's relations and interactions. Including four Police authorities, two Border Agencies, one Police School, SMEs, and researchers from SSH and ICT, RITHMS Consortium is well positioned to offer solutions to counteract more effectively organisations involved in illicit activities against cultural heritage.

5.5.2. ENIGMA

101094237

ENDORSING SAFEGUARDING, PROTECTION, AND PROVENANCE MANAGEMENT OF CULTURAL HERITAGE

ARISTOTELIO PANEPISTIMIO THESSALONIKIS (EL)

€3,993,500

01/01/2023 – 31/12/2025

https://doi.org/10.3030/101094237

https://eu-enigma.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL2-2022-HERITAGE-01-07 - Protection of artefacts and cultural goods from anthropogenic threats

anthropogenic threats, identification, traceability, provenance

https://www.youtube.com/@EnigmaProjectEU

Objective

ENIGMA shall achieve excellence in the protection of CGs and artefacts from man-made threats by contributing to identification, traceability, and provenance research of CGs as well as by safeguarding and monitoring of endangered heritage sites. ENIGMA objectives are designed to help the involved stakeholders better respond to this complex, and multi-dimensional problem, and leverage active collaboration by fostering and enabling interlinking of databases, and evidence-based deployment of preventative measures.

Reported results

The UAI, advanced ML/AI tools for metadata analysis and interlinking of disparate databases, the earth observation toolkit and the Alert system are non-destructive methods and digital tolls that will highly contribute to the protection and safeguarding of both movable and immovable CGs. Furthermore, in addition to these tools the enhanced documentation workflows combined with the developed methodologies for information sharing among various databases, the provenance research tools the collaborative stakeholder platform, and the public engagement API will contribute to the protection, tracing, restitution and safeguarding, as well as **provenance research of European endangered cultural heritage**.

The results of the testing, evaluation and validation of all the developed tools and the ENIGMA platform through the implementation of the pilot cases in the second half of the project, will lead to the creation of policy recommendations and briefs thus producing evidence-based research to support the deployment of preventative measures against looting and illicit trade of cultural goods.

5.5.3. AURORA

101094245

Artwork Unique RecognitiOn and tRacking through chemicAl encoded data, miniaturized devices and blockchain alliance

AVVALE S.P.A. (IT)

€3,466,354

01/01/2023 – 31/12/2025

https://doi.org/10.3030/101094245

https://www.aurora-euproject.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL2-2022-HERITAGE-01-07 - Protection of artefacts and cultural goods from anthropogenic threats

chemical marker, artwork digital ID, artwork tracking, blockchain, cultural identities, technological democratization, anti-counterfeiting, embedded tracking, artwork authenticity verification

Objective

Artifacts are constantly threatened by anthropogenic actions. Looting, smuggling, and illicit trade of cultural goods pose major risks to our cultural goods and our cultural identity. One of the major consequences of these illegal activities is the dissociation, meaning the loss of information associated with an object, such as provenance, identification, or location information, without which the object loses significance or is lost. Thus, art property marking and tracking are crucial to ensure the object identification, collection management and protection, while maintaining their state of conservation. Several actions have been adopted during the years to strengthen the protection of cultural property and cultural heritage, but they often lack effectiveness. The AURORA project wants to fill this gap by demonstrating how chemical marking, miniaturized device, art deep-scan technique, cloud and blockchain based platforms can be combined to create a cost-effective, non-destructive, and non-invasive countermeasure against illegal activities while preserving artifacts. The technologies researched and implemented in AURORA will converge in a digital tool allowing relevant stakeholders, art dealers, curators, auction houses, logistic services, and enforcement agents, to easily verify artwork authenticity and provenance, enabling art piece movement tracking. In close collaboration with relevant cultural institutions partners, AURORA will showcase real setting applications to validate the non-invasiveness, low cost, long-term stability, confidentiality, and data security features of the proposed solutions. In parallel, AURORA will foster technological democratization among cultural institutions, by creating a knowledge bridge between technical competencies and cultural heritage professionals. Specific awareness-raising activities will demonstrate how the AURORA solutions can be a means to preserve cultural identities mixtures through cultural heritage preservation.

Reported results

During the M1-M16 phase of the AURORA project, significant activities were undertaken to shape the technological and functional requirements necessary to achieve the project's ambitions. Efforts were concentrated on evaluating artwork categories most susceptible to illicit trafficking and compatible with AURORA's technological constraints, such as size and material composition. An

extensive comparative study of the latest distributed ledger solutions was conducted to assess their limitations and advantages in terms of cultural heritage applicability for the development of the digital ledger platform. Additionally, an in-depth analysis of IoT technologies was performed to define the main properties of the embedded artwork device. These foundational studies set constraints for each AURORA technological stream. During M1-M16, AURORA achieved complete instrumental and software testing for XRF maps acquisition, with the matching algorithm nearing finalization. Preliminary solutions for the **embedded device tracking were designed and tested in a museum lab environment**, allowing for the evaluation of selected hardware and communication protocols. The AURORA digital tool was designed according to cultural heritage institutions' requirements and software capabilities, fully defining system architecture, user roles, and functionalities. Software development for this tool commenced as planned. The chemical research activities were focused on testing multiple formulations. In this first period, **AURORA also collaborated with ANCHISE and ENIGMA** projects through specific meetings and engaged cultural heritage stakeholders via survey and conferences, gaining important insights on the technological preparedness of various cultural institutions. As AURORA is entering the second stage (M17-M36), all technologies are in full development or finalized, as in case of the **XRF mapping**. The efforts will focus on constructing the AURORA digital infrastructure and refining platform operativity regarding cultural heritage professionals' needs and the artwork authenticity procedure. Selected chemical materials will be tested under different stress conditions, and the IoT embedded device will be prototyped and tested in its final version. The project will conclude with testing both the tracker and the authenticity verification process through demonstrators on real artworks.

5.5.4. ANCHISE

101094824

Applying New solutions for Cultural Heritage protection by Innovative, Scientific, social and economic Engagement

ECOLE FRANCAISE D'ATHENES (EL)

€4,037,259

01/02/2023 – 31/01/2026

https://doi.org/10.3030/101094824

Programme
HORIZON EUROPE

Topic title
HORIZON-CL2-2022-HERITAGE-01-07 – Protection of artefacts and cultural goods from anthropogenic threats

3D, photogrammetry, archaeological site monitoring, data engineering, AI, border control, heritage collection protection, spectral fluorescence signature, object authentication, humanities

Objective

ANCHISE - 'Applying New solutions for Cultural Heritage protection by Innovative, Scientific, social and economic Engagement' aims at offering to European societies efficient methods, knowledge and toolkit to enhance the protection of cultural heritage against looting and illicit trafficking. Crossing the methodology of networking that has proved its efficiency in the H2020 NETCHER project with the innovative results of new technologies developments (3D/photogrammetry for site monitoring, data engineering and AI for border control object identification and heritage collection protection, spectral fluorescence signature for object authentication), ANCHISE will create an operational set of tools applicable for European contexts and replicable in other situations abroad.

The aim of the project is to bring coordinated solutions to the key existing needs in the domain of Cultural heritage protection: 1) Understand, 2) Prevent, 3) Act, 4) Repair. ANCHISE will consist of: 1) a hub of social science, politics and economics (for in-depth results likely to lead to structural evolutions in heritage protection), 2) a large-scale evaluation of technologies and needs, 3) a toolkit of innovative solutions, 4) pilot experimentation areas (museums, border control, archaeological sites), and 5) a unique and wide network of practitioners. 9 demonstrations will involve at list 150 practitioners. Guidelines and procedures on the use of ANCHISES tools will be largely disseminated (INTERPOL, PANDORA, institutions in charge of cultural heritage).

The Consortium is a strong partnership of leading stakeholders in the fight against illicit trafficking of cultural objects: companies (PARCS, ICONEM, INOV), research centres (FRAUNHOFER, ICCS, Cyprus Institute), universities (French School at Athens, University Lyon 2, University of Poitiers, EUI Florence, Cyprus University of Technology), practitioners (France's National Police College, ICOM), and associations (Michael Culture).

6. Risk analysis and predictive intelligence

6.1. Introduction

The European Union (EU) and its Member States are progressively adopting artificial intelligence and advanced digital technologies to improve risk and threat detection in border management. These tools can be used both before migrants reach EU territories and at physical border checkpoints, reflecting a shift toward predictive, data-driven, and automated security practices.

Artificial intelligence (AI) and emerging technologies are transforming border management and customs controls by enhancing the accuracy, speed, and efficiency of risk and threat analysis. These technologies provide border agencies with advanced tools to monitor, assess, and respond to potential threats with greater precision and fewer resources. AI algorithms can analyse vast datasets—including travel patterns, criminal databases, and historical incidents—to predict potential risks. Machine learning models assess passenger and cargo data in real time to flag anomalies or patterns associated with illegal activities, such as trafficking or smuggling. This enables proactive risk-based screening rather than relying on random or manual checks. AI-driven video analytics enhance surveillance by automatically detecting suspicious behaviours or unauthorized movements across borders. Drones and smart sensors equipped with AI can monitor remote or difficult-to-patrol areas, alerting border agents to unusual activities without the need for constant human supervision. Natural Language processing technologies can monitor communications, social media, and intercepted messages to detect potential threats based on keywords, sentiment, or emerging narratives. This helps authorities stay ahead of coordinated threats or cross-border criminal activity. AI-based tools can detect anomalies in network behaviour, prevent data breaches, and ensure the integrity of sensitive border control systems.

Pre-Arrival Risk Forecasting

Technologies such as machine learning algorithms and agent-based models are used to forecast migratory flows and displacement patterns to improve border security and migration management and meet the anticipated challenges. Important examples include:

- The European Union Agency for Asylum (EUAA)'s Early Warning and Preparedness System (EWPS), which integrates data from sources such as GDELT, Google Trends, Frontex, and asylum applications to forecast migration patterns up to four weeks in advance.
- National initiatives like Germany's PREVIEW, Sweden's PREDICT, and the Danish Refugee Council's Foresight Project, which use supervised learning and Bayesian network analysis to forecast conflict-driven migration over varying timescales (Ozkul, 2023).

While there are many attempts to predict migration trends, Angenendt et al. (2023) suggest that there is a gap between the expectations that political decision-makers place in quantitative tools, in particular AI-based migration predictions, and their actual usefulness.

Situational Awareness and exchange of intelligence

Technological advancements in surveillance have resulted in systems like the European Border Surveillance System (EUROSUR), which is the EU's operational platform specifically for border surveillance and control at sea and on land. It connects Member States' border control authorities with each other and with Frontex. These systems serve dual purposes: facilitating real-time operational awareness and providing archival data for strategic risk assessments. Tools such as

thermal cameras, radar systems, and unmanned aircraft swarms contribute to persistent intelligence, surveillance, and reconnaissance (ISR), improving response coordination at borders. AI systems also facilitate inter-agency data sharing and intelligence-led border management, with passenger data and behavioural indicators exchanged among EU Member States. These systems enhance the detection of unknown individuals of interest through pattern recognition and classification based on pre-defined risk factors.

Algorithmic Profiling

Automated border control technologies—e-gates and biometric verification systems include AI capabilities and support both identity confirmation and various types of profiling for risk analysis, often cross-referencing traveller data with databases such as the Visa Information System (VIS) and the European Travel Information and Authorisation System (ETIAS) (Dumbrava, 2021). The use of other big data sets, such as for example PNR (Passenger Name Record) or API (Advanced Passenger Information), allow gathering and analysis of data from a very large number of passengers.

The use of remote biometric identification technologies for individual profiling in Europe is mainly experimental. In 2017 Brussels International Airport acquired four cameras that were connected to a facial recognition software for use by the airport police. Subsequently, the use was ruled out as unlawful. In Kortrijk, Belgium, and Marbella, Spain, the local police used "body recognition" technology (Kayser-Bril, 2020). The software relied on "appearance search": estimates for "unique facial traits, the colour of a person's clothes, age, shape, gender and hair colour". These features of individuals could be used to search for suspects fitting a particular profile (Ragazzi, et al., 2021). In the US the Defence Advanced Research Projects Agency (DARPA) initiated the development of biometric technologies for screening passengers. The Project 'Hostile Intent' (later renamed Future Attribute Screening Technology, FAST) explored remote, non-invasive, automated sensor technology capable of real-time detection of deception.[9] Cameras and sensors were used to remotely monitor, record and analyse video, audio, respiration, cardiovascular reactivity, bodily secretions, eye movement, facial features and facial expressions, and readings of the skin's electrical resistance. The system also included the possibility for the scanning of micro-facial expressions. The development of portable 'bad intentions' scanners was also envisaged (Selter and Kolbel, 2010). The system consists of sensors, which use infrared rays to check body temperature, skin moisture, pulse and a data analysis system (Selter and Kolbel, 2010).

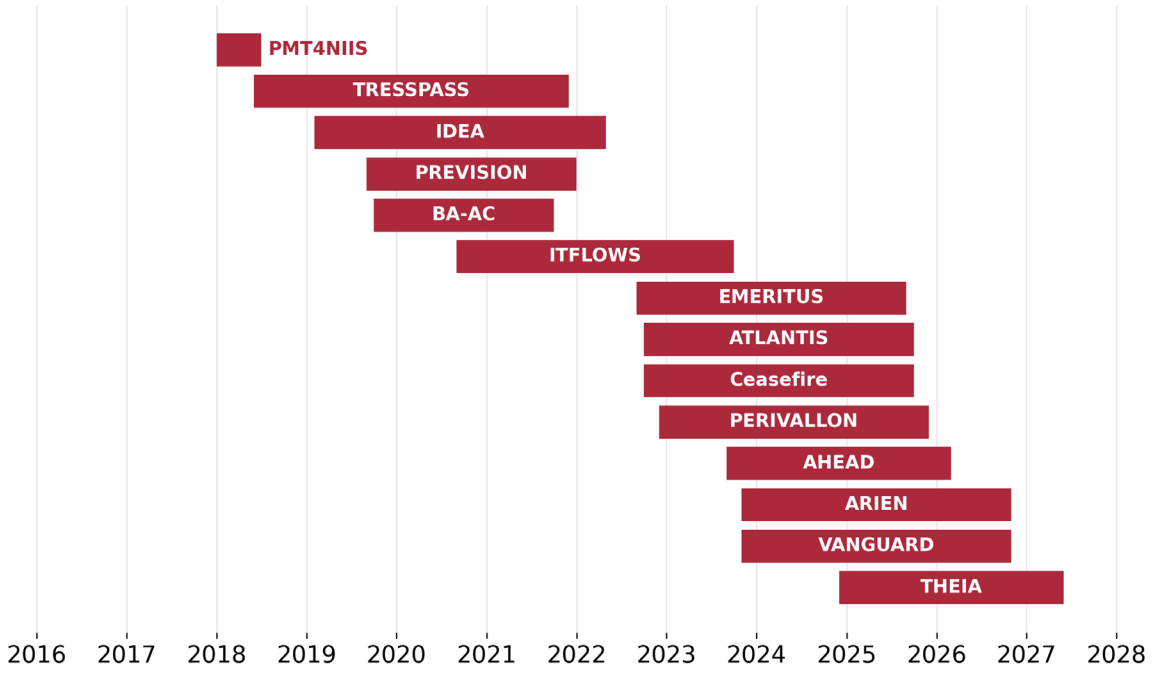
6.2. EU funded projects

We identified 14 EU-funded projects that developed work on risk analysis and predictive intelligence in the context of AI for border management and customs controls (see Figure 17). Several EU-funded projects focus on improving the ability to **anticipate security threats** before they materialise. *TRESSPASS* developed a risk-based screening framework for passengers and cargo data at border crossing points. The system was tested in operational environments across air, land, and sea BCPs. Building on the predictive dimension, *ITFLOWS* designed AI-enabled tool to model not only migration numbers but also public sentiment and risks of societal tension. The tool comprised two complementary approaches to simulation and prediction: one based on agent-based modelling, and

[9] See more at https://www.dhs.gov/science-and-technology/deception-detection. Accessed on 29/9/2025.

the other on artificial intelligence. Projects such as *ATLANTIS* and *AHEAD* extend the scope from borders to critical infrastructure and foresight capacities, aiming to improve resilience against systemic risks and to equip civil security actors with forward-looking methodologies. In parallel, *THEIA* enhances situational awareness during crises by integrating data fusion and geospatial AI. Together, these projects highlight the EU's effort to embed predictive intelligence into both border management and wider security domains.

Figure 17. EU projects on risk analysis and predictive intelligence, in the context of border management and customs controls.



Source: JRC

Other projects concentrate on detecting and **disrupting criminal networks** that exploit borders. *IDEA* improved access to, and use of, INTERPOL databases within the EU, providing a stronger basis for cross-border investigations. *PREVISION* developed AI-driven analytics to support law enforcement facing complex threats such as terrorism and cybercrime. The project provided tools for near-real-time support with multiple big-data streams from online social networks, the open web, the Darknet, CCTV and video surveillance systems, traffic and financial data sources. Concerning environmental crime, *EMERITUS* developed a protocol for effective environmental crime investigation, leveraging on the integration of innovative monitoring and analysis technologies (e.g. drones, satellite data, virtual sensors, and geo-intelligence data). *PERIVALLON* aimed at a comprehensive intelligence picture of organised environmental crime and developing tools for detecting and preventing such activities. Similarly, *Ceasefire* targets illicit firearms trafficking through interconnected AI technologies and national focal points, while *ARIEN* applies machine learning and data integration to combat drug production and trafficking. Finally, *VANGUARD* brings a societal dimension by combining advanced technologies with awareness-raising to counter human trafficking.

6.3. Anticipating threats

6.3.1. TRESSPASS

787120

robusT Risk basEd Screening and alert System for PASSengers and luggage

NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS" (EL)

€7,901,471

01/06/2018 – 30/11/2021

https://doi.org/10.3030/787120

Programme
HORIZON 2020

Topic title
SEC-15-BES-2017 – Risk-based screening at border crossing

Risk based screening

Objective
TRESSPASS project will develop a solution that aims to effectively combat border-related crimes and various other significant threats while also integrating with other risk-based border management systems. The project's approach involves the development of an innovative border management solution, which will be showcased through three pilot programmes and simulations. The outcome will empower authorities to effectively screen and monitor potential instances of smuggling, terrorism, cross-border crimes and more.

This proposal includes all tiers of the four-tier access model: (1) measures undertaken in, or jointly with third countries or service providers; (2) cooperation with neighbouring countries; (3) border control and counter-smuggling measures, and (4) control measures within the area of free movement. TRESSPASS will (1)develop a single cohesive risk-based border management concept (2)develop three pivoting pilot demonstrators (3)demonstrate the validity of the single cohesive risk-based border management concept by using red teaming and simulations(4)prepare for the further development of this concept beyond this project by linking to other known risk-based border management projects (in- and outside EU, within EU research frameworks and on national levels), and describe how their results contribute to a single cohesive risk-based border management concept.

Reported results

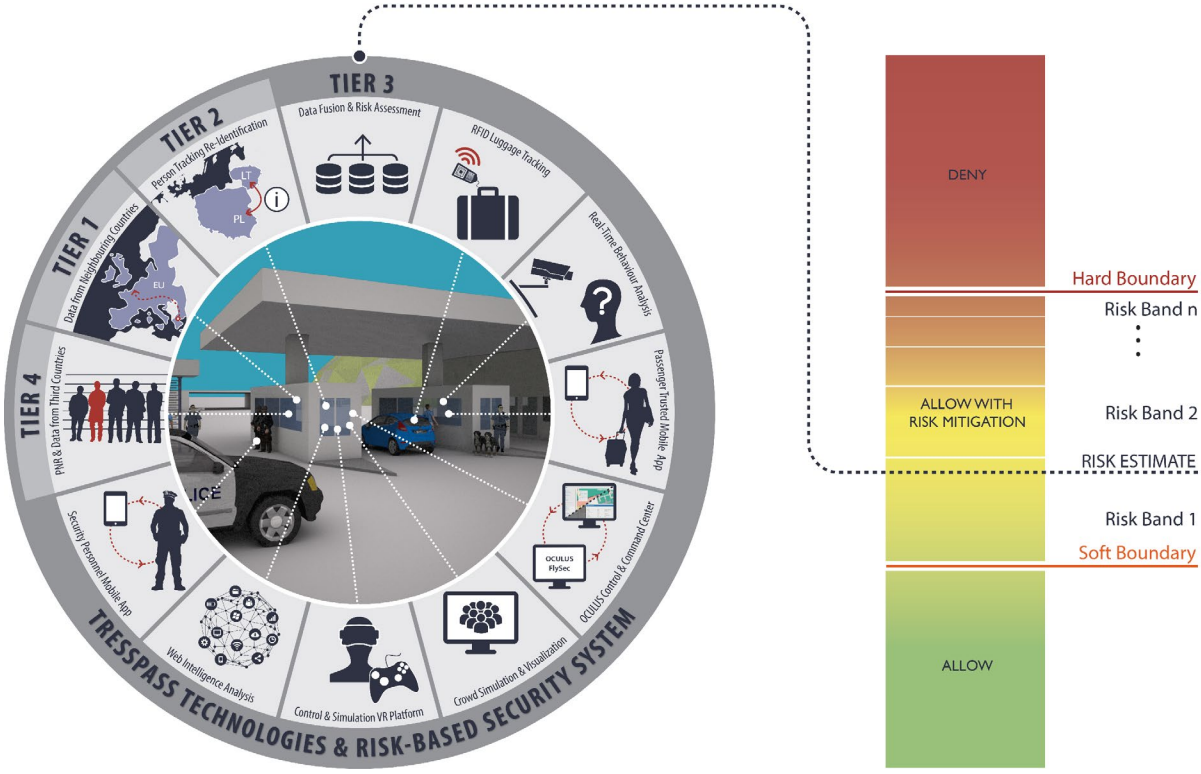
TRESSPASS developed risk-based border management systems to analyse passenger and cargo data at border crossing points and to assign dynamic risk scores, enabling selective screening based on real-time threat evaluation rather than random checks. AI-driven simulations and "red teaming" methods were used to test and validate these systems. The TRESSPASS project defined and provided the Risk-Based Border Management concept for BCPs, along with relevant models and CONOPS, including multinational risk-based cooperation. Pilot tests were successful, and the TRESSPASS system was tested in operational environments across Air, Land, and Sea BCPs. Ethical Impact Evaluation ensured the protection of ethical, legal, data, and security aspects. The validation framework included Pilot Planning & Red Teaming scenarios and a Simulation Framework. Exploitation

and further development of the concept included a sustainability report (Roadmap) and Policy Options, including a white paper on standardisation.

Selected publications

Kowalski (2020) explored the outcomes of investigations on the detection of various face presentation attacks in thermal infrared in various conditions including thermal heating of masks and various states of subjects. The paper provides a thorough analysis of presentation attacks using printed and displayed facial photographs, 3D-printed, custom flexible 3D-latex and silicone masks. The paper also includes an intensity analysis of thermal energy distribution for specific facial landmarks during long-lasting experiments. Kowalski and Mierzejewski (2021) also studied detection of 3D face masks using thermal infrared imaging and deep learning techniques. The paper demonstrates results of a two-step neural network-featured method for detecting presentation attacks.

Figure 18. TRESSPASS risk-based security concept for border crossing points.



Source: Cordis

6.3.2. ITFLOWS

882986

IT tools and methods for managing migration FLOWS

UNIVERSITAT AUTONOMA DE BARCELONA (ES)

€4,871,831

01/09/2020 – 30/09/2023

https://doi.org/10.3030/882986

https://www.itflows.eu

Programme
HORIZON 2020

Topic title
SU-BES01-2018-2019-2020 – Human factors, and social, societal, and organisational aspects of border and external security

Prediction, migration flows, models, management, digital solutions, tool, research, policy recommendations, tension reduction, public sentiment, asylum, practitioners

Objective

Prediction and management of migration are a major challenge for the European Union and also for many different stakeholder groups that provide resources and assistance to migrants along their journey. Yet, in order to develop better strategies and instruments among policy makers and practitioners in the field of migration, we still need a deeper understanding of this phenomenon.

ITFLOWS will generate novel insights on migration. The purpose of ITFLOWS is to provide accurate predictions and adequate management solutions of migration flows in the European Union in the phases of reception, relocation, settlement and integration of migration, according to a wide range of human factors and using multiple sources of information. These insights will be provided by an evidence-based ICT enabled solution (the EUMigraTool) and precise models. All solutions will have fitness for purpose continually validated by policy-makers and practitioners in cooperation with civil-society organisations in a dynamic and iterative process.

ITFLOWS will propose tailor-made solutions for practitioners and policy makers for managing migration. On the one hand, the EUMigraTool targets first-line-practitioners, second-level reception organisations and municipalities. It will provide modular solutions based on the prediction of migration flows and the identification of risks of tensions between migrants and EU citizens. On the other hand, an in-depth analysis on drivers, patterns and choices of migration as well as public sentiment towards migration will lead to the drafting of adequate recommendations and good practices for policy makers, governments and EU institutions.

Reported results

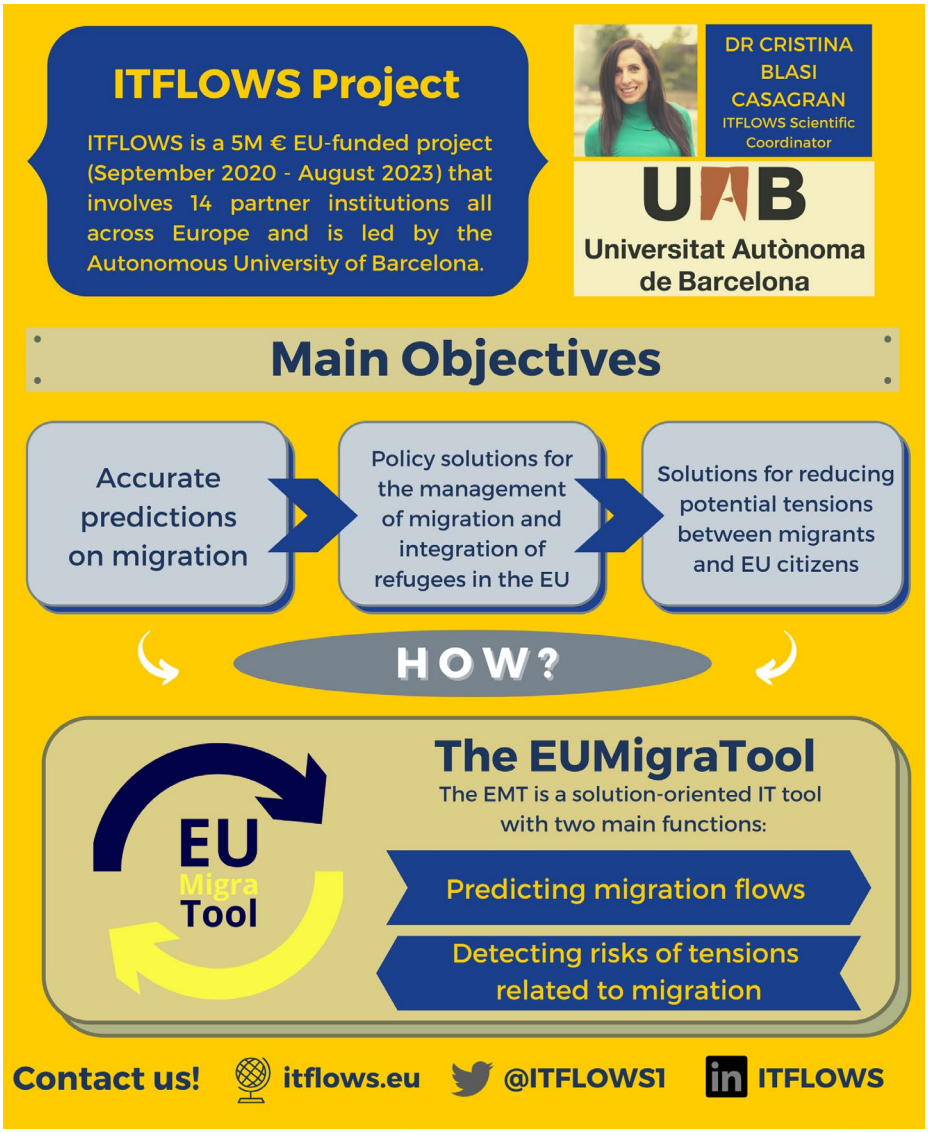
The ITFLOWS project's EUMigraTool (EMT) leverages deep learning and diverse datasets – including social media, conflict databases, and online news – to model not only migration numbers but also public sentiment and risks of societal tension. The EMT aims to provide accurate predictions and adequate management solutions for migratory and refugee movements arriving in the EU. The backend of the EMT comprises two complementary approaches to simulation and prediction: one based on agent-based modelling, and the other on artificial intelligence. The EMT generates predictions using algorithms that address two key challenges in migration forecasting: (a) the appropriate selection of relevant data sources, and (b) the correct identification of potential drivers to be monitored and setting of warning thresholds. ITFLOWS also collected, annotated, and processed

data, and built models to analyse migration intent in tweets. The GitHub repository provides a detailed description of the project, from environment setup to the implementation of the software prototype. It includes models developed for Iraqi, Egyptian, and Tunisian dialects. Future work will expand this methodology to include additional Arabic dialects and other low-resourced languages.

Selected publications

Boss et al. (2023) developed monthly refugee flow forecasting models for 150 origin countries to the EU27, using machine learning and high-dimensional data, including digital trace data from Google Trends. Blasi Casagran et al. (2021) carried out an in-depth examination of selected existing predictive tools in the migration field and their impact on the governance of migratory flows. The study looked at how and why these instruments continue to face limitations that in turn affect migration management, especially as it relates to increasing EU institutional and stakeholder efforts to forecast or predict mixed migration.

Figure 19. Poster illustrating the concept of the ITFLOWS project.



Source: Cordis

6.3.3. ATLANTIS

101073909

Improved resilience of Critical Infratsructures AgainsT LArge scale transNational and sysTemic rISks

ENGINEERING - INGEGNERIA INFORMATICA SPA (IT)

€9,998,535

01/10/2022 – 30/09/2025

https://doi.org/10.3030/101073909

https://www.atlantis-horizon.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-INFRA-01-01 – European infrastructures and their autonomy safeguarded against systemic risks

Large scale systemic risk management, Hybrid threat, Resilience, Threat intelligence, Situational awareness

https://www.youtube.com/@Atlantis-eu

Objective

ATLANTIS aims at enhancing resilience and Cyber-Physical-Human (CPH) security of the key EU Critical Infrastructures, going beyond the scope of distinct assets, systems, and single CI, by addressing resilience at the systemic level against major natural hazards and complex attacks that could potentially disrupt vital functions of the society.

The mission of ATLANTIS, which involves 39 European partners with complementary roles and skills, is to improve the resilience and the protection capabilities of interconnected ECI exposed to evolving systemic risks due to existing and emerging large-scale, combined, cyber-physical threats and hazards, guarantee the continuity of operations, while minimizing cascading effects in the infrastructure itself, the environment, other CIs, and the involved population, enabling public and private actors to meet current and emerging challenges by adopting sustainable security solutions.

The mission of ATLANTIS will be achieved by - Improving knowledge on large-scale, vulnerability assessment and long-term systemic risks. - Improving the systemic resilience of ECI, through novel, adaptive, flexible, and customizable security measures (“by design”) and tools (“by innovation”). - Improving effective cooperation among CI operators and government security stakeholders, while preserving CI autonomy and sovereignty. - Delivering an open TRL-7 technological framework that will provide the ECIs with AI -based solutions for increased AWARENESS, CAPABILITY and COOPERATION in managing systemic threats.

The ATLANTIS solution will be validated and demonstrated in 3 large-scale cross-border and cross-sector pilots: - LSP#1: Cross-Border/Cross Domain Large Scale Pilot in Transport, Energy and Telecoms (Slovenia, Croatia, Italy and France);- LSP#2: Cross Domain Large Scale Pilot in Health, Logistics/Supply Chain and Border control (Cyprus, Greece and Croatia);- LSP#3: Cross-Country Large-Scale Pilot in FinTech/Financial (Spain, Germany, Cyprus).

Reported results

The ATLANTIS project developed an Earth Observation-based technological solution focused on enhancing risk assessment and damage monitoring capabilities for various hazards such as floods, landslides, wildfires, and earthquakes. A comprehensive review of available technologies for building a resilient IT architecture was conducted, comparing equivalent solutions based on partner expertise. A laboratory testbed capable of handling GNSS interference signals was set up. This involved configuring equipment to inject jamming signals into live GNSS feeds and recording the resulting data.

Another key objective of ATLANTIS was to establish a secure communication layer ensuring traceability of information and metadata, while enabling zero-knowledge proof techniques. Various distributed ledger technologies (DLT) were assessed to determine the most suitable for effective inter-DLT communication. Additionally, work was carried out on integrating security systems across critical infrastructure using a layered approach that included Detection, Adaptation, and Data Analytics. This involved standardising alerts using the Universal Alert Format and analysing them for threat categorisation to support risk assessment. Furthermore, pre-developed components were integrated to establish communication flow between RRIM and SAFER, and to incorporate the Knowledge Graph into the Systemic Risks Foresight and Incidents Detection DSS.

The TrulyMedia platform for network and sentiment analysis, along with the HiViC platform for human engagement in CI protection, was developed. A key component, the Threat Intelligence Tool, was created to detect web-based threats. The Entropy-Based Risk Assessment tool was also developed to generate entropy levels for network nodes based on topology and analysed network traffic. Subsequently, efforts focused on developing an Explainable AI toolset tailored to stakeholders' needs, aiming to provide interpretable explanations that support risk anticipation and incident summarisation, thereby enhancing CPH protection.

Finally, integration tools and environments for ATLANTIS components were set up. CI/CD pipelines were configured following the DevSecOps approach, and the ATLANTIS Integrated Framework was scripted for on-premise deployment.

Selected publications

Spartalis et al. (2024) reviewed the trade-offs between explainability, privacy, and security in the context of using eXplainable AI (XAI) within critical infrastructure. While XAI improves transparency and trust, it also increases the risk of privacy breaches and vulnerability to adversarial attacks. The authors categorize these challenges by XAI taxonomy, examine interactions with privacy defences, and offer insights for advancing trustworthy AI in critical domains.

6.3.4. AHEAD

101121338

TowArd sustainable foresigHt capabilitiEs for increAseD Civil Security

MINISTERE DE L'INTERIEUR (FR)

€1,499,920

01/09/2023 – 28/02/2026

https://doi.org/10.3030/101121338

https://he-ahead-project.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-SSRI-01-01 – Increased foresight capacity for security

futur, foresight framework, holistic approach, civil security, LEAs capabilities, technology, industry, trends, megatrends, threats, innovation

Objective

With the rise of potentially disruptive technologies such as artificial intelligence and virtual reality, a particular challenge has emerged for LEAs and other civil security practitioners. Tasked with fostering safe societies and protecting European citizens from new forms of criminality, they need to pre-empt technological evolutions, turning potential threats into opportunities, and anticipate the operational future of policing. Technology foresight is one possible anticipatory process. However, pure technology watch-based approaches are not helpful for decision-making if not embedded in a qualitative assessment of threats and capabilities. Hence the need to move towards a comprehensive foresight framework which is the rationale behind AHEAD.

Standing for TowArd TecHnological ForEsight for IncreAseD Security, AHEAD is a LEA-led project which will adopt a 360-degree approach accounting for megatrends but also contextual elements (i.e. legal, ethical, societal and economic) to design and operationalise a sustainable capability-based civil security foresight framework. Tested through iterative Foresight Exercises addressing the five Destinations of the Horizon Europe Cluster 3 (Fighting crime and terrorism, Border management, Resilient infrastructure, Increased cybersecurity and Disaster-resilient society), this framework will be tailored to the needs of civil security practitioners to generate “evidence-based” capability roadmaps. Such roadmaps will translate the foresight process’ outcomes into concrete courses of actions to support decision- and policy-makers at EU and national levels when deciding on strategies and policy actions in the field of civil security, including the programming of security research and innovation investments. By enhancing a forward-looking culture in civil security, AHEAD will contribute to the strategy put forward in the European Commission's Foresight Agenda to embed in a more systematic way strategic foresight into policy-making.

Reported results

The AHEAD project supports law enforcement agencies in forecasting technological and criminal trends using AI-powered foresight tools. This enables proactive strategy development instead of reactive policing, including in border-related contexts. The AHEAD project concluded two foresight cycles and initiated a third. It established a Foresight Experts database and developed a comprehensive methodology, including scenario building, STEEPL and LEA workshops featuring a 'serious game', and roadmap generation.

6.3.5. THEIA

101190051

Enhancing Copernicus Security Services - EU governmental crisis management hub for forced population displacement

GEOSYSTEMS HELLAS IT KAI EFARMOGESGEOPLIROFORIAKON SYSTIMATON ANONIMIETAIREIA (EL)

€3,997,608

01/12/2024 – 31/05/2027

https://doi.org/10.3030/101190051

https://theia-project.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL4-2024-SPACE-01-36 – Copernicus for Security

Copernicus Security Services (CSS), Forced population displacement, Very-High-Resolution (VHR), Radio Frequency (RF), Machine Learning (ML), Platform

Objective

Addressing critical challenges such as population displacement due to conflicts, exacerbated by factors like climate change, extreme weather events, food shortages, and poverty, remains paramount. The implementation of THEIA, integrating data fusion, processing, and analysis, particularly leveraging Geospatial Artificial Intelligence (GeoAI) and Machine Learning, is poised to enhance the efficacy of existing services significantly. Through the amalgamation of multi-temporal data and diverse datasets, THEIA empowers better decision-making and adapts to evolving policy and user needs. This technological advancement, bolstered by GeoAI, augments detection capabilities and ensures timely access to crucial information, bridging the gap between capabilities and stringent security demands.

By integrating non-space data and end-user intelligence, THEIA's supply chains add value not only at the operational level but also at regional and local levels, facilitating improved coordination. Furthermore, THEIA catalyzes fostering EU-independent capabilities and technologies, thereby bolstering the European space ecosystem's consolidation and ensuring the sustainable coexistence of legacy and New-Space solutions. Its services cater to a wide array of end-users, including EU entities such as SatCen and Frontex, Member State Ministries of Defence, Intelligence Agencies, Police Forces, NATO, and potentially Extra-EU National and Supranational Entities such as the United Nations.

6.4. Disrupting criminal networks

6.4.1. IDEA

817963

INTERPOL Databases Enhanced Access in European Union

THE INTERNATIONAL CRIMINAL POLICE ORGANIZATION (FR)

€1,936,831

01/02/2019 – 30/04/2022

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/31077817/817963/ISFP

Programme
Internal Security Fund Police

Topic title
ISFP-2017-AG-IBA-SLTD – SLTD

improved border management systems SLTD unique global repository with over 75 million data from 176 countries playing pivotal role in enhance border security

Objective

Stolen and lost passports are among the most valuable assets for terrorists and international criminals. In several major terrorist attacks over recent decades, the perpetrators had traveled internationally using invalid passports to conceal their identity. Lost, stolen and fake travel documents are used by fugitives to escape justice, by foreign fighters to travel to conflict zones, and by organized criminal networks to traffic vulnerable individuals and illicit commodities across borders.

INTERPOL SLTD – a unique global repository, created in 2002 by INTERPOL, the world's largest international police organization, as a tool for law enforcement to detect and prevent the use of fraudulent identity documents. Today, the database includes more than 75 million records from 176 countries. In 2016 alone, the database was checked more than 1,59 billion times, generating more than 100,000 alerts where individuals tried to use travel documents registered as stolen or lost.

The widespread and systematic use of SLTD has the potential to stop criminals and terrorist in their track and so protect global public safety. Global interchange of information on lost or stolen travel documents provides improved border integrity and helps identify identity theft either at the border or in other situation where travel documents are presented as a form of identification.

The Project aims to enhance with the support and using INTERPOL policing capacities the cooperation among EU MC for fighting terrorism and transnational crime through effective border management. This project aims to prevent and detect movement of terrorist and FTFs and fight against transnational crimes through the assistance to the least performing EU Member Countries (MC) by improving their use and population of INTERPOL's Stolen and Lost Travel Document (SLTD), Travel Document Associated With Notices (TDAWN) and NOMinal databases.

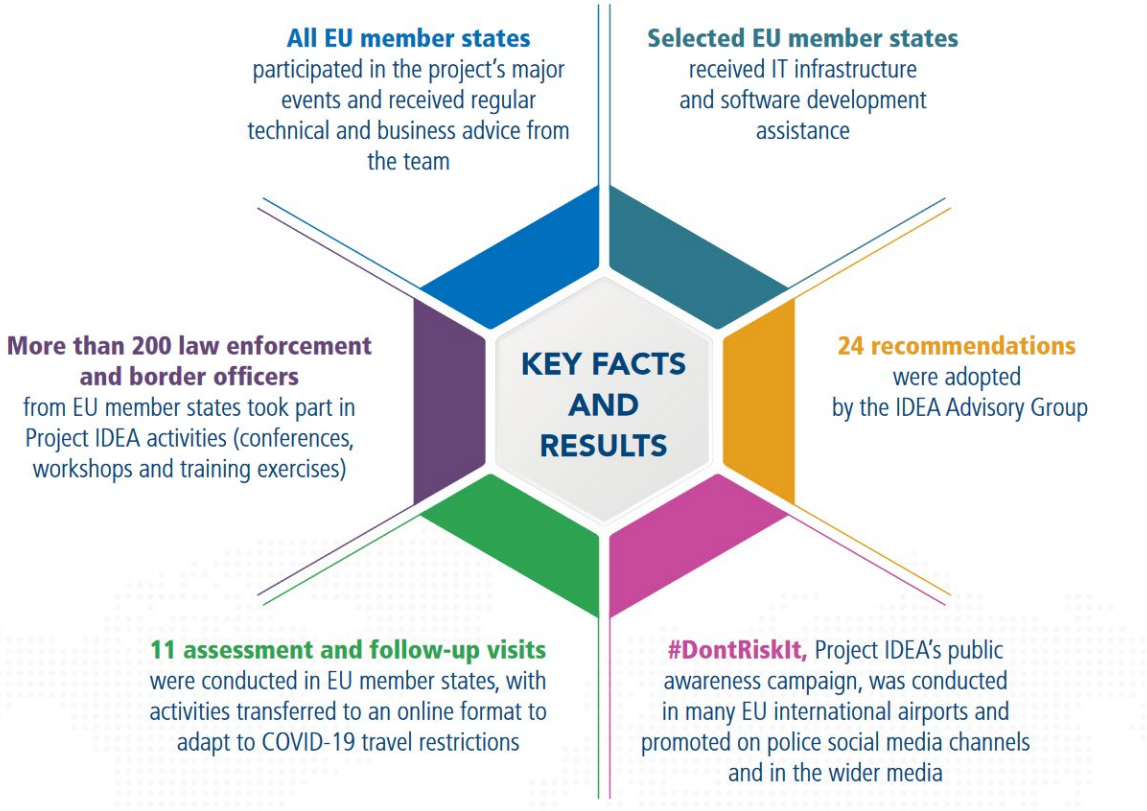
Reported results

IDEA integrated INTERPOL's lost/stolen passport databases with national systems, using AI to accelerate identity checks and flag potential threats at entry points. Around 200 law enforcement and border officers from EU Member States participated in Project IDEA's activities and key events.

An INTERPOL e-learning course on policing capabilities for frontline border officers was also developed, offering additional education on INTERPOL tools applicable to border control procedures. In addition, the INTERPOL e-learning course on policing capabilities for front line border police officers was developed to provide additional training on INTERPOL capabilities that can be used during border control procedures.

In addition, Project IDEA launched a Public Awareness Campaign for EU Member States, which was about (1) informing passport holders about the consequences of attempting to travel with documents previously reported lost or stolen, and (2) reducing the number of "administrative" hits against the SLTD database by increasing public engagement across Europe. The campaign was also publicised on social media, with the creation of the hashtag #DontRiskIt and the translation of campaign messaging for participating EU Member States.

Figure 20. Key facts and results of the IDEA project.



Source: Interpol

6.4.2. PREVISION

833115

Prediction and Visual Intelligence for Security Information

EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON (EL)

€8,001,180

01/09/2019 – 31/12/2021

https://doi.org/10.3030/833115

https://prevision-h2020.eu

Programme
HORIZON 2020

Topic title
SU-FCT03-2018-2019-2020 – Information and data stream management to fight against (cyber)crime and terrorism

https://www.youtube.com/watch?v=iFe1BJz5oj0

Objective

The mission of PREVISION is to empower the analysts and investigators of LEAs with tools and solutions not commercially available today, to handle and capitalize on the massive heterogeneous data streams that must be processed during complex crime investigations and threat risk assessments. With criminals being ever more determined to use new and advanced technology for their cause, the aim is to establish PREVISION as an open and future-proof platform for providing cutting-edge practical support to LEAs in their fight against terrorism, organised crime and cybercrime, which represent three major cross-border security challenges that are often interlinked.

PREVISION provides advanced near-real-time analytical support for multiple big data streams (coming from online social networks, the open web, the Darknet, CCTV and video surveillance systems, traffic and financial data sources, and many more), subsequently allowing their semantic integration into dynamic and self-learning knowledge graphs that capture the structure, interrelations and trends of terrorist groups and individuals, cybercriminal organisations and organised crime groups, giving rise to enhanced situational awareness in these fields.

PREVISION has a pan-European engagement and support agenda for LEAs: ten (10) different LEAs and practitioners take part in its consortium, while additional ones (including Europol) have joined its external advisory board. A strong inter-disciplinary dimension, combining technological expertise with sociological, psychological, linguistic and data science models, will lead to a common strategic approach for predicting abnormal and deviant behaviour, radicalisation potential, threat risks for soft targets, and cybercrime trends at different timescales.

PREVISION will conduct demonstrations on five representative and complementary use cases, under real-life operational conditions, in full compliance with fundamental rights and applicable legislation.

Reported results

The PREVISION platform integrates big data streams (e.g., darknet, telecom, financial data) using machine learning and knowledge graphs to predict and visualise threat networks. AI models analysed behavioural patterns and links across various data types. The PREVISION project has achieved the following results:

— A collection of over 20 data analytics tools capable of processing more than 37 different types of heterogeneous data (including video, web, dark web, telecom, network, financial, and traffic).

— A robust ontology comprising nearly 1,000 classes for representing crime events and actors, paired with an extensible Semantic Reasoner to infer semantic information.

— An integrated, scalable platform linking all developed tools, allowing for workflow and service bundle composition.

— A unified, web-based Human-Machine Interaction environment integrating and harmonising visual outputs from different tools.

— A high-TRL tool named ARTE-Fact for crawling, analysis, and detection of stolen cultural objects.

— Demonstration and evaluation across five representative use cases (anti-radicalisation, public space protection, cyber-enabled crime, identification of fraudulent companies, illicit trafficking of cultural goods), tested in multiple iterations.

— Organisation of 17 workshops (online, physical, or hybrid) for demonstration, feedback, and evaluation.

— Launch of an online training portal.

— Collaboration with 12 other security research projects, publication of 17 peer-reviewed papers, release of a white paper on knowledge engineering, and participation in over 25 dissemination events.

— Development of ethics guidelines and publication of a multi-stage societal acceptance study as a policy paper.

Selected publications

Demestichas et al. (2021) combined algorithms and tools and used them to correlate different pieces of data leading to the discovery and recording of forensic evidence. The collected data were, then, combined to handle inconsistencies, whereas machine learning techniques were applied to detect trends and outliers.

Zarras et al. (2019) explored whether Machine Learning services could imitate human (cognitive and creative) tasks to an extent that their behaviour remains indistinguishable from human behaviour. They evaluated a number of publicly available ML services and considered implications for security.

6.4.3. EMERITUS

101073874

Environmental crimes' intelligence and investigation protocol based on multiple data sources

GMV AEROSPACE AND DEFENCE SA (ES)

€4,634,193

01/09/2022 – 31/08/2025

https://doi.org/10.3030/101073874

https://emeritusproject.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-FCT-01-09 - Fight against organised environmental crime

investigation protocol, platform, satellites, drones, integration layer, lawful court-proof collection of crime evidences

https://www.youtube.com/@emeritus_EU

Objective

The aim of the EMERITUS project is to realize and implement a protocol for effective environmental crime investigation, leveraging on the integration of innovative monitoring and analysis technologies (drones, satellite data, virtual sensors, geo-intelligence data, etc.), and on a complementary training programme aimed at fostering environmental enforcement authorities (e.g. Police Authorities and Border Guards Authorities) intelligence and investigation capabilities, at national and cross-border level. In order to achieve this goal, the EMERITUS consortium has gathered 8 Police/Border Guard authorities from 5 countries, 4 security experts, 2 training specialists and 6 technological partners, that will be the core of the platform.

The consortium will focus on the co-creation of an investigation protocol centered on environmental crimes investigation, that will orient to the creation of the geo-intelligence platform. This platform will integrate several monitoring and investigation technologies and data sources, in a single view to support Police Authorities and decision-makers. Secondly, a new training programme for Police Authorities and other security practitioners, focused on environmental (waste-related) crimes investigation and prevention, will be delivered combining both theoretical aspects and hand-on simulations to empower end-users to actually use the platform and related technologies. Such exercises will serve to validate both the platform and the protocol, via simulations based on 4 realistic use cases (heterogeneous and of growing complexity), taking advantage of the shared will of Police Authorities and Border Guards to focus on ad hoc demonstration sites, with the additional availability of partners' simulation facilities. The validation of the platform and of the protocol will result in the successful elaboration of a set of evidence-based recommendations for policy authorities and decision-makers.

6.4.4. Ceasefire

101073876

Advanced versatile artificial intelligence technologies and interconnected cross-sectoral fully-operational national focal points for combating illicit firearms trafficking

ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS (EL)

€4,999,809

01/10/2022 – 30/09/2025

https://doi.org/10.3030/101073876

https://ceasefire-project.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-FCT-01-10 – Fight against firearms trafficking

Artificial intelligence, national focal points, incident tracking tool, seizure registration, dark Web marketplaces, scanning technologies, 3D-printed firearm blueprints, cryptocurrencies

Objective

Ceasefire targets the development of a highly innovative, holistic, multi-disciplinary, high-tech and versatile approach for significantly increasing/broadening the operational capabilities of EU Law Enforcement Agencies (LEAs) in their struggle to detect, **analyse and track cross-border illicit firearms trafficking related activities**. In particular, Ceasefire efforts will concentrate on the following major driving axes: a) The development of advanced Artificial Intelligence (AI) technologies for significantly facilitating the everyday work of the involved practitioners, and b) The establishment of fully-operational National Focal Points (NFPs), by targeting to alleviate from current organizational, operational, cooperation, legal, cross-jurisdictional, trans-border and information exchange challenges.

The former corresponds to a wide set of AI-enabled tools, including solutions for cyber patrolling, Web data gathering, on-the-spot detection of firearms, advanced Big Data analytics, **cryptocurrency analysis**, large-scale information fusion, visual analytics and firearms-related intelligence collection. On the other hand, particular attention will be given towards defining models, protocols and procedures for making NFPs fully functional, complemented by activities for mapping the various legal frameworks (within different countries) and procedures (within different agencies) for harmonizing investigations, cooperation, lawful evidence collection and forensics analysis.

The development of efficient information sharing mechanisms among LEA (police, border guard, customs), judicial and forensics authorities will further reinforce the system capabilities. Moreover, the tasks of Ceasefire will complement the objectives and activities of the EU Policy Cycle (EMPACT) – Firearms (where partner MIRPN acts as Driver and partner FMI as Co-driver), while continuous guidance from the European Firearms Experts (EFE) group (where partner PSP acts as the Chair) is foreseen.

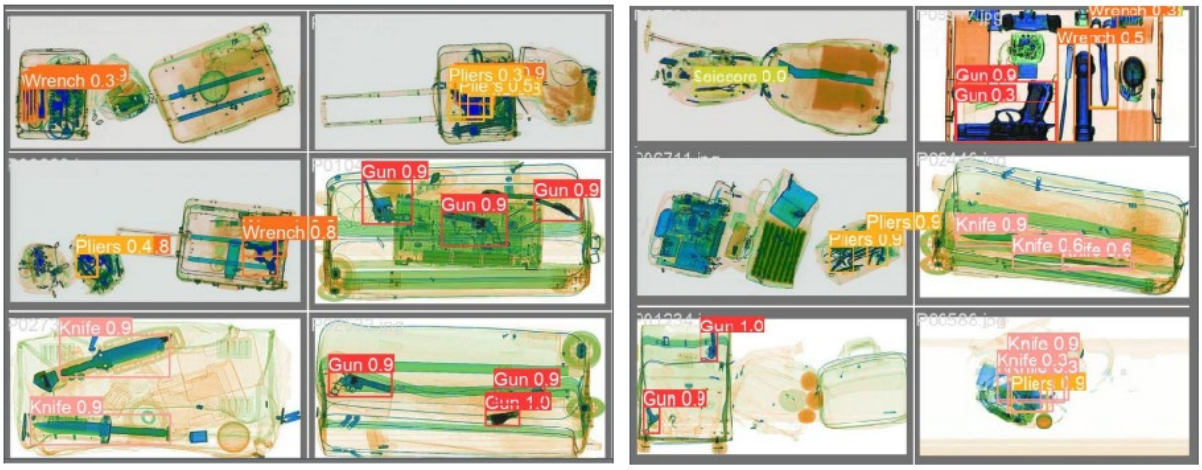
Reported results

During the first half of the project (first reporting period) the consortium focused on designing the proposed solutions, by consulting with the involved end-users, to define detailed requirements on how the proposed solutions could assist them in their day-to-day efforts to combat illegal firearm transactions. After the user requirements were defined, the system (functional and non-functional) ones were exported, as well as the individual and overall architectures (for the applications and the CEASEFIRE platform respectively). In addition, the fundamental methods and algorithms (AI, **image and text analysis** etc.) were developed by the relevant partners. As the project progressed, the first prototypes of the various applications were developed and showcased initially within the consortium, and later to relevant external stakeholders. Moreover, the pilot demonstrations of the project have been planned and are under way to be performed during the 2nd period, including several training sessions, hackathons and 2 public events, that will further push the outreach of the project. It is worth mentioning that the consortium has established a communication channel with EMPACT firearms, which has proven very useful for the technical developments as well, since the consortium has been able to receive valuable feedback from more end users.

Selected publications

Mademlis et al. (2023) discusses the role of automated X-ray image analysis in detecting contraband, which is crucial for enhancing public safety at high-traffic locations like airports and post offices. Given the large volume of data during peak times, this becomes a Big Data challenge. Modern computer vision techniques, especially those using Deep Neural Networks (DNNs) – **YOLO** and **transformers algorithms**, can handle this task effectively, even in constrained environments, like embedded systems using fast object detectors. The findings highlight the effectiveness of transformers detectors, suggest that some auxiliary neural modules developed recently are now outdated for security applications, and showcase the efficiency of the CSP-DarkNet backbone in convolutional neural networks.

Figure 21. Predictions on the SIXray test subset.



Source: Mademlis et al. (2023)

6.4.5. PERIVALLON

101073952

Protecting the EuRopean terrItory from organised enVironmentAl crime through inteLLigent threat detectiON tools

ETRA INVESTIGACION Y DESARROLLO SA (ES)

€4,670,654

01/12/2022 – 30/11/2025

https://doi.org/10.3030/101073952

https://perivallon-he.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-FCT-01-09 – Fight against organised environmental crime

Geospatial intelligence, AI multimodal analytics, intelligence picture, environmental crime observatory, HFCs, waste disposal and trafficking, smuggling routes, training

https://www.youtube.com/@PERIVALLON_HE

Objective

PERIVALLON aims to provide an improved and comprehensive intelligence picture of organised environmental crime and develop effective and efficient tools and solutions for detecting and preventing such types of criminal activities and for assessing their environmental impact based on geospatial intelligence, remote sensing, scanning, online monitoring, analysis, correlation, risk assessment, and predictive analytics technologies, by leveraging the latest advancements in Artificial Intelligence (AI) in the fields of computer vision and multimodal analytics. As a result, enhanced investigation processes and methodologies will be derived through the capabilities provided by the developed tools and solutions, and the insights obtained though the proposed Environmental Crime Observatory.

The capacity of end users (including Police Authorities and Border Guards) will also be improved and will enable them to tackle such criminal activities in an effective manner based on advanced tools and solutions and also on the innovative training curricula developed using physical and/or digital twins of relevant environmental crime scenarios. Moreover, improved international cooperation will be facilitated through improved data sharing enabled by blockchain technologies, while improved regulation shaping and tuning will be supported through relevant policy recommendations.

PERIVALLON will be validated in field tests and demonstrations in four operational use cases. Extensive training, hands-on experience, joint exercises, and training material will boost the uptake of PERIVALLON tools and technologies. With a Consortium 5 Police and Border Guard Authorities, 3 authorities related to environmental protection, 6 Research/Academic institutions, 8 industry partners (including seven SMEs), one EU Agency, and one Foundation, PERIVALLON delivers a strong representation of the challenges, requirements and tools to meet its objectives.

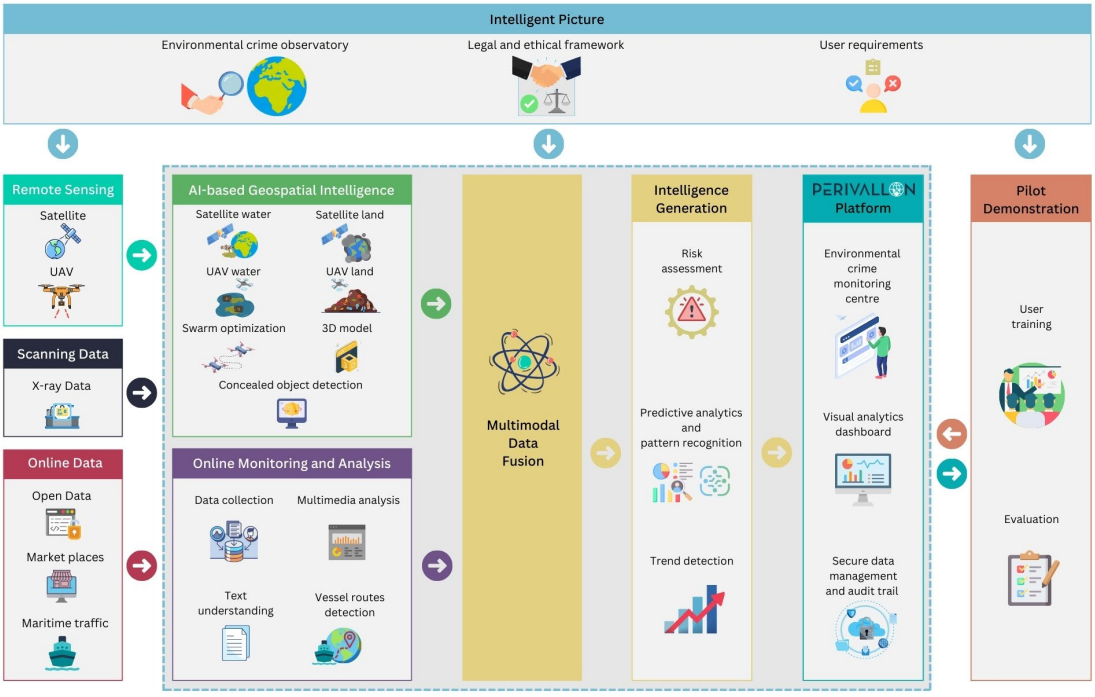
Reported results

The PERIVALLON project established partnerships with Police Authorities, Border Guards, and National/Regional Authorities who co-created four pilot use cases, 76 Functional Requirements, 25 Non-Functional Requirements, and 13 Security Requirements. To meet Law Enforcement Authorities' needs, an integrated platform architecture with 28 components and over 130 technical requirements was designed. In parallel, the Environmental Crime Observatory developed an improved intelligence picture of organised environmental crime through case studies in six countries. These efforts supported the development of 16 AI methods and more than five datasets for training and evaluation, all of which are part of the first PERIVALLON integrated platform prototype. This prototype is being demonstrated and evaluated in 11 pilot demonstrations: four in Italy, one each in Greece, Sweden, and Belgium, two in Romania, and two transnational pilots. Waste codes were also collected and initially categorised based on the European Waste Catalogue, resulting in 105 distinct codes potentially detectable using the PERIVALLON tools.

Selected publications

Villamor Medina et al. (2025) worked on delivering an environmental crime observatory aiming to provide an improved and comprehensive intelligence picture of organised environmental crime and an investigation platform. Villamor Medina et al. (2025) also worked on a framework for predicting the likelihood of terrorist incidents within a specific period in a target country based on machine learning and deep learning models. The models were trained on news data for the target country sourced from the Global Database of Events, Language, and Tone (GDELT), while the terrorist incidents recorded in the Global Terrorist Database (GTD) serve as the ground-truth

Figure 22. PERIVALLON concept diagram.



Source: Cordis

6.4.6. ARIEN

101121329

ARtificial IntelligencE in fighting illicit drugs production and traffickiNg

ENGINEERING - INGEGNERIA INFORMATICA SPA (IT)

€4,268,160

01/11/2023 – 31/10/2026

https://doi.org/10.3030/101121329

https://arien-euproject.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-FCT-01-06 – Effective fight against illicit drugs production and trafficking

Intelligence picture, Artificial Intelligence, International cooperation, Information exchange, Criminological analysis, Police authorities, Border and coastguards

Objective

ARIEN proposes the execution of a comprehensive workplan that thoroughly addresses the call topic aligned to the 2021-2025 EU Drugs Action Plan (DAP) achieved through a holistic innovation action that builds a real-time intelligence picture illegal drug production and trafficking incidence across the EU bolstered by AI-driven tools to understand, detect, analyse and track the entire drugs trafficking chain, its financial flows and the role of black-markets in the global darknet and their relationships with of social media.

ARIEN builds on a strong platform of multi-disciplinary research, following a co-creation approach, across the criminological, societal, and legislative landscapes that will make recommendations for harmonization and drive an agenda for the development of modular solutions embedding modern Artificial Intelligence (AI) techniques and best practices for international cooperation in data-rich environments in a legally and ethically compliant manner.

Aiming at improving the investigative capability of LEAs on illegal drugs production and trafficking, ARIEN will leverage on innovative AI techniques to monitor online illicit drugs markets and collecting precious information on physical drug-dealing hotspots through social media. Additional layers of analysis, supported by explainable AI (XAI) methods, will provide actionable knowledge on modus operandi and emerging threats and trends both on drugs and drug-related organized crime networks through integrations with criminal records data and detection of online activities.

Moreover, ARIEN will promote effective strategies on international cooperation among EU and non-EU LEAs, Customs and Border Guards Authorities and policy makers to enhance investigative activities against criminal groups, capitalising on effective AI techniques.

6.4.7. VANGUARD

101121282

adVANced technoloGical solutions coupled with societal-oriented Understanding and AwaReness for Disrupting trafficking in human beings

ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS (EL)

€4,536,682

01/11/2023 – 31/10/2026

https://doi.org/10.3030/101121282

https://vanguard-horizon.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-FCT-01-07 – Effective fight against trafficking in human beings

intelligence picture, technological solutions, AI, computer vision, multimodal analytics, awareness raising, serious games, policy recommendations

Objective

VANGUARD aims to strengthen the fight against trafficking in human beings (THB) at the nexus of advanced technological solutions, understanding, awareness raising, and training in order to disrupt the trafficking chain at an early stage and address the culture of impunity. In particular, VANGUARD aims to provide an improved intelligence picture of THB, with particular focus on THB for purposes of sexual exploitation, labour exploitation, and forced criminality. This will be achieved through developing a modular and trustworthy suite of tools for detecting, identifying, investigating, and preventing online-facilitated THB activities and THB-related activities at (border) checkpoints based on the analysis of online multimedia content and multimodal streams, by leveraging the latest advancements in Artificial Intelligence (AI).

It will also address the societal dimensions of THB through improved understanding and engagement with relevant actors, improved strategies of international cooperation, increased awareness, and better evidence-based policymaking. The capacity of end users (including Police and Border Guard Authorities) will be further enhanced and will enable them to tackle such criminal activities in an effective manner based on advanced tools and solutions and on innovative training curricula.

VANGUARD will be validated in field tests and demonstrations in three operational use cases. Extensive training, hands-on experience, joint exercises, and training material will boost the uptake of VANGUARD tools and technologies. With a Consortium of seven Police and Border Guard Authorities (including from countries of origin and transit of THB networks and countries neighbouring Ukraine), one Police Academy, eight research/academic institutions, four industry partners (including two SMEs), and two CSOs, VANGUARD delivers a strong representation of the challenges, requirements, and tools to meet its objectives.

6.5. Operations research

6.5.1. PMT4NIIS

790798

Predictive Maintenance Tool for Non-Intrusive Inspection Systems

DANLEX EOOD (BG)

€50,000

01/01/2018 – 30/06/2018

https://doi.org/10.3030/790798

Programme
HORIZON 2020

Topic title
SMEInst-13-2016-2017 – Engaging SMEs in security research and development

predictive, maintenance, inspection, detection, system, security, customs, airport, port, border, service, critical infrastructure, x-ray

Objective
The importance of providing for effective, time and cost-saving maintenance of Non-Intrusive Inspection Systems (NIIS) cannot be overstated. For the purpose of our proposal, NIIS are X-ray systems deployed to inspect all types of parcels, baggage, cargo, vehicles, containers, and trains, in a non-intrusive manner. Over the past 10 years Danlex has realised sales of NIIS equipment and NIIS maintenance services totalling more than EUR 40 million in Bulgaria, Macedonia, and Kosovo. Airports, customs administrations, border police, harbours, and other authorities responsible for protecting the freedom and security of societies worldwide (security organisations) in the European Union and worldwide have been charged with: preventing global terrorist threats and illegal human trafficking; detecting and identifying illicit or undeclared goods; increasing security to the global supply chain. To tackle these challenges security organisations have increasingly been utilising NIIS equipment over the last two decades, as the most effective equipment for fulfilling their duties in compliance with the applicable regulations and standards. With the average NIIS lifespan of 10 years the maintenance has proved to be high and lead to much higher than anticipated running costs over the equipment's lifetime, totalling to 100% of NIIS purchase price. To ensure value for money, it is important that NIIS operate with maximum efficiency and at the lowest possible level of maintenance and life-cycle costs. NIIS maintenance solution enabled by PMT (PMT4NIIS) will allow us to equip the security organisations and market stakeholders with the next generation of NIIS maintenance. Independent reports estimate that with reliability-centered maintenance powered by predictive models, stakeholders in other markets spend 15% of their overall maintenance time on Predictive Maintenance, yet achieve an impressive 50% decrease in maintenance costs.

6.5.2. BA-AC

880158

Cloud-based Airport Operations Planning tool with holistic forecasting of airport allocation planning

COPENHAGEN OPTIMIZATION APS (DK)

€2,153,901

01/10/2019 – 30/09/2021

https://doi.org/10.3030/880158

Programme
HORIZON 2020

Topic title
EIC-SMEInst-2018-2020 – SME instrument

Objective
Airports are facing an increasing challenge of high sustained demand for passenger air travel, expected to increase by 15% by 2025. However, 40% of EU airports will not be able to meet this demand. A new aircraft stand (where aircraft are parked) has a cost between €5m and €15m, in addition to the cost of disruption while constructing. At the same time, operating an airport is a very complex task involving many stakeholders, making it difficult to identify options for increasing efficiencies. Better Airport® is the first cloud-based, holistic, Airport Operations Plan solution, providing airport operators with an AI-driven, easy-to-use solution which efficiently and effectively creates optimised operational plans, enabling informed decision making across airport stakeholders. Using advanced mathematical models and based on years of consultancy within airport operations, Better Airport® covers the entire value chain, from passenger check-in, security and border control to baggage handling and aircraft allocation. Our cloud-based approach allows all stakeholders to access information wherever, whenever, increasing efficiency across operations and ultimately saving airports at least €10m/year, while also driving the EU goal of a Single European Sky. Better Airport® will allow airport operators to forecast, plan and interact airports in between, connecting all airports with one system. The BA-AC project builds upon upscaling our validated prototype, most of our efforts will focus on technology maturation, piloting and market maturation, specifically on the maturation and integration of Better Aircraft, into Better Airport®, and perform large scale piloting at 3 major airports, advancing our current prototype from TRL7 to TRL 9. With a total market of 575 airports, we expect 5 years post project, to reach a market share of 17.8%, totalling 89 client airports, an annual recurring revenue of €20m, accumulated revenue of €65.78m, creating 47 new full-time employee positions.

Reported results

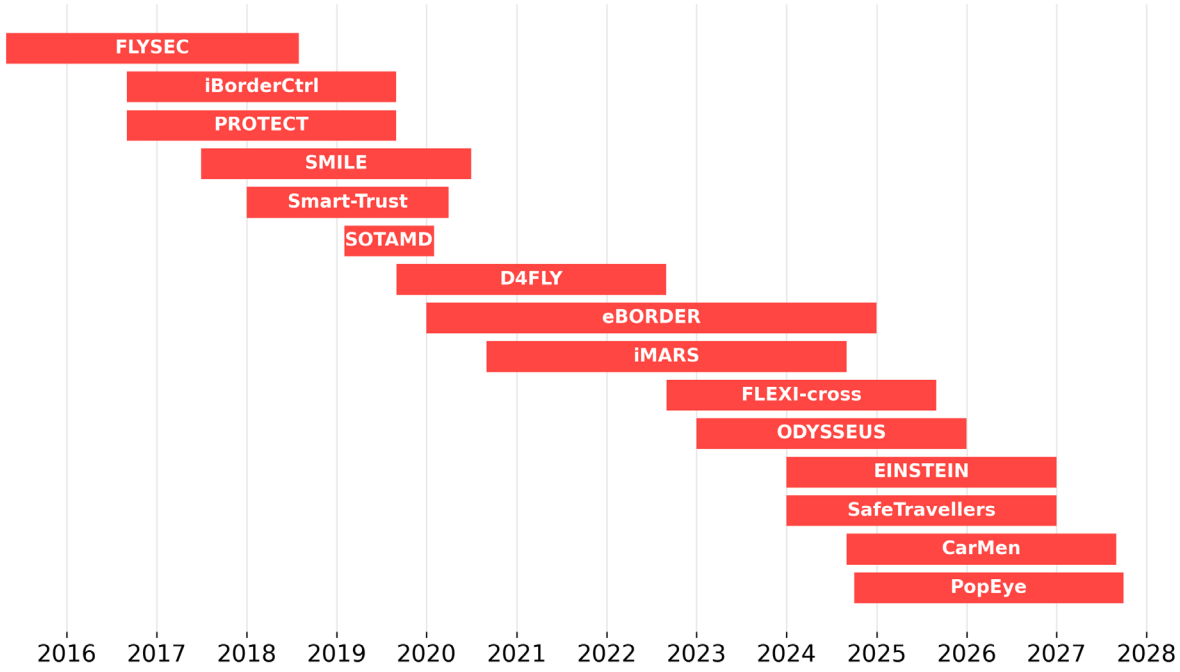
Better Stand & Gate entered its testing phase period before becoming the final piece completing our Better Airport® software package. The product interface and the AI algorithm have been implemented and integrated into our Better Airport software. Furthermore, the product has been refined during the pilot phase where we received positive feedback from our pilot airports. We consistently implement the feedback received allowing us to advance our current prototype from TRL7 to TRL9. By the end of summer 2021, Better Stand & Gate became a key component of Better Airport, reaching commercialization. We are proud to say that more than 500 million passengers per anno go through airports planned in Better Airport® during normal traffic levels (pre-COVID).

7. Identity, biometrics, and document security

7.1. Introduction

We identified 15 projects focussing on AI for identify, biometrics and document security. These projects range from integrating biometric data of border-crossing individuals to detecting fraudulent documents and crossings to prevent large influxes of people at the borders. A timeline of these projects is shown in Figure 23. The aim of all the projects is to provide technologies (including AI technologies) for a better border crossing flow on land, sea, and air.

Figure 23. EU projects featuring AI for identity, biometrics and document security.



Source: JRC

Biometric features used for person identification include **fingerprints**, **3D face** (incl. morphed face detection), iris, periocular, somatotype, palm vein, etc. Some solutions are deployed as **mobile applications** for board guards.

It is noticeable that several projects are working on **on-the-move detection** (6 out of 15 projects) and **fraud detection on identity documents** (6 out of 15 projects).

Some projects have the main objective analysing **privacy and data protection, ethical and social issues**, including any potential concern that may occur during contactless and multimodal biometric identification mechanisms. The projects presented in this chapter are varied. However, common features between the projects are gathered and presented in Table 6.

Table 6. Summary of topics addressed by each project.

Category	Project	Biometric traits ([1])	Person counting ([2])	Identity verification on-the-move	Identity documents fraud detection	Wireless mobile app	Focus	Tested / deployed
On-the-move and multimodal biometrics	**FLYSEC**	face	X			X	Radio frequency identification for carry-on luggage tracking and quick unattended luggage handling	X (Luxembourg Airport Terminal)
	iBorderCtrl	face; palm vein scanning; lie detection using micro-expressions	X		X		*Cancellable biometric system* for a better privacy of the biometrics data	
	PROTECT	face; fingerprints; finger vein; thermal face recognition	X	X		X	Investigation of privacy and data protection, ethical and social issues.	X (Kętrzyn, Poland)
	SMILE	facial hair (moustache, beard); eyes colour; skin colour (ethnicity); gender; age group (child, adult, senior); eyewear						X (Nădlac border between Hungary and Romania; Ruse border between Romania and Bulgaria)
	eBORDER	face; fingerprints; finger vein				X	Security on wireless communication, availability, confidentiality, authentication, and integrity.	

Category	Project	Biometric traits ([1])	Person counting ([2])	Identity verification on-the-move	Identity documents fraud detection	Wireless mobile app	Focus	Tested / deployed
	FLEXI-cross	face; fingerprint				X		
	ODYSSEUS	face	X					
	CarMen	face; iris; periocular	X	X	X		"on-the-move" persons counting.	X (UK-France border)
	PopEye	face; gait		X				
Document authentication and fraud detection	**Smart-Trust**	face	X			X		X (international airports of Lisbon and Schiphol)
	SOTAMD	face					Morphed face image detection.	
	D4FLY	face	X	X	X			
	iMARS	face; fingerprints; periocular			X	X	Face morphing attack detection. Detecting double-identity fingerprint.	
	EINSTEIN	face; iris		X	X	X		X
	SafeTravellers	? ([3])		X	X			

1 See Table 7 for the comprehensive list of biometric traits analysed by the projects.

2 Person counting / detecting passenger trends and anomalies in real-time.

3 Existing, but not mentioned which ones.

Source: JRC

Although many of the biometrics are used in at least one project analysed in this chapter, some biometrics found in general literature are not. Table 7 shows the breakdown of the biometrics traits into features used and not used by at least one analysed project.

Table 7. Summary of biometrics analysed and not analysed by the projects.

Category	**Biometrics included in the projects**	**Biometrics found in literature, but not included in the projects**
Face 	2D face (NIR, VIS) ([1]) 3D face Thermal face	
Eye 	Iris (NIR, VIS) Periocular (NIR, VIS) Retina scan	Pupil dynamics
Hand 	Fingerprints Finger Veins Hand Veins	Palm print Hand geometry
Voice and Speech 	Voice	Speech patterns Speaker recognition Linguistic features (e.g., accent, dialect)
Body 	Anthropometrics Gait Posture Body shape and size	
Behavioural 	Video analytics Anomaly detection ([2])	Keystroke dynamics Mouse dynamics Signature recognition
Other 		DNA Ear shape Footprint Odour recognition (e.g., scent, pheromones)
Multimodal 	Face-voice combination Face-iris combination Fingerprint-vein combination	Other multimodal combinations

1 NIR – Near infrared; VIS – Visible.

2 Research done in the ISOLA project, part of the "Maritime and border surveillance" chapter.

Source: JRC

Even if **dialect recognition** technology is not part of the analysed projects, this technology is already used by the German authorities. Federal Office for Migration and Refugees (BAMF) in Germany is using IT assistance systems consisting of dialect recognition software and a **transliteration assistant**. A unified language/dialect detection system for all EU countries would be very useful for all the EU countries facing many asylum-seekers requests. These initiatives on national levels can be the foundation for the unified system for all EU countries.

Anomaly detection of individuals' behaviour using video analytics can be useful for monitoring people's flows at the checkpoints in airports and land borders. Regarding behavioural features existing in the literature for the border-context, the **signature recognition** is feasible, as many EU national ID cards contains or are in the process of adding the signature of the document holder. **Mouse dynamics** and **keystroke dynamics** (how a person types on a keyboard or touchscreen) are a modern authentication method part of the behavioural biometrics. These techniques can't be used for border-crossing, but they can be used for the pre-border crossing online procedures. For instance, in the case of the future **ETIAS** (European Travel Information and Authorisation System) that will require filling in a form in advance from the moment of border crossing. In this case, mouse and keystroke dynamics can be a useful tool in the online application process.

The very sensitive biometric traits such as **DNA** are strictly regulated by European and international legal documents in the context of **privacy and human rights**. The following documents shapes the individuals' rights in the border crossing context: *Schengen Borders Code (Regulation (EU) 2016/399)*, *Regulation (EU) 2016/794 on Europol*, *Charter of Fundamental Rights of the European Union*, *General Data Protection Regulation (GDPR)*. All these documents shall be respected during the implementation of any EU-founded project.

Other biometric traits such as odour recognition (scent, pheromones) can hardly be integrated for the continuous flows of civils at borders. These traits might be used regarding high-risk civils with a previous criminal record. This shall be analysed by the relevant authorities.

All the biometrics-related technologies are evolving, and new biometric traits appear. At the same time, border crossing methodologies, legislation, and individual privacy rights are **continuously changing and adapting**. For instance, in the package agreed at **EU-United Kingdom Summit** in May 2025 it was decided that British citizens can now use electronic gates for border control in the EU airports. Another example is the **EU Digital Identity Wallet**. The ongoing projects and the future projects shall integrate in their research and deployment the future EU Digital Identity Wallet, which will be available to every European citizen, resident and business by the end of 2026. All these changes should be addressed fast and any previous research on technologies is very useful. Therefore, **continuous research** on the biometrics and identity topic is needed to maintain safe, strong, and resilient European borders.

7.2. On-the-move and multimodal biometrics

7.2.1. FLYSEC

Optimising time-to-FLY and enhancing airport SECurity — 653879

NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS" (EL)

€4,089,500

01/05/2015 – 31/07/2018

https://doi.org/10.3030/653879

http://www.fly-sec.eu/

Programme
HORIZON 2020

Topic title
DRS-16-2014 – Critical Infrastructure Protection topic 6: Improving the aviation security chain

airport security process for passengers

https://youtu.be/A42Ee0fsey0?si=j7BTup3t3TUMVmn5

Objective

Complementing the ACI/IATA efforts, FLYSEC project aims to develop and demonstrate an innovative integrated and end-to-end airport security process for passengers, enabling a guided and streamlined procedure from the landside to airside and into the boarding gates, and offering for the first time an operationally validated innovative concept for end-to-end aviation security. On the technical side, FLYSEC achieves its ambitious goals by integrating new technologies on video surveillance, intelligent remote image processing and biometrics combined with big data analysis, open-source intelligence and crowdsourcing.

Repurposing existing technologies is also in the FLYSEC objectives, such as mobile application technologies for improved passenger experience and positive boarding applications (i.e. services to facilitate boarding and landside/airside wayfinding) as well as RFID for carry-on luggage tracking and quick unattended luggage handling. Besides more efficient background checks and passenger profiling, FLYSEC aims to implement a seamless risk-based security process within FLYSEC combining the aforementioned technologies with behavioural analysis and innovative cognitive algorithms.

A key aspect in the design of FLYSEC risk-based security is applying ethical-by-design patterns, maximizing the efficiency of security controls through passenger differentiation ranging from “unknown” to “trusted”, while remaining ethical and fair in the process. Policy, regulatory and standardisation aspects will also be examined in the context of FLYSEC innovative security concept.

Reported results

The FLYSEC final field test was the first deployment of the innovative airport risk-based screening and security concept in an actual operational airport environment with Lux-Airport security personnel, practitioners and security managers operating the FLYSEC system and components.

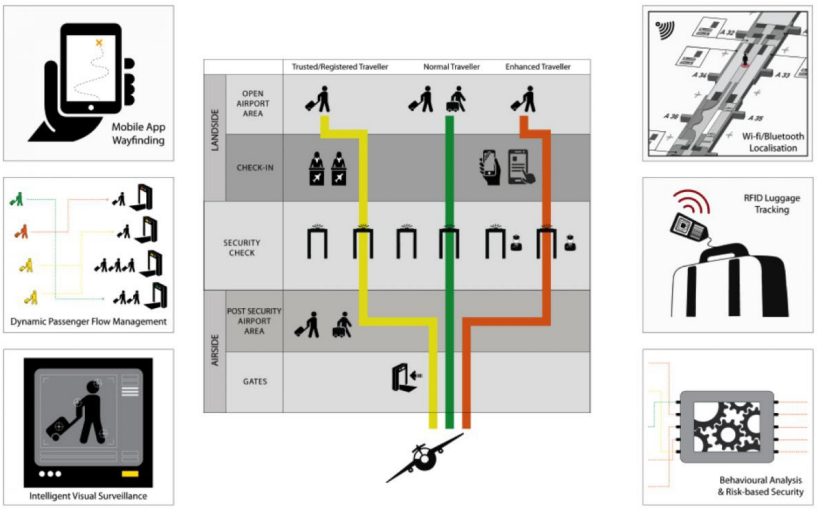
The integrated FLYSEC System successfully deployed and demonstrated:

- A mobile application for passengers, including features such as routing and navigation, positive boarding and assisted security walkthrough, assisted shopping and time management.
- A mobile application for security personnel with location-based alerts, incident handling capabilities and notifications, queue monitoring and assisted passenger and behaviour screening.
- Intelligent Visual Sensors for behaviour analytics and risk detection.
- The TravelDoc mobile kiosk and terminal for travel document scanning, enhanced validity verification and integration of PNR data in the FLYSEC risk classification service.
- A SmartQ component for smart security queue management integrated with transparent risk classification assessment algorithms.
- The RFID carry-on luggage tracking system for unattended luggage detection and fast retrieval service.
- A Beacon-based localization system for enabling location-based passenger services, also offering location correlation capabilities to the security fusion algorithms.
- A web-based command and control portal, providing holistic airport real time monitoring and security checkpoint management as well as communication capabilities which include integrated passenger simulation module for training and decision support.
- Risk Classification and Enhanced situational awareness fusion and machine learning services, correlating input from heterogeneous sources, behaviour and risk indicators and generating relevant alerts.
- The FLYSEC simulation platform for airport operational flow and passenger/personnel crowd simulation including realistic 3D visualization of Luxembourg Airport Terminal.

Selected publications

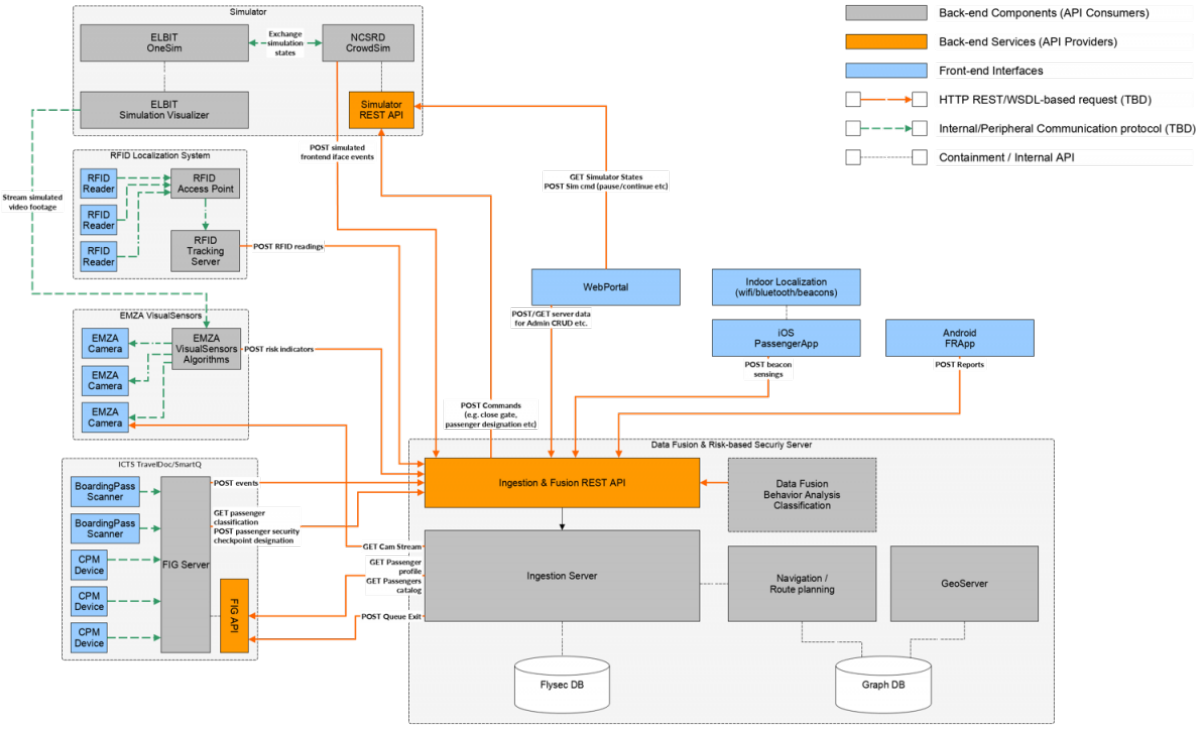
Thomopoulos et al. (2018) details the general structure of the FLYSEC project. Figure 24 shows the airport flow management using technologies such as Wi-Fi and Bluetooth localisation, RFID luggage tracking, behavioural analysis, risk-based security, video surveillance. This information is segregated and available for airport officers in a dedicated app. Figure 25 shows the entire architecture (backend and frontend) of the mobile app and web portal developed by the project.

Figure 24. FLYSEC Secure Tunnels scenario.



Source: Thomopoulos et al. (2018)

Figure 25. The backend and frontend chart of the FLYSEC mobile app and web portal.

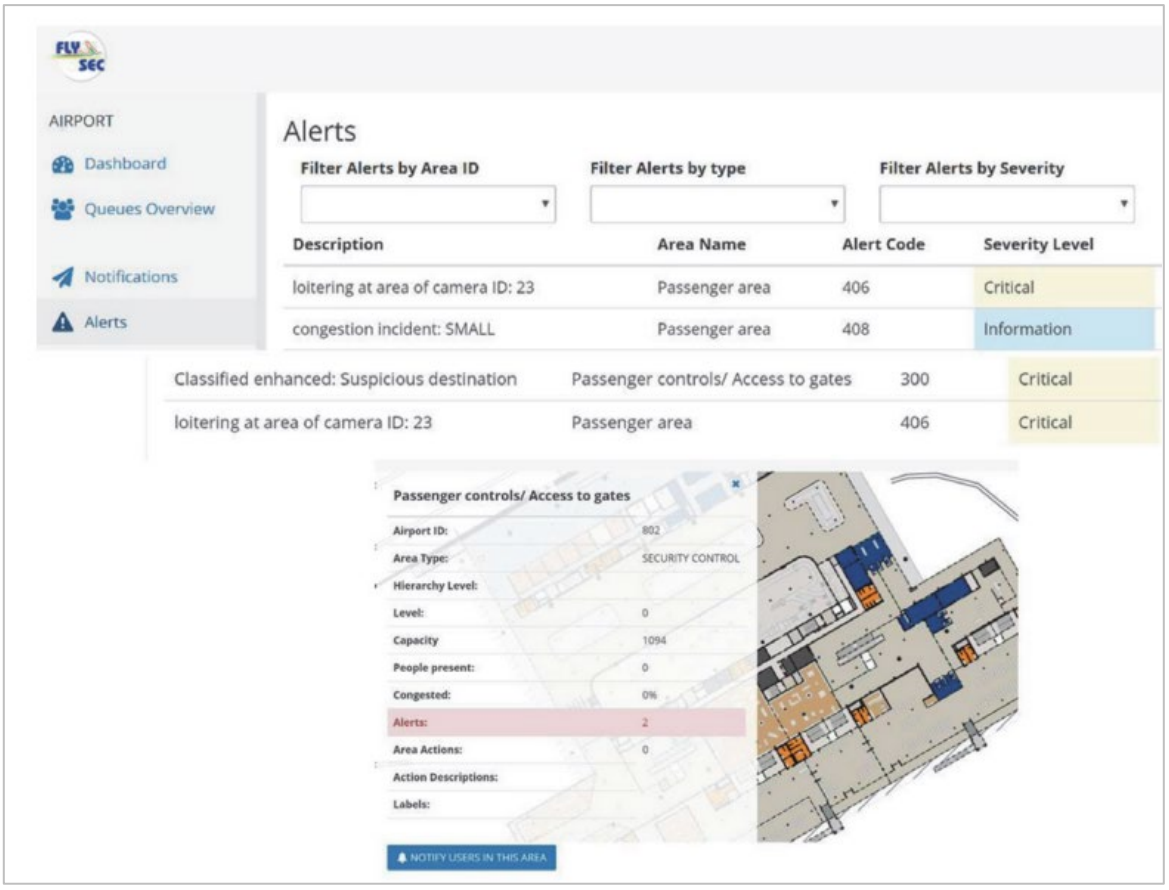


Source: https://cordis.europa.eu/project/id/653879/results (Deliverable D8.2)

The entire project (including the passengers flow management and the app/web portal) was aligned with the *ACI (Airports Council International) and IATA (International Air Transport Association) Smart Security programme* of 2016, the programme that evolved into the current *Smart Security Strategy* in 2019. This reveals the importance of complying not only with the EU regulations, but also with the **relevant international associations and bodies in the aviation field**.

Thomopoulos et al. (2018) presents, among others, AI techniques for **anomaly detection from passenger trajectories** using the **iCrowd Simulator agent-based simulation platform**. Samples of the results of the project, namely screenshots from the FLYSEC web portal used by the airport officers are presented in Figure 26.

Figure 26. FLYSEC web portal screenshots.



Source: FLYSEC website

7.2.2. iBorderCtrl

700626
Intelligent Portable Border Control System
EUROPEAN DYNAMICS LUXEMBOURG SA (LU)
€4,501,878 01/09/2016 – 31/08/2019 https://doi.org/10.3030/700626	Programme HORIZON 2020 Topic title BES-05-2015 – Border crossing points topic 1: Novel mobility concepts for land border security
Risk based approach, deception detection, anti-hacking cloud based architecture

Objective

iCROSS envisages to enable faster thorough border control for third country nationals crossing the borders of EU, with technologies that adopt the future development of the Schengen Border Management. The project will present an optimal mixture of an enhanced, voluntary form of a Registered Traveller Programme and an auxiliary solution for the Entry/Exit System based on involving bona fide travellers.

iCROSS designs and implements a system that adopts mobility concepts and consists of a two-stage procedure, designed to reduce cost/time spent per traveller at the crossing station. It leverages software and hardware technologies ranging from portable readers/scanners, various emerging and novel subsystems for automatic controls, wireless networking for mobile controls, and secure backend storage and processing. The two-stage procedure includes: (A) the registration before the travel to gather initial personal, travel document and vehicle data, perform a short, automated, non-invasive interview with an avatar, subject to lie detection and link the traveller to any pre-existing authority data. Utilizing multifactor analytics and risk-based approach, the data registered is processed and correlated with publicly open data or external systems such as the SIS II. Processing will need the travellers consent as set in EU legislation and national law. (B) the actual control at the border that complements pre-registered information with results of security controls that are performed with a portable, wireless connected iCROSS unit that can be used inside buses/trains or any point.

Multiple technologies check validity and authenticity of parameters (e.g. travel documents, visa, face recognition of traveller using passport picture, real-time automated non-invasive lie detection in interview by officer, etc.). The data collected are encrypted, securely transferred and analysed in real time, providing an automated decision support system for the border control officers.

Reported results

Beside the functionalities related to current border checks (check of travel document or validation of traveller identity) iBorderCtrl would offer additional ones and innovation provided by the modules that compose the system and the combinations between them. These modules used technologies at different stages of maturity and could be further developed or tested either individually or in combination. They include:

— Pre-arrival registration of travellers previous to their arrival at the border: the checks take place in advance, allowing for much more computationally expensive methods to be deployed.

— Assessment of attempts by the travellers to deceive border agents regarding the data indicated during the pre-arrival registration (purpose of their trip, length of stay etc).

— Personalized Avatar agent: the avatar agent that performs the interview can be personalized to gender and language of the traveller.

— Inclusion of new advanced biometric technologies: face matching and palm vein scanning.

— Forecast of traveller flow at the border crossing points: statistics, trends and models help with better planning and resources allocation.

— Analysis of historical data for border management improvement and identification of patterns of illicit border crossings.

— Adaptive risk assessment of travellers: a multi-factor risk assessment, following a robust Multi-Criteria Decision Analysis methodology to support the decision-making of border guards.

Selected publications

Crockett et al. (2017) approaches interesting biometrics topics, such as *cancellable biometric system*, a method that ensure a better data privacy of the biometric stored in large systems. This is a method that transforms a user's biometric data, such as fingerprints or facial features, into a new format using a password or key specific to that user. This transformed data is then stored in a database, rather than the original biometric information. When a user tries to access a system, their biometric data is transformed in the same way and compared to the stored template. This approach allows a user to have multiple, unique biometric templates for different applications. The concept of transformed biometrics was first introduced by Soutar et al. (1998), but it wasn't until that Ratha et al. (2007) fully developed the idea of cancellable biometrics, which has since become a key area of research in the field.

7.2.3. PROTECT

Pervasive and UseR Focused BiomeTrics BordEr ProjeCT	700259
THE UNIVERSITY OF READING (UK)	
€4,981,752 01/09/2016 – 31/08/2019 https://doi.org/10.3030/700259	Programme HORIZON 2020 Topic title BES-06-2015 – Border crossing points topic 2: Exploring new modalities in biometric-based border checks
https://youtu.be/XLGOr7sYLpo?si=QVHXgj43RFVN4vLc	

Objective

The goal of the PROTECT project is an enhanced biometric-based person identification system that works robustly across a range of border crossing types and that has strong user-centric features. The system will be deployed in Automated Border Control (ABC) areas supporting border guards to facilitate smooth and non-intrusive rapid crossing by travellers based on deployment of the next generation of biometric identification detection methods. The ability for the system to efficiently process low-risk travellers, combined with increased levels of accuracy, security and privacy standards and enabling border guards to concentrate resource on higher-risk travellers, are central ambitions of the project.

To achieve these goals, a multi-biometric enrolment and verification system is envisaged, taking into account current and next-generation e-Passport chips, mobile equipment and person identification 'on the move'. Research will be undertaken into optimization of currently deployed biometric modalities, application of emerging biometrics (including contactless finger vein, speaker recognition and anthropometrics), multi-modal biometrics and counter-spoofing, for border control scenarios. An integral part of the project is collection and dissemination of new border-realistic biometric datasets, and systematic evaluation of the developed biometric methods including vulnerability and privacy assessment.

The PROTECT project is strongly user-driven and demonstration of the developed biometric system will be conducted at two different border crossing sites. Finally, the PROTECT project will make contributions to facilitating border crossing of bona-fide non-EU citizens as well as evolving standards in biometric systems.

Reported results

A key innovation, the *biometric capture area* was conceived, specified and developed incorporating a multimodal biometric sensor network applicable to travellers on-foot (principally at air/sea borders) and land with travellers in vehicles (at land borders). Additionally, for the air/sea border use case, a novel person re-identification system was developed to support tracking of travellers within the *biometric capture area*.

Mobile passports (on smartphones), and advanced passports incorporating multimodal biometrics, were researched and developed enabling identity verification on-the-move. Individual biometric

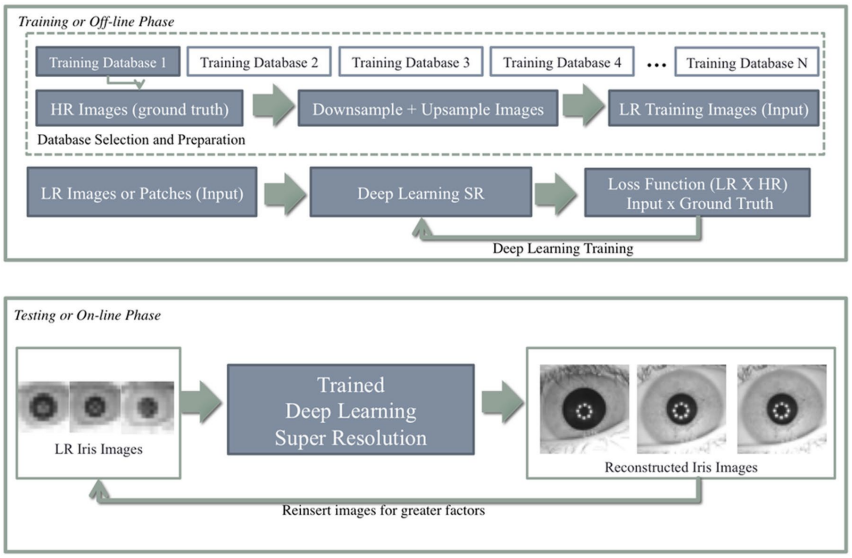
devices were further developed, and new privacy enhancing (especially template protection) and counter spoofing methods were researched.

A key part of the project work was also to undertake a thorough investigation of privacy and data protection, ethical and social issues raised by contactless and multimodal biometric systems.

Selected publications

Ribeiro et al. (2019) explores improving iris recognition from low-resolution images, such as those from mobile phones and surveillance cameras, using single image Super Resolution techniques with Convolutional Neural Networks (CNNs). The goal is to enhance image detail and realism, maintaining fidelity to the original low-resolution images to improve recognition accuracy. The authors test various CNN architectures and training databases on infrared and mobile phone image datasets. They assess the quality of the improved images and conduct recognition experiments, finding that deeper architectures trained with well-balanced texture databases boost recognition performance. Figure 27 summarises the pipeline for the reconstruction of the iris images using super resolution.

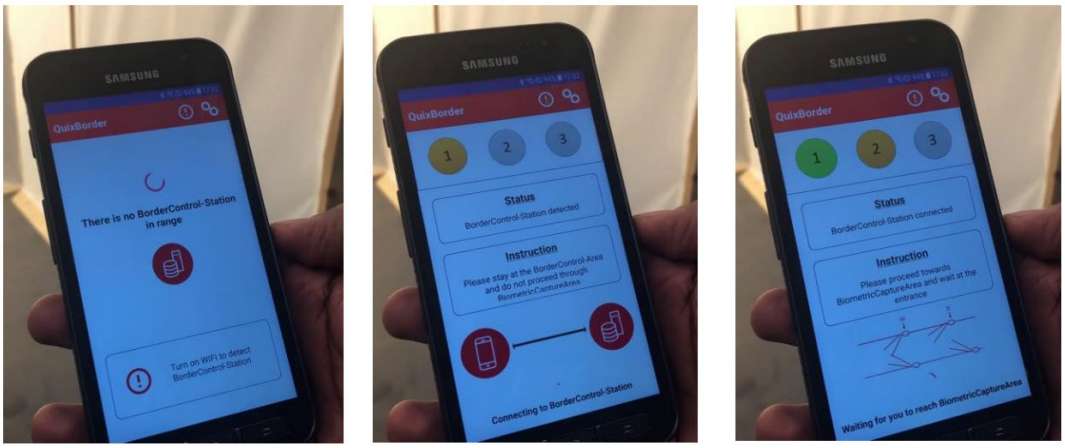
Figure 27. Overview of the training and reconstruction method for the Iris Super Resolution using CNNs.



Source: Ribeiro et al. (2019)

Regarding the deliverables of PROTECT project, it is important to mention the two demonstrations of the project results. Figure 28 reveals some screen shots from the mobile app developed, while Figure 29 shows the biometrics used during the final demonstration of the project at the Polish Border Guard Training Centre in Kętrzyn.

Figure 28. Digital Travel Credentials stored in a mobile phone.



Source: https://cordis.europa.eu/project/id/700259/results (Deliverable D8.2)

Figure 29. Biometrics used at the Polish Border Guard Training Centre in Kętrzyn.

Demonstrators	Type A Air / Sea Border		Type B Land Border	
Sensor Type	**Fixed**	**Mobile**	**Fixed**	**Mobile**
2D Face VIS	✓	-	-	-
2D Face NIR	✓	-	-	-
3D Face	-	-	✓	-
Thermal Face	-	-	✓	-
Iris NIR	-	-	-	-
Iris VIS	-	-	-	-
Periocular VIS	✓	-	-	-
Periocular NIR	✓	-	-	-
Finger Veins	-	-	✓	-
Hand Veins	-	-	-	-
Voice	-	-	-	-
Anthropometrics	✓	-	-	-

Source: https://cordis.europa.eu/project/id/700259/results (Deliverable D9.4)

7.2.4. SMILE

740931

SMart mobILity at the European land borders

ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS (EL)

€4,999,276

01/07/2017 – 30/06/2020

https://doi.org/10.3030/740931

Programme
HORIZON 2020

Topic title
SEC-14-BES-2016 - Towards reducing the cost of technologies in land border security applications

access control, architectural design, colors, ethnology, ecosystems

Objective

SMILE proposes a novel mobility concept that addresses the aforementioned challenges by designing, implementing and evaluating in relevant environments (TRL6) prototype management architecture, for the accurate verification, automated control, monitoring and optimization of people' flows at Land Border Infrastructures. It leverages the capabilities of the smart mobile devices in biometric control for secure and trusted authentication, and elaborates on their exploitation as part of a multimodal biometric verification process that supplements / complements existing approaches. Furthermore, SMILE's mobility concept builds upon Private Cloud Infrastructure technologies which communicate with remote SMILE handhelds through a secure gateway. SMILE ecosystem will target EU land borders which will be the beneficiaries of the proposed solutions. In fact, the proposed technology and business framework developed in SMILE will be validated through pan-European demonstrations in 3 BCPs. The operational properties of the technologies and overall solution will be validated and evaluated against cost, performance, effectiveness and usability indicators. Use cases will be supported by different architectural designs, which will be classified according to the operation mode. BCPs participating in the project's pilots will deploy and evaluate the solution at business as usual and emergency situations across various status operations. SMILE aims to (1) minimise the exposure of BCPs to security risks and threats, and (2) help them successfully respond to security incidents, while relieving them from all unnecessary and costly efforts of identifying, acquiring and using the appropriate technology. To this CNBP, HBP & RBP BCP partners will deploy and validate the proposed secure & reliable ecosystem in two use cases (Romania Bulgaria), in which the adaptation of SMILE framework to focused applications will be performed.

Reported results **Travellers' registration**: SMILE introduces the concept of pre-registration for land border crossings. Using the pre-registration SMILE application, a traveller (or a group of travellers) will be able to pre-input their information in the SMILE system, such as passport information, biometrics, border crossing point and time of arrival, vehicle information, VISA information (if applicable). This way, the information checking is performed offline, while the traveller(s) are travelling to the border crossing point, and when they arrive the checks have been performed, thus allowing them (provided that no alarms were raised) to cross the border with a very fast identity checking.

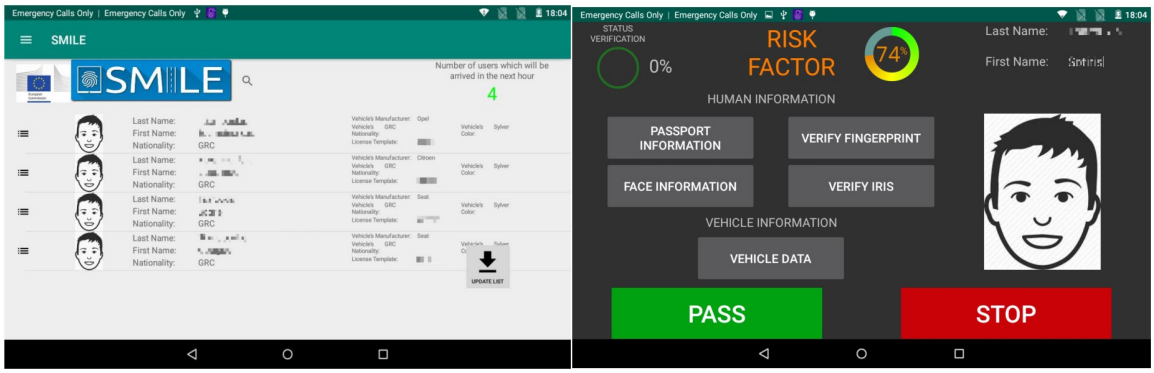
Multimodal & Multifactor biometrics: Multimodal biometrics will be used in order to enhance the recognition performance and avoid cases where a biometric could fail (e.g. the traveller's finger

is injured thus failing the fingerprint recognition). By relying on multiple biometric traits, the false rejection and the false acceptance rates will be significantly lowered.

Soft Biometrics for security enhancement, recognition boosting and group formation: Soft-biometrics will be introduced for enhancing the performance of hard biometric modules as well as to assist in group identification. Using soft-biometrics, SMILE system will be able to recognise groups of travellers and identify them faster and more reliable. The soft-biometrics traits that are currently being used are facial hair (moustache, beard), eyes colour, skin colour (ethnicity), gender, age group (child, adult, senior) and eyewear.

Security Mechanisms in SMILE infrastructure: Security is a very important aspect of SMILE project. In order to ensure data integrity and privacy, a set of security mechanisms will be used. Besides the common security elements, like encryption, vpn, user authentication, a secure element using a Photonic Physical Unclonable Function (pPuf) is introduced. The pPuf element, will be connected to the dedicated SMILE smart gateway, and provide un-replicable encryption keys between the various SMILE devices.

Figure 30. SMILE Border Control Posts (BCP) officers' application.



Source: Cordis

7.2.5. eBORDER

872878

Secure and Wireless Multimodal Biometric Scanning Device for Passenger Verification Targeting Land and Sea Border Control

INSTITUTO DE TELECOMUNICACOES (PT)

€1,025,800

01/01/2020 – 31/12/2024

https://doi.org/10.3030/872878

http://h2020-eborder.eu/

Programme
HORIZON 2020

Topic title
MSCA-RISE-2019 – Research and Innovation Staff Exchange

Biometric verification, facial recognition, fingerprint recognition, data fusion, wireless security

Objective

Every year a large number of visitors enter the European Union (EU) by cars, buses, trains or ships through land-border crossing points or sea ports bringing significant economic benefits; in particular, 9% of Europe's GDP comes from tourism. This trend is set to continue, as Europe further considers visa liberalization agreements in a bid to sustain their competitive stance in the global tourism market. This will create major implications on the security and border management in the EU, as well as diminishing the Quality of Experience for passenger travel. Existing directives for controlling border crossing is to conduct checks using biometric data in conjunction with VIS (Visa Information Systems), as specified in the Schengen Border Code.

However, current biometric systems rely on one mode of biometric data, that can at times can lead to false identification, in particular under uncontrolled conditions during live sample acquisition. Moreover, the current crop of scanning devices are limited in coverage and typically have fixed links, thus confining the range of border control applications and compromising the comfort of passengers.

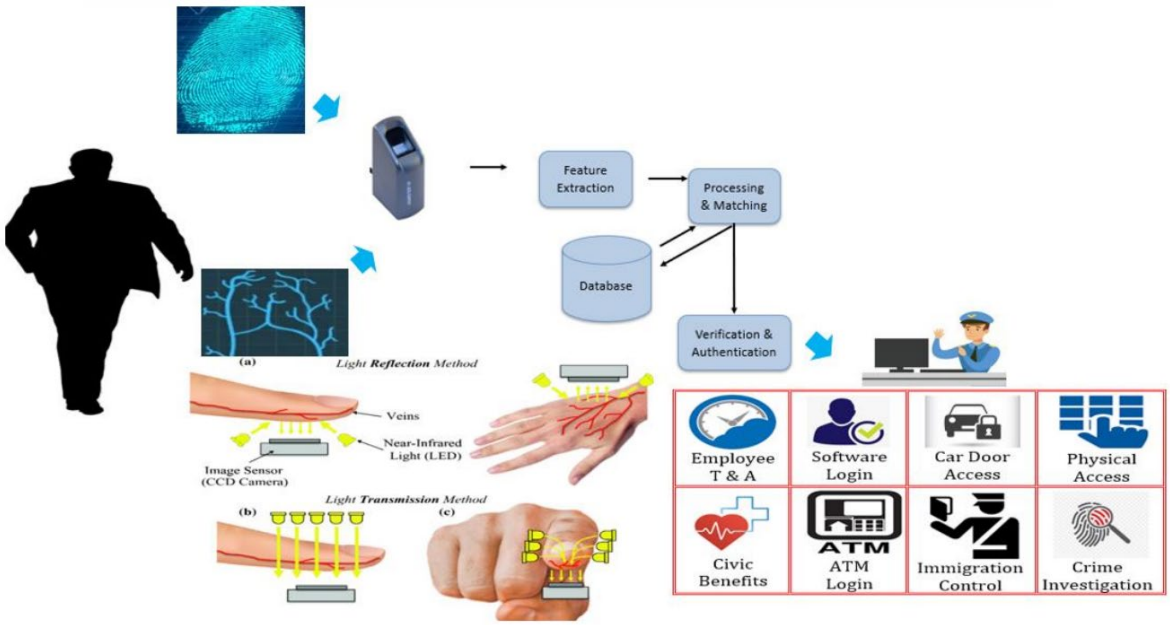
We aim to go beyond and propose an innovative passenger-centric biometric system that authenticates "on the fly". We combine the scientific field of wireless security, biometric recognition and data fusion to develop a secure and wireless multimodal biometric scanning device for passenger verification targeting land and sea border control applications. The research approach is the basis for a RISE training action that we refer to as eBORDER, that is foreseen to enhance both the career potential of staff members and innovation capacity at the institutional level.

In this context, this action brings together key players from industry and academia with complementary expertise on security, biometrics, data fusion and practical experimentation to train staff by research through intersectorial and international staff exchanges.

Selected publications

Kyeremeh et al. (2022) explores the transition from traditional password authentication to biometric systems, emphasizing security and recognition accuracy. Biometric identification utilizes an individual's physiological and behavioural traits for security and identification, with fingerprint biometrics historically being common. However, due to vulnerabilities such as spoofing, finger vein recognition (FVR) has emerged as a potentially safer alternative. FVR analyses the patterns of a person's finger veins for identity verification. The article reviews various aspects of both fingerprint and finger vein recognition systems, including image acquisition, pre-processing, feature extraction, and matching, highlighting their benefits and drawbacks. For developing an automated system that integrates both fingerprint and finger vein identification, extracting fingerprint minutiae is crucial. Figure 31 presents the integration of finger vein detection mode of operation in the eBORDER system. A very important aspect for every project regarding biometrics technologies is the cost-effectiveness relationship. According to Kyeremeh et al. (2021), iris recognition is the most accurate approach, but also the costliest one.

Figure 31. eBORDER proposed system.



Source: Kyeremeh et al. (2022)

7.2.6. FLEXI-cross

101073879

Flexible and Improved Border-Crossing Experience for Passengers and Authorities

ENGINEERING - INGEGNERIA INFORMATICA SPA (IT)

€3,974,400

01/09/2022 – 31/08/2025

https://doi.org/10.3030/101073879

https://flexicross-project.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-03 - Improved border checks for travel facilitation across external borders and improved experiences for both passengers and border authorities' staff

Secure border-crossing, Anti-trafficking protection, Anti-smuggling protection, Ad-hoc deployment of border-check point, Biometrics, safe and private data management

https://www.youtube.com/@flexi-cross3480

Objective

The FLEXI-cross project aims to increased security and reliability of EU border checks for people and goods, while enhancing and improving the border management capabilities, through the development, deployment and validation of a toolkit of innovative border-checking solutions, in real operational environments, addressing road, rail and port borders. The resulting flexibility and dynamicity of border check planning will offer novel capabilities such as dynamic deployment of checkpoints and support via mobile applications for border personnel, while guaranteeing high level of security, privacy of personal data and protection of people's fundamental rights. Through the FLEXI-cross solutions the following functionalities will be delivered: i) anti-trafficking and anti-smuggling protection via predictive risk assessment of vehicle and people, ii) enhanced border security through portable biometric based checks, iii) secure person verification through real-time multi-source cross-referencing, iv) flexible, fast and cost-effective deployment of ad-hoc Border Check Points (BCP), v) secure, private and traceable sensitive / personal data exchange based on blockchain technology and vi) increased safety and improved experience for border-personnel based on advanced Human Machine Interfaces and enhanced situational awareness via Augmented Reality (AR). To achieve these ambitious goals, the FLEXI-cross project engages key European border authorities and end-users (police, port/rail operators, etc.) and brings them together with leading SW and technology developers, research organizations and innovative SMEs, to co-design the requirements and the solutions and to validate the project outcomes. This collaboration will drive the deployment of three state-of-the-art border-checking trial facilities, to enable real life trials in diverse operational environments, namely, vehicle-based road border crossing, rail-based border-crossing and port embarkation/disembarkation.

Reported results

The FLEXI-cross solutions will deliver efficient, secure, safe and flexible border-checking operation which will significantly improve the traveller's experience by reducing the waiting and processing time of the border-checking process via: (i) traveller pre-registration capability, (ii) targeted in-depth searches instead of randomized ones, (iii) increased detection rate and detection speed of risky

incidents, (iv) predictive risk assessment of travellers and vehicles and (v) flexible and optimized checking process based on real-time data and with optimized use of human and other resources. Moreover, the border crossing experience for both travellers and border personnel will be significantly improved thanks to the following features delivered by FLEXI-cross: (vi) traveller interaction and information provision via the FLEXI-Cross portal (e.g. HMI via smartphone) and (vii) enhanced awareness for the border personnel via AR enabled solutions.

The results delivered during the first period are:

- the first drop of the FLEXI-cross toolkit (more detail will come later when the modules will be dropped in their final version)
- the almost final version of the data model for the interoperability among different authorities
- the first version of the GUI for the Visuali Analytics Dashboard
- the first version of the Data Export
- the use cases scenarios identified with steps for the validation and the related association of KPI and metrics that will be measured during the trials

In the second reporting period more information will be presented to explain the final scientific achievement of the technical modules.

7.2.7. ODYSSEUS

101073910

Unobtrusive Technologies for Secure and Seamless Border Crossing for Travel Facilitation

SOFTWARE IMAGINATION & VISION SRL (RO)

€3,457,062

01/01/2023 – 31/12/2025

https://doi.org/10.3030/101073910

https://odysseusproject.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2021-BM-01-03 – Improved border checks for travel facilitation across external borders and improved experiences for both passengers and border authorities' staff

land and sea borders, secure identity verification, unobtrusive vehicle, luggage, cargo checks, X-Ray scanners mounted in UAVs, identity verification, border crossing with the help of smartphones

https://www.youtube.com/@odysseusproject

Objective

The World Economic Forum suggests that transnational travel will increase by 50% over the next ten years, bringing the number of arrivals to 1.8 bn by 2030 before the COVID-19 pandemic. Given the ever-increasing travel both on land and sea, delays and queues are often observed at cases where people are crossing border. These delays create huge frustration for the citizens and at the same time put extra burden to the border authorities' staff. In Europe, airports have managed to improve to a certain extend the border crossing through the facial recognition booths for European citizens that own an RFID enabled passport, however there are still huge queues observed. For non-European citizens, the situation is even worse. The border control for land and sea are placing all the effort to the border authorities staff requiring from them to check the passports of the travellers as well as to control the luggage and baggage. Reasonably, even without huge number of travellers, large delays and queues are observed.

ODYSSEUS will enable citizens to seamlessly cross the border without stopping, while reducing the workload and improving productivity for Border Guard Authorities by providing highly detailed risk analysis. ODYSSEUS will leverage the power of digital technology offering to citizens the appropriate tools to cross land and sea borders in a secure and seamless manner without stopping while equipping the border authorities with novel tools for secure identity verification and unobtrusive vehicle/luggage/cargo checks, eliminating long delays at border areas. The proposed technologies will allow citizens to cross borders without any intervention by just leveraging their smartphones combined with strong and continuous identity verification. Remote vehicle/luggage scanning through X-Rays and UAVs, will allow authorities to remotely perform the appropriate controls.

7.2.8. CarMen

CarMen 101168325

YNCREA MEDITERRANEE (FR)

€2,998,348

01/09/2024 – 31/08/2027

https://doi.org/10.3030/101168325

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-BM-01-03 – Beyond the state-of-the-art "biometrics on the move" for border checks

Objective
Challenges related to "on the move" biometrics are 1/ lower quality live biometric data, and 2/ no time to read the ePassport. Also, fully automatic biometric border control solutions, even with a stop allowed, are currently deployed only for pedestrians in controlled environments. Carmen innovates biometric solutions for non-stop border control, suitable for pedestrians and vehicles, in uncontrolled environmental conditions.

Travellers' authentication is achieved in two steps: 1) the biometric data (face, iris, periocular) of travellers is securely stored in their smartphones, thanks to a DTC (Digital Traveller Credential); 2) the biometric data is securely transferred from the DTC to the border authority infrastructure, and compared to the live biometric data collected as the traveller crosses the border control point.

To address a variety of environmental conditions, Carmen uses both NIR and RGB live biometric images and compares them to the reference images that have been acquired with one type of lighting only. To make biometrics more robust, Carmen has a multimodal approach on face iris and periocular regions. In fraud detection, Carmen detects presentation attacks on moving travellers and detects anomalous traveller behaviour. To address small and large border crossing points, Carmen enables the use of fix and body-worn cameras. To address travellers in cars or lorries, Carmen detects face images in slowly moving vehicles, through the windows. Travellers in coaches are controlled by border guards walking through the coach, using dedicated portable equipment as they walk.

Carmen addresses the robustness of DTC via data injection attacks. Carmen complies with the existing legal and ethical standards, and privacy is a central concern. Carmen solutions will be demonstrated in operational conditions over the UK – France border, as French and UK border authorities are part of Carmen consortium, using the infrastructure proposed by the partner Brittany Ferries.

Reported results

Carmen innovates biometric solutions for non-stop border control, suitable for pedestrians and vehicles, in uncontrolled environmental conditions. Travellers' authentication is achieved in two steps: 1) the biometric data (face, iris, periocular) of travellers is securely stored in their smartphones, thanks to a DTC (Digital Traveller Credential); 2) the biometric data is securely transferred from the DTC to the border authority infrastructure and compared to the live biometric data collected as the traveller crosses the border control point.

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7.2.9. PopEye

101168317

robust Privacy-preserving biOmetric technologies for Passengers' identification and verification at EU external borders maximising the accuracY, reliability and throughput of the rEcognition

AIT AUSTRIAN INSTITUTE OF TECHNOLOGY GMBH (AT)

€3,212,590

01/10/2024 – 30/09/2027

https://doi.org/10.3030/101168317

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-BM-01-03 – Beyond the state-of-the-art "biometrics on the move" for border checks

Objective
PopEye unique value proposition focuses on a) strengthening the security at the external EU borders through fusing robust biometric technologies for verification and identification, b) maximising the travellers' experience when crossing the external EU borders through unobtrusive on-the-move biometric technologies, c) improving the productivity of Border Authorities and Law Enforcement Agencies in performing identification and verification of people on the move and without stopping them, at up to 200m distance, when multiple travellers cross borders, on foot or inside the same vehicle, in different nature e.g. open-air conditions, night, time, time constraints, space constraints, etc.

PopEye will introduce novel and robust biometric technologies in the five areas identified by the aforementioned Frontex report combined with gait recognition, in order to maximise the impact and the return on investment in short term. PopEye through the proposed technologies will support both EU and Third Country Nationals (TCN) citizens, will be compliant with existing and upcoming regulations such as GDPR and AI Act, will comply with the 2019 Ethics Guidelines on Trustworthy AI, will empower the EU values and fundamental rights, including on data protection and avoid bias and discrimination.

7.3. Document authentication and fraud detection

7.3.1. Smart-Trust

778571

Smart Trust: Secure Mobile ID for Trusted Smart Borders

VISION BOX - SOLUCOES DE VISAO PORCOMPUTADOR SA (PT)

€2,093,700

01/01/2018 – 31/03/2020

https://doi.org/10.3030/778571

Programme
HORIZON 2020

Topic title
SMEInst-13-2016-2017 – Engaging SMEs in security research and development

Objective

Smart-Trust introduces a new technological enabler for Mobile ID which drastically increases the reliability and trust levels of identity verification at European borders, thus increasing the security of member states. The platform is compatible with a seamless self-service experience based on biometrics on the move which provides freedom of movements to citizens while ensuring their privacy safeguarded according to Privacy by Design principles.

Smart-Trust will simultaneously address the business needs of governments, airports, airlines and border police in major pilots in the international airports of Lisbon and Schiphol. After establishing requirements sought by the various stakeholders and integrating the Smart-Trust platform into the airport and border control infrastructure, we will define KPIs for assessing the business cases and design experiments to measure them.

Smart-Trust will be implemented by delivering an agile, highly configurable, modular and open architecture platform with the following robust, versatile and scalable core components:

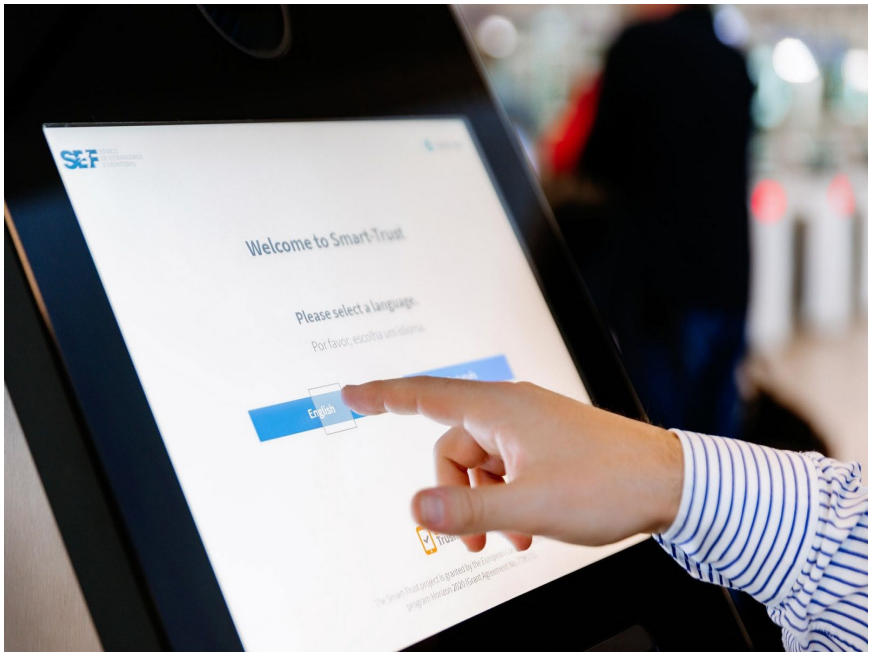
- Mobile ID services and app, mobile identity enrolment and verification services
- TrustChain, a blockchain-based infra-structure to assert citizen transactions in a distributed and self-regulatory fashion
- Workflow Orchestration, a workflow management component which caters for stakeholder-driven business rules
- Analytics and Risk-Assessment, an industrial-strength real time analytics platform able to deliver professional customized monitoring dashboards as well as detecting passenger trends and anomalies.

These components will be developed as part of Vision-Box's overarching Services Platform and hence compatible with standard Vision-Box services such as Automated Border Control, including Advance Passenger Information Systems (APIS)and Entry-Exit System (EES), and Happy Flow, the first 100% self-service passenger processing system allowing passengers to go from curb to boarding by simple presenting their face.

Reported results

Research and Development. During the 27 months of Smart-Trust project has developed an agile, modular, and highly configurable open architecture platform with the following robust, versatile, and scalable core components: Mobile ID services and app: mobile identity enrolment and verification services. TrustChain: a blockchain-based infrastructure to assert citizen transactions in a distributed and self-regulatory fashion. Workflow Orchestration: a management component catering to stakeholder-driven business rules. Analytics and risk-assessment algorithms: deliver professional customized monitoring dashboards as well as detecting passenger trends and anomalies in real-time. These components were developed to be compatible with other established Vision-Box services including Automated Border Control, Advance Passenger Information Systems (APIS) and the forthcoming Schengen Area required Entry-Exit System (EES).

Figure 32. Smart-Trust portal for secure mobile ID for trusted smart borders.



Source: Cordis (© Smart-Trust)

In the first year, the mobile ID concept has been tested in internal pilots with hundreds of people, yielding thousands of individual tests, and the seamless flow concept (excluding border) has been tested in pilots in one major international airport. Face recognition was used as a test in a scenario for positive identification in a normal airport environment, with cooperative non-habituated users. The evaluation was conducted in accordance with the "Best Practices in Testing and Reporting Performance of Biometric Devices" and performed 6,275 transactions over a 40-day period. The pilot results consider: Failure to Enrol and Failure to Acquire Rates. Trade-off between matching errors and between decision errors over a range of decision criteria. Throughput rates of users in the live application and of the matching algorithm in offline processing.

7.3.2. SOTAMD

State of the art of Morphing Detection

842250

MINISTERIE VAN BINNENLANDSE ZAKEN EN KONINKRIJKSRELATIES (NL)

€1,000,000

01/02/2019 – 31/01/2020

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/31077833/842250/ISFB

Programme
Internal Security Fund Borders and Visa

Topic title
ISFB-2018-AG-IBA-MORP –

Morphing; morphed face image; algorithms; border control gate; identity documents; application process; biometric

Objective

With the widespread deployment of biometric recognition systems, in particular the EU Smart Borders initiative, the interest in attacking these systems is increasing. In the recent past and as one of the main results of the EU funded research project FIDELITY, the vulnerabilities of biometric systems and automated border control applications to so-called morphed face image attacks have been unveiled to some governments within the EU. In such attacks, biometric samples of multiple subjects are merged, in order to allow an unlawful but successful verification of all contributing subjects against the created artificial identity. This is a rather young area of research and only a few European research institutions have developed detection mechanisms for such attacks. These labs are co-applicants of this project.

The objective of the SOTAMD project is to identify the state-of-the-art of morphed face image detection mechanisms by collecting in a distributed effort a database of morphed face images, for which image quality according to ICAO and EU Regulation 2252/2004 is ensured. Furthermore the project will evaluate the existing morphed face image detection mechanisms based on a common established benchmark protocol and morphed face image detection metrics. The project will focus on the scenario of differential morphing detection, where a bona fide facial image (captured in a trusted environment) is compared against the target of investigation (e.g. the printed image in the passport application process) in a signal to signal analysis mode.

The beneficiaries of the project results will be all European members states with their respective agencies that are in charge of passport application processes (facial enrolment) and/or of the border control processes (facial verification). Depending on the level of accuracy of the evaluated differential detection mechanisms, a short term deployment of identified mechanism can be recommended to the member states.

7.3.3. D4FLY

833704
Detecting Document frauD and iDentity on the fly
VERIDOS GMBH (DE)
€6,984,728 01/09/2019 – 31/08/2022 https://doi.org/10.3030/833704 https://d4fly.eu
Programme HORIZON 2020 Topic title SU-BES02-2018-2019-2020 – Technologies to enhance border and external security
on-the-move lightfield document-verification-system
https://youtu.be/IhdwavK3rfo

Objective

The D4FLY project will augment the current capabilities and capacities of border authorities in countering emerging threats in document and identity verification (e.g., forged documents, impostor fraud, morphed faces) at manual and highly automated border control points and in the issuance process of genuine documents. The confluence of D4FLY set of tools and systems will improve the quality of verification and reduce major time sinks in the processes thus enabling real on-the-move border crossing experience for travelers.

Novel sensor hardware based on advanced lightfield cameras and novel algorithms developed in the project will enhance verification accuracy and robustness via the combined usage of 2D+thermal face, 3D face, iris and somatotype biometrics. Analytical means to identify known criminals based on somatotype and 3D face data generated from mugshots and observation data will be developed. Various operational needs of end-users with different threat landscapes constitute the backbone of D4FLY development efforts. D4FLY will create a resilient document verification system that can verify a multitude of physical and electronic security features (e.g. Kinegrams®, MLIs, CLIs), detect complex forms of electronic fraud and advanced morphing, and identify fraud in breeder documents. The potential benefit of blockchain technology in identity verification will also be investigated.

The D4FLY solution will consist of a border control kiosk geared with enhanced enrolment, verification and detection capabilities; smartphones applications for improved performance and verification capabilities; and a non-stop on-the-move system for biometric verification. The innovation will be validated against European societal values, fundamental rights, privacy, data protection and applicable legislation. Four different border control points and one document fraud expertise center will form the project's testing and demonstration ground.

Reported results

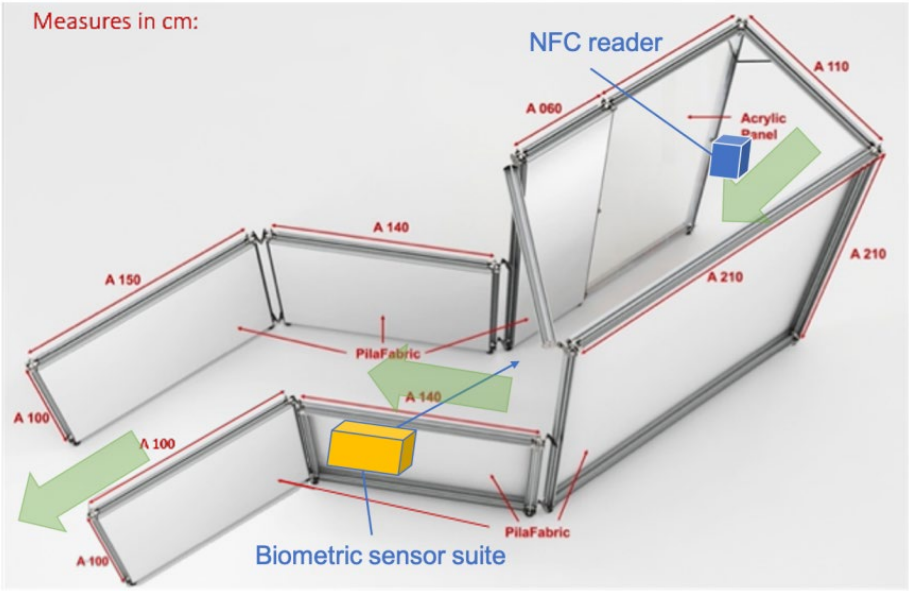
The EU-funded D4FLY project developed tools for automated analysis of breeder documents, helping experts to detect manipulations and prevent issuance of documents containing false identities, which could later on be used to illegally cross borders. D4FLY investigated new technologies to enhance the document and identity verification process, laying the foundations for smoother border crossing

experiences for travellers. D4FLY helped improve the quality of identity verification in border crossings. Project partners collaborated with end users, gathering useful feedback on social aspects to validate the proposed technologies and better meet end user needs.

Selected publications

Danelakis & Theoharis (2020) explores the use of somatotype, derived from full-body images, as a new soft biometric feature for person recognition at a distance and while on-the-move. Somatotype is effective for identity recognition even with changes in apparel and can be easily captured with standard 2D cameras. The study evaluates somatotype in two scenarios: identification and verification, utilizing deep learning networks and publicly available datasets. Results demonstrate somatotype's potential as a valuable biometric trait for remote recognition, particularly for non-collaborative individuals, and as part of a broader biometric fusion system. The paper contrasts this biometric approach with Re-Identification (Re-ID – detecting and tracking a person and then using features such as appearance, body shape, and clothing to match their identity in different frames) techniques, noting that somatotype-based methods can outperform Re-ID in verification and are competitive in identification when using a fixed gallery. This highlights somatotype's promise for applications in high-security settings like border control. Chen et al. (2021) details more about the D4FLY project and presents the design of the on-the-move biometric verification corridor (see Figure 33).

Figure 33. D4FLY design of the on-the-move biometric verification corridor for border control.



Source: Chen et al. (2021)

7.3.4. iMARS

image Manipulation Attack Resolving Solutions	883356
IDEMIA IDENTITY & SECURITY FRANCE (FR)	
€6,988,521 01/09/2020 – 31/08/2024 https://doi.org/10.3030/883356 https://imars-project.eu	Programme HORIZON 2020 Topic title SU-BES02-2018-2019-2020 – Technologies to enhance border and external security
Image manipulation, morphing, ID document fraud, detection techniques, face sample quality	

Objective

Though difficult to quantify, the variety of possible ID document frauds at borders is a reality that threatens politicians holding responsibility for borders as well as EU citizens. Based on the needs expressed by practitioners, the iMARS consortium has shaped a project that will:1/ improve the operational capacity of passport application and border control operators by providing both - short-term ad hoc solutions ensuring reliable passport application procedures - and mid-term solutions with no-reference and differential analysis solutions that can in particular detect manipulated and morphed passport images as well as document fraud. iMARS will provide:

- Image morphing and manipulation attack detection solutions to assess ID documents validity against document fraud, during enrolment and renewal steps, as well as at the border crossing stations
- Document verification and fraud detection solutions to support border guards in the verification process by providing mobile tools that can check document's integrity.

To reach the targeted TRL6, these technologies will be validated in laboratory conditions using operational data collected from six border control sites (with border to countries outside EU).2/ ensure the uptake of its outcomes after the project's end.

iMARS will:

- promote usability/ergonomic aspects by providing training, guidelines and best practices to border guards and passport application officers
- contribute to the ongoing standardisation efforts in the field of Presentation Attack Detection and face image quality
- provide open access benchmarks (2 datasets with multiple enrolment morphed face images and border gate probe images) on a specific testing platform serving follow-on research activities
- ensure that the technologies developed are accepted by citizens and respect privacy and legal EU regulations.

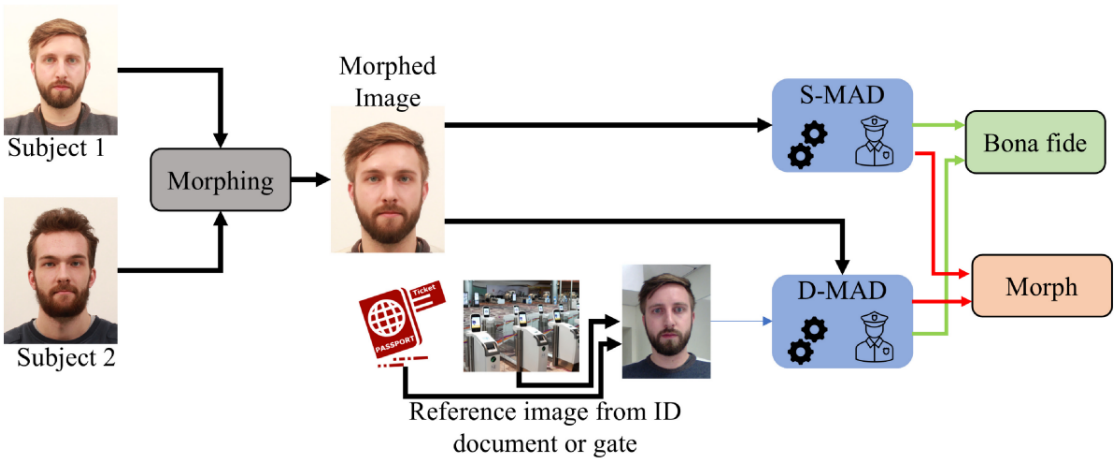
Reported results

The project focused on two types of morphing attack detection (MAD): single image morphing attack detection (S-MAD); and differential morphing attack detection (D-MAD). In S-MAD, the technical solution consists of one input (a potentially morphed face image of an ID document or to be printed on an ID document), and one output (a score representing the likelihood that the input image is morphed). In D-MAD, the person holds two inputs: a potentially morphed face image of an ID document and another face image corresponding to the face of the person holding the ID document. The system then delivers the score of the likelihood the first input is morphed. Through iMARS, the team created many algorithms to support these detection mechanisms. They gathered data to learn and test the algorithms, which were then iterated until effective. The project also created a MAD training programme for border guards and ID document experts, to support the verification of the algorithms. The project also raised awareness to European stakeholders of the risk of morphing. At the end of the project, the work of iMARS continued under two other Horizon-funded projects: EINSTEIN and SafeTravellers.

Selected publications

Godage et al. (2023) examines the ability of human observers to detect facial morphing attacks (see Figure 34), where images of two or more faces are digitally combined into one. Such morphed images challenge both automated Face-Recognition Systems (FRS) and human observers due to their realistic quality. While prior research focused on FRS vulnerabilities, this study assesses human observer proficiency, especially those who regularly verify ID documents. A new benchmark dataset with 400 morphed images from 48 subjects and an additional set of 180 images was created. Observations were conducted with 469 governmental employees and 103 control group members. Results showed that human observers are less accurate at detecting morphed images compared to four automated detection algorithms, highlighting the need for enhanced training programs to better equip observers in preventing security breaches with altered images.

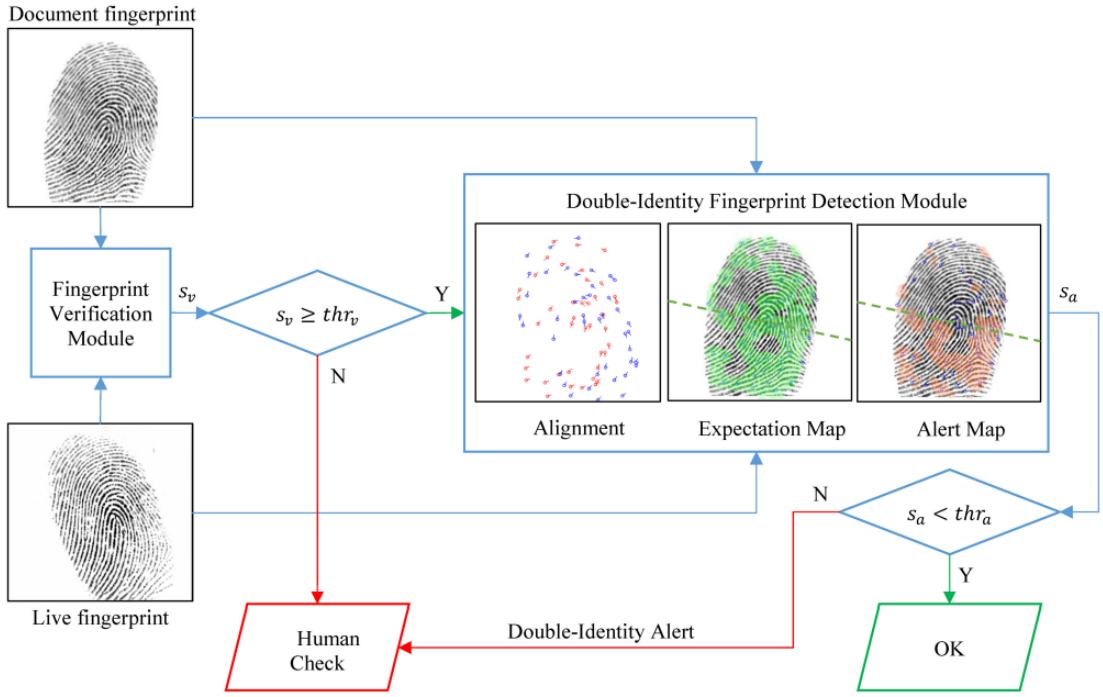
Figure 34. Pipeline of Single-Image Morphing Attack Detection (S-MAD) and Differential MAD (D-MAD).



Source: Godage et al. (2023)

Ferrara et al. (2023) addresses the threat posed by double-identity biometrics, where features of two individuals are combined into a single template to deceive biometric systems. Such well-synthesized samples can trick verification systems into wrongly accepting both identities. The study introduces one of the first methods (see Figure 32) to counteract double-identity fingerprint attacks. The approach involves inspecting regions where two aligned fingerprints overlap and checking if minutiae can't be consistently paired. If these regions are of high quality, thus reducing false detections, the alarm score is increased. Experiments conducted on two fingerprint databases, using different techniques for creating double-identity fingerprints, validate the effectiveness of this method.

Figure 35. Proposed fingerprint verification procedure (including the double-identity fingerprint detection module) performed at ABC gates.



Source: Ferrara et al. (2023)

7.3.5. EINSTEIN

101121280

Interoperable applications suite to enhance European identity and document Security and fraud detection

ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS (EL)

€5,379,425

01/01/2024 – 31/12/2026

https://doi.org/10.3030/101121280

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-02 – Enhanced security of, and combating the frauds on, identity management and identity and travel documents

biometrics, MAD, PAD, fast track, interoperability

Objective

Combatting frauds on identity and travel documents is a key mission of law enforcement agencies and border guards. Industries have been working on new means to combat frauds on identity, and public authorities are using numerous technologies to accomplish their mission, but permanent innovation is required to fight highly skilled defrauders. A number of European innovation projects have tested new ways to combat fraud but few of them reached an operational level.

EINSTEIN objectives is to enhance significantly existing public authorities' means through innovation, building on technologies proven in the labs but not mature for an operational usage yet. EINSTEIN will deliver six applications essential to fight identity frauds: 1) online ID issuance using a secure cloud-based server for real-time biometric quality checks and fraud detection, 2) mobile document and identity checks using commercially available smartphones, 3) document authentication module to detect fraudulent documents, 4) pre-registration for land-border crossings including biometrics and DTC, 5) EES kiosk with advanced fraud detection using video surveillance, 6) fast track for enrolled travelers using on-the-move face and iris.

To ensure TRL7 at a minimum, practitioners will run six different pilot use cases in their own environment. A key objective is also to ensure interoperability and flexibility of all the components developed in the project so that they can be reused in different contexts, possibly by different providers. Design of open, well-defined, standardized interfaces will allow achieving this objective.

Privacy being an essential concern of all European citizens and governments, EINSTEIN will take into account privacy-by-design principles, developing flexible components to ease their customization in order to meet not only the European legislation on data protection but also national legislations on this matter, which vary significantly from one country to the other.

7.3.6. SafeTravellers

101121269

SafeTravellers: Secure and Frictionless Identity for EU and Third Country National Citizens

INOV INSTITUTO DE ENGENHARIA DE SISTEMAS E COMPUTADORES INOVACAO (PT)

€6,081,079

01/01/2024 – 31/12/2026

https://doi.org/10.3030/101121269

https://safetravellers-project.eu

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2022-BM-01-02 – Enhanced security of, and combating the frauds on, identity management and identity and travel documents

Securing identity and travel documents, Biometrics, Machine learning, Visual analytics, Fraud detection

Objective

Identity theft is rapidly expanding, causing substantial financial loss to millions of people all around the world. This invisible crime is also widespread across EU countries, where a growing number of citizens is targeted by sophisticated fraudulent attacks each year, both offline and online. 56% of Europeans have experienced at least one type of fraud in the last two years. European security officials speak of an “epidemic” created by a spike in demand from asylum-seekers and from terrorists carrying counterfeit documents to enter the EU. Security documents are increasingly being counterfeited or tampered with by criminals to facilitate transnational crime. The continued vulnerability of different types of identity and travel documents makes it extremely difficult to combat this problem.

SafeTravellers value proposition aims at a) strengthening the security at the borders, b) improving the productivity of the Border Authorities and LEAs by providing them with the appropriate tools to combat identity fraud at the hardware, identity and travel document, and biometrics level, while c) offering a frictionless border crossing experience for EU/TCN citizens as they will not have to stop at the border checkpoints.

SafeTravellers is both proposing a new way of citizen identification based on multiple biometrics instead of the problematic identity document, as well as an enhancement of the current way of identity verification at the borders through a set of tools that will detect attacks at the biometric hardware, identity and travel document fraud and attempts to falsify biometrics.

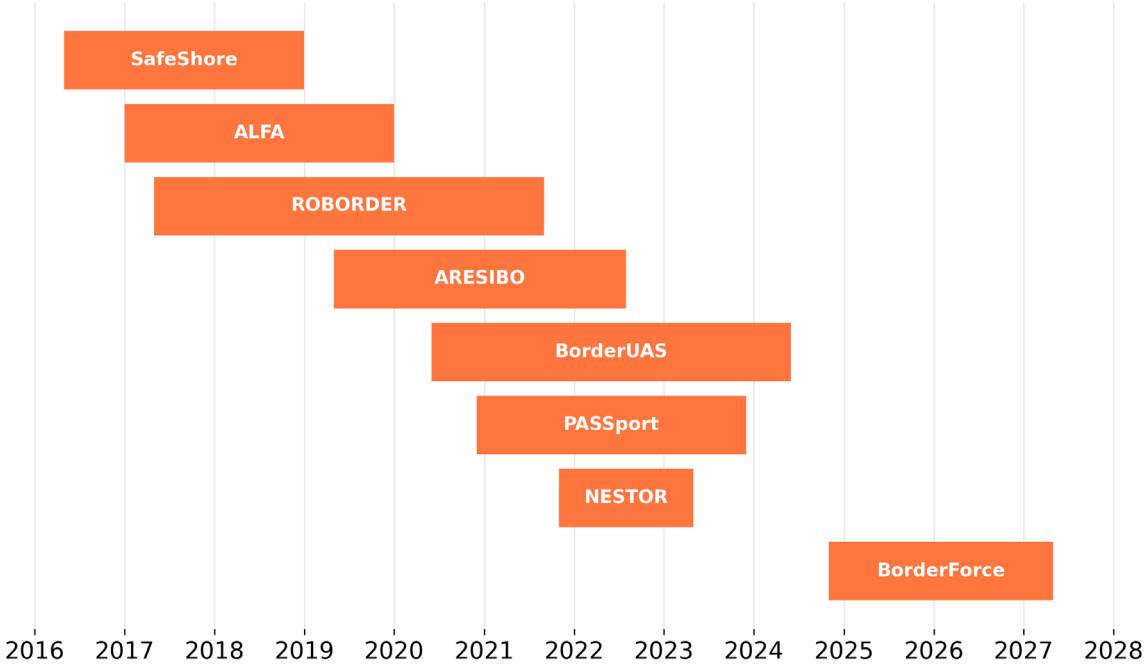
The proposed solution is GDPR compliant and introduces various privacy-preserving mechanisms to safeguard the citizens' rights. Through the distributed European Multi-Biometric Data Space offered by SafeTravellers, each Member State will keep in its jurisdiction the personal data of its country nationals while allowing cross-border identity checks without transferring or revealing any biometric data.

8. Unmanned vehicles and autonomous systems

8.1. Introduction

The evolving complexity of cross-border threats has spurred a wave of EU projects aimed at enhancing situational awareness and responsiveness through the **use of unmanned vehicles**. The term *unmanned vehicles* (UxVs) comprises unmanned aerial vehicles (UAVs), and/or unmanned surface vehicles (USVs), and/or unmanned underwater vehicles (UUVs). In the context of AI for border management and customs controls, eight EU-funded projects are focussed on the use of unmanned vehicles (see timeline in Figure 35).

Figure 36. EU projects with AI for unmanned vehicles and autonomous systems.



Source: JRC

This chapter features eight EU projects exploring the use of unmanned vehicles (UxVs), i.e., unmanned aerial vehicles (UAVs), unmanned surface vehicles (USVs), and unmanned underwater vehicles (UUVs) platforms – for border and customs control.

- **Detection of small aerial threats**: *SafeShore* and *ALFA* developed LIDAR, radar, and video analytics to detect low-flying drones and small aircraft used in smuggling.
- **Multi-domain surveillance networks**: *ROBORDER*, *ARESIBO*, and *PASSPort* integrated aerial, surface, and underwater vehicles into coordinated fleets, coupled with data fusion and augmented reality to support operators.
- **Persistent, large-area coverage**: *BorderUAS*, *NESTOR*, and *BorderForce* introduced endurance platforms (lighter-than-air vehicles, tethered UAVs) and integrated sensors to generate a real-time pre-frontier intelligence picture.

The projects, their application area and their use of AI are summarised in Table 8.

Table 8. Summary of projects featuring unmanned vehicles and AI for border management.

Category	Project	Main Focus	Vehicle(s)	Example of AI
Detection of small aerial threats	SafeShore	Coastal surveillance, drug smuggling, irregular migration	UAV detection	Video analytics for drone–bird discrimination
	ALFA	Drug trafficking via small planes	UAV detection	ResNet-50 deep neural network
Multi-domain surveillance networks	ROBORDER	Multi-domain border surveillance	UAV, USVs, UUVs	Data fusion, swarm intelligence, deep CNNs, semantic fusion
	ARESIBO	Augmented reality situational awareness	UAV, USVs, UUVs	Deep learning for sensor fusion, AR for C2
	PASSPort	Port safety & security	UAV + UUVs (port)	GNSS-based autonomy + analytics
Persistent, large-area coverage	BorderUAS	Persistent surveillance & S&R	Lighter-than-air UAV	YOLO-based vision, acoustic AI, ontology
	NESTOR	Pre-frontier intelligence picture	UAV (including tethered and swarms), UUVs	Deep learning for multiple object tracking
	BorderForce	Extendable border surveillance	UAV + satellite	OSINT analysis, autonomous monitoring sensors

Source: JRC

These projects demonstrate a shift from hardware-focused detection prototypes toward AI-driven autonomy, swarm intelligence, and operator support systems, reflecting the growing complexity of cross-border threats.

8.2. Featured projects

8.2.1. SafeShore

700643

System for detection of Threat Agents in Maritime Border Environment

ECOLE ROYALE MILITAIRE - KONINKLIJKE MILITAIRE SCHOOL (BE)

€ 5,133,583

1/5/2016 – 31/12/2018

https://cordis.europa.eu/project/id/700643

https://safeshore.eu/

Programme
HORIZON 2020

Topic title
BES-02-2015 – Maritime Border Security topic 2: affordable and easily deployable technologies for EU coastal border surveillance with reduced impact on the environment

control systems; optical sensors; aircraft; radar; drones

https://youtu.be/3gQwhHrsYLk

Objective

The main objective of the SafeShore project is to cover existing gaps in coastal border surveillance, increasing internal security by preventing cross-border crime such trafficking in human beings and the smuggling of drugs. It is designed to be integrated with existing systems and create a continuous detection line along the border. One of the threats to the maritime coast are small Remotely Piloted Aircraft Systems (RPAS) which can carry explosives or which can be used for smuggling drugs, boats and human intruders on the sea shore.

The SafeShore core solution for detecting small targets that are flying at low attitude is to use a 3D LIDAR that scans the sky and creates above the protected area a virtual dome shield. SafeShore will also integrate the 3D LIDAR with passive acoustic sensors, passive radio detection and video analytics. One of SafeShore specific objectives will be to cover 1500-m1800m along the coastal border. This will be done with 3 mobile platforms. Each one of those will cover itself with a dome-shaped virtual detection shield with a radius of about 250m to 300m. There will be approximately 50 meters overlapping between the platforms. The overlapping will create a continuous detection shield along the shore.

The SafeShore objective will be to demonstrate the detection capabilities in the missing detection gaps of other existing systems such as costal radars, thereby also ensuring the fusion of information and increasing the situational awareness and better implementation of the European Maritime Security Strategy based on the information exchange frameworks, EUROSUR and EUCISE 2020. All SafeShore objectives are measurable, realistic and achievable within the duration of the project, as the SafeShore project will build 3 prototypes that are going to assess the accomplishment of the objectives. Three end-user scripted and end-user validated field trials will be set up to validate these systems: in the North Sea, Black Sea and Mediterranean.

Reported results

The project produced three fully functional prototype systems – each containing multiple sensors for detection and the data fusion engine. The prototype detectors, which can be chained in succession to create a continuous detection line across the coast, have been tested in three locations: the North Sea (Belgium), the Black Sea (Romania), and the Mediterranean.

Project partners employed a wide range of threat agents to field-test the detection system, including drones, various small vessels, jet skis, kitesurfers, swimmers and windsurfers. Each test revealed the SafeShore detection system could, to a certain degree, detect these small-scale threat agents both in the air and on water. Installed for lengthy periods at beaches across Europe, the systems also proved their versatility in rough conditions.

Figure 37. SafeShore detector installed at the beach in Belgium during the North Sea trial.



Source: Cordis

Selected publications

SafeShore promoted an international workshop within the 14th edition of the IEEE International Conference on Advanced Video and Signal based Surveillance, held in Lecce, Italy, from 29 August to 1 September 2017. About 20 different international groups joined the "Drone-vs-Bird Detection Challenge", a competition launched by the SafeShore Consortium to award the most effective algorithm able to automatically discriminate between RPAS and birds under diverse conditions in real environments (Coluccia, 2017).

8.2.2. ALFA

700002

Advanced Low Flying Aircrafts Detection and Tracking

TECHNIKON FORSCHUNGS- UND PLANUNGSGESELLSCHAFT MBH (AT)

€ 4,613,831

1/1/2017 – 31/12/2019

https://doi.org/10.3030/700002

https://alfa-h2020.technikon.com/

Programme
HORIZON 2020

Topic title
BES-04-2015 – Maritime Border Security topic 4: Detection of low flying aircraft at near shore air space

https://www.youtube.com/watch?v=KI8H2B2mMyY

https://youtu.be/epki5PojrtY

Objective

The last decades witnessed the ever growing effectiveness of Europe-subsidized border protection projects like SIVE and SIVICC. As maritime smuggling from Morocco towards the European borders of Portugal and Spain was combatted more effectively, the criminal modus operandi changed drastically, approaching air routes with cheap and small planes. New drone technology opens opportunity to both manned and unmanned airborne drug transports. Launched from any location and moving at low altitude and speed to mask their presence with the present clutter environment, drones can autonomously reach any landing site under nearly all circumstances.

The ALFA system bridges this detection capability gap by drastically improving the situational awareness through the detection of LSS (Low, Small and Slow) manned and unmanned aircraft. ALFA is future-ready as technologies for drone detection will be a part of the system, which will use heterogeneous, easy-to-deploy mobile sensors based on several novel technologies. All sensor data, augmented by other existing sources of information, will be combined using evolved data fusion, providing accurate positional data for targets including eventual indication of the air vehicle type and reliable prediction of its landing site. This information will be communicated to the regional law enforcement units using a secure communication link and mobile device application, drastically improving the reaction time.

Final ALFA capabilities will be demonstrated in a realistic operational context using relevant targets and in close cooperation with two principal end users taking part in the consortium. With their active participation, the ALFA system will make a significant contribution to the development of EUROSUR (in particular, cooperating with SIVE and SIVICC) and be suitable for a range of other missions and scenarios such as homeland and event protection and the protection of critical infrastructure.

Selected publications

Baptista et al. (2020) proposed a static camera-based surveillance system for detecting, tracking, and classifying aerial targets like drones, birds, and aircraft. Moving objects are detected using background subtraction and tracked using a Kalman filter and the Jonker-Volgenant algorithm. Suspicious objects are zoomed in on and classified using a retrained ResNet-50 deep neural network. Real-world testing showed good tracking accuracy and high classification performance. In the frame of combating drug trafficking by detecting low-flying suspicious vehicles, Fernandes et al. (2019) describe a system that tracks and classifies aerial objects using a combination of classical computer vision and deep learning. A Kalman filter and Hungarian algorithm handle multi-object tracking, while a neural network classifies targets into aircraft, drones, birds, or clouds.

van den Broek et al. (2019) describes a study outlines a threat detection framework for border surveillance, especially targeting low-flying threats like drones or smugglers. It focuses on integrating uncertain and incomplete threat indicators using techniques like Bayesian Reasoning, Fuzzy Logic, and primarily Dempster-Shafer Theory. The system produces a probability and uncertainty score for each target to inform human operators.

Krueckemeier et al. (2019a) presents a passive detection and localization system for unmanned aircraft. It combines RF signal detection from emissions (like telemetry or phones) and reflections of ambient radio signals (like FM radio) to identify and locate targets, enabling stealthy and power-efficient surveillance. Krueckemeier et al. (2019b) tackles the technical challenge of synchronizing multiple USRP X310 SDRs for applications like direction-of-arrival (DoA) and passive radar. It discusses phase coherence and how even with a shared clock, phase ambiguities arise. The authors demonstrate a solution with preliminary results showing direction-finding capability and phase stability.

Figure 38. ALFA aimed to detect, identify and track aerial targets in real time.



Source: Cordis © Technikon

8.2.3. Roborder

740593

autonomous swarm of heterogeneous RObots for BORDER surveillance

ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS (EL)

€7,999,316

1/5/2017 – 31/8/2021

https://doi.org/10.3030/740593

https://roborder.eu/

Programme
HORIZON 2020

Topic title
SEC-20-BES-2016 – Border Security: autonomous systems and control systems

Objective

Border authorities and Law Enforcement Agencies (LEAs) across Europe face important challenges in how they patrol and protect the borders. Their work becomes more problematic considering the heterogeneity of threats, the wideness of the surveyed area, the adverse weather conditions and the wide range of terrains. Although there are several research tools and works targeting these areas independently for border surveillance, nowadays border authorities do not have access to an intelligent holistic solution providing all aforementioned functionalities.

Towards delivering such a solution, ROBORDER aims at developing and demonstrating a fully-functional autonomous border surveillance system with unmanned mobile robots including aerial, water surface, underwater and ground vehicles, capable of functioning both as standalone and in swarms, which will incorporate multimodal sensors as part of an interoperable network. The system will be equipped with adaptable sensing and robotic technologies that can operate in a wide range of operational and environmental settings. To provide a complete and detailed situational awareness picture that supports highly efficient operations, the network of sensors will include static networked sensors such as border surveillance radars, as well as mobile sensors customised and installed on board unmanned vehicles. To succeed implementing an operational solution, a number of supplementary technologies will also be applied that will enable the establishment of robust communication links between the command and control unit and the heterogeneous robots.

On top of this, detection capabilities for early identification of criminal activities and hazardous incidents will be developed. This information will be forwarded to the command and control unit that will enable the integration of large volumes of heterogeneous sensor data and the provision of a quick overview of the situation at a glance to the operators, supporting them in their decisions.

Selected publications

In terms of **situational awareness and surveillance**, Orfanidis et al. (2019) developed an integrated surveillance architecture that deploys heterogeneous unmanned vehicles (UxVs) equipped with visual, thermal, radar, and RF sensors to detect and characterize threats. Their system fuses data from these sources and applies semantic interpretation, enhancing situational awareness for operators in diverse environments. Orfanidis et al. (2021a) describes a computer vision-enabled robotic swarm system for border surveillance. They implemented autonomous navigation based on swarm intelligence and enriched the captured data with semantic layers, offering a responsive

solution against increasingly sophisticated cross-border crime. Maresca et al. (2019) demonstrated a photonics-based MIMO radar system capable of coherent multi-target tracking in outdoor environments. They showed how photonic technologies ensure long-term phase coherence and antenna remoting via optical fibers, providing a powerful and scalable tool for advanced surveillance applications.

Leveraging **AI and computer vision for maritime monitoring**, Orfanidis et al. (2018) trained a deep convolutional neural network (DCNN) to segment oil spills in satellite Synthetic Aperture Radar (SAR) images. Using Sentinel-1 data and EMSA records, they developed a fully automated detection system that accurately identifies oil-contaminated areas, demonstrating the DCNN's robustness under complex maritime conditions. Orfanidis et al. (2021b) modified the Single Shot Detector (SSD) object detection model to operate efficiently in low-resource environments. They proposed lighter SSD variants that preserved detection accuracy while significantly improving frame rates, making the approach suitable for real-time deployment in constrained robotic platforms.

In the field of **autonomous systems and swarm robotics**, Salavasidis et al. (2018) designed a trajectory planning algorithm for cooperative seafloor mapping using teams of Autonomous Underwater Vehicles (AUVs). They adapted the Parametrized Cognitive-based Adaptive Optimization (PCAO) method to address the challenges of unknown seafloor topographies. Their experiments revealed the algorithm's scalability, efficiency, and superior performance compared to gradient-based and one-step-ahead methods.

In terms of **data fusion and interoperability**, Riga et al. (2020) created EUCISE-OWL, an ontology-based formalization of the Common Information Sharing Environment (CISE) for maritime monitoring. They replaced XML schemas with a semantic model to support richer reasoning, better interoperability, and automated cross-border data exchange, addressing longstanding limitations in maritime crisis coordination. Papataxiarhis & Hadjiefthymiades (2018) implemented an event correlation algorithm using a variable-order Markov model to analyze real-time sensor data in maritime networks. They validated the model with large-scale fleet sensor streams and showed how it effectively detects, correlates, and forecasts events for operational awareness.

Finally, in the area of **cyber-physical security**, Loukas et al. (2019) published a foundational reference for securing autonomous vehicles against cyber-physical attacks. They surveyed intrusion detection systems tailored for mobile platforms like drones, ships, and cars, and classified detection strategies based on resource constraints, system diversity, and physical behavior. Their work highlighted open research gaps and practical considerations for vehicle-specific security.

8.2.4. ARESIBO

833805

Augmented Reality Enriched Situation awareness for Border security

AIRBUS DEFENCE AND SPACE SAS (FR)

€6,999,882

01/05/2019 – 31/07/2022

https://doi.org/10.3030/833805

https://aresibo.eu

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 – Technologies to enhance border and external security

Situation awareness, augmented reality, sensors correlation, Command and Control

Objective

ARESIBO aims at improving the efficiency of the border surveillance systems by providing the operational teams and the tactical command and control level with an accurate and comprehensive information. The pillars of research in ARESIBO are three-fold: 1. Set-up a complete configuration at tactical and execution level to optimise the collaboration between human and sensors (fixed and mobile), 2. Improve situation awareness by enhancing the understanding of the situation through adapted processing of sensor data, correlation between heterogeneous data and information and creation of knowledge through deep learning techniques and 3. Create a situation awareness capability at C2 level that will combine reports on previous missions, real time situation understanding and threat analysis for future actions. This capability will be used to optimise the operations (teams deployment and sensor positioning) as well as an online briefing tool for the teams that will be able to access to the results of the previous missions while in the field.

ARESIBO integrates research activities in the domain of 1. surveillance platforms (air, ground, surface, underwater) to optimise the collaborative capabilities of the platforms and their positioning (between themselves and with the teams), 2. Sensor processing to interpret, fuse and correlate all the data to produce information and knowledge and 3. Augmented reality techniques to elaborate and provide to the operators a situation awareness picture which is fit for their missions (minimum information for maximal understanding) both as team level and tactical C2 level.

The ARESIBO system will be developed incrementally during the 3 years with two major versions that will lead to sub-versions for land and maritime borders. The system will be tested and assessed in 1. a controlled environment enabling testing at any time without pre-requisite authorisations and 2. in real conditions in Finland, Greece, Romania and Portugal for the 2 versions.

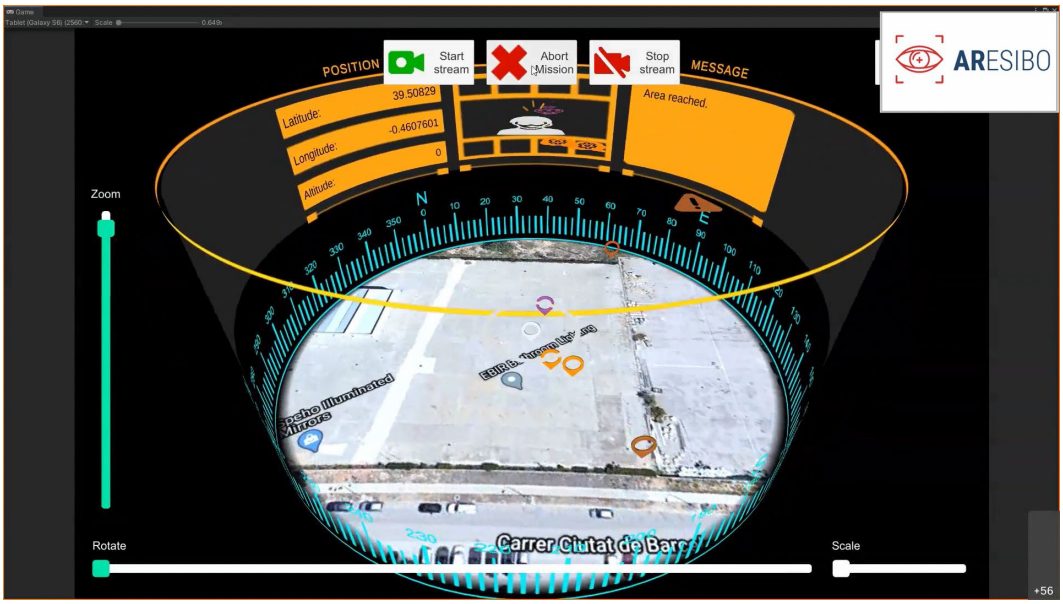
Selected publications

Chatzidakis and Hadjiefthymiades (2022) addressed the security challenges in mobile ad-hoc networks by proposing a lightweight trust-based mechanism for detecting and managing malicious nodes. The system not only identifies threats but also includes a recovery process for misclassified or restored nodes, with tuneable parameters to balance sensitivity and detection speed.

Apostolidis et al. (2022) presented a mission planning platform that enables efficient deployment of unmanned aircraft swarms for remote sensing applications, such as precision agriculture and search and rescue. Leveraging a Simulated Annealing-based optimization for coverage path planning and the DARP task allocation algorithm, the system significantly improves area coverage and mission duration, even in environments with no-fly zones and heterogeneous unmanned aircraft capabilities. The platform's real-world applicability is demonstrated through successful field tests and a publicly available implementation.

Karatzinis et al. (2022) developed a unmanned aircraft swarm-based monitoring system for detecting hazardous CH gas plumes in complex indoor environments. Using a novel path-planning algorithm (BCD-CAO), the unmanned aircraft swarm dynamically adapts to maximize gas density measurements, outperforming a benchmark EGO method across multiple simulations. The solution demonstrates the potential of autonomous multi-agent systems in early hazard detection, offering promising implications for personnel safety in sensitive or high-risk border facilities.

Figure 39. ARESIBO augmented reality view for command and control.



Source: Cordis

8.2.5. BorderUAS

883272

Semi-autonomous border surveillance platform combining next generation unmanned aerial vehicles with ultra-high-resolution multi-sensor surveillance payload

SOFTWARE IMAGINATION & VISION SRL (RO)

€6,997,332

1/6/2020 – 31/5/2024

https://doi.org/10.3030/883272

Programme
HORIZON 2020

Topic title
SU-BES02-2018-2019-2020 – Technologies to enhance border and external security

https://www.youtube.com/@borderuas4440

Objective

The project will combine for the first time a multi-role lighter-than-air (LTA) unmanned aerial vehicle (UAV) with an ultra-high resolution multi-sensor surveillance payload supporting border surveillance as well as search & rescue applications, and specifically rough terrain detection. The sensor payload will include synthetic aperture radar (SAR), laser detection and ranging (LADAR), shortwave/longwave infrared (SWIR/LWIR) and acoustic cameras for direct target detection, as well as optical and hyperspectral cameras for indirect detection (via vegetation disturbance).

The project will use the ground-based infrastructure of border police units (command & control centres), innovative data models (to identify illegal crossing patterns and preferred routes) and advanced audio/video analytics and storage (to provide additional detection capabilities). The technology concepts will be validated in the field by 6 border police units (Greece, Bulgaria, Romania, Moldova, Ukraine, Belarus) covering 3 major illegal migration routes into Europe (Eastern Mediterranean, Western Balkan and Eastern Borders Routes), which represent 58% of all illegal border crossings detected and are also the most used for smuggling of drugs, weapons and stolen vehicles. The combined solution will provide high coverage, resolution and revisit time with a lower cost (4 EUR/kg/hr) than satellites and higher endurance (100 kg payload for 12 hours) than drones. Based on the field trial results, the consortium expects to develop a solution that can be deployed further by European border polices after project completion.

The project will also involve the contribution of NGOs working with illegal migration and human right protection issues, as well as regulatory experts dealing with the ethics and privacy requirements of border surveillance solutions.

Selected publications

Concerning the holistic **multisensor unmanned aircraft surveillance**, Kampas et al. (2022) define the technical requirements for the *BorderUAS* aerial platform. They describe a structured methodology for identifying sensor specifications, ensuring that UAV payloads meet the operational needs of law enforcement agencies involved in border control. Salom et al. (2023) design a novel acoustic camera to be mounted on an airship. The camera features 64 microphones, a 2-meter aperture, and a dedicated processing system capable of detecting low-level, far-field acoustic events from altitudes up to 300 meters. This work fills a key technological gap in aerial acoustic surveillance. Athanasakis et al. (2025) describe the concept for the *BorderUAS* platform, a semiautonomous surveillance system based on a Lighter-Than-Air (LTA) unmanned aircraft carrying an ultra-high-resolution multisensor payload.

In the field of **object detection in aerial imagery**, Antonakakis et al. (2022) address the computational challenge of object detection in 8K-resolution aerial imagery. They deploy a YOLOv5-based pipeline on an Nvidia Jetson AGX Xavier, splitting images into tiles processed in parallel Docker containers. Their method achieves near-real-time detection and is suitable for energy-autonomous embedded systems. Tzedakis et al. (2023) evaluate the generalization performance of YOLOv5s, YOLOv5su, and YOLOv8s on pedestrian and car detection from aerial static video frames. Using the Stanford Drone Dataset, they demonstrate that YOLOv5s outperforms the newer models in both accuracy and generalization, particularly in handling background complexity and unseen test scenes. Antonakakis et al. (2023) propose the Two-Phase ResNet, a modified deep learning architecture for detecting small objects in aerial imagery. By adding a mirrored FPN structure, they improve performance over baseline ResNet and rival leading object detectors in terms of mean average precision. Zachariadis et al. (2023) develop the ShRe-Xception model for wild forest fire detection using RGB and infrared unmanned aircraft imagery. Their model significantly outperforms standard Xception variants, achieving over 99% accuracy on IR data and 90% on RGB. The work demonstrates strong potential for early wildfire detection in aerial surveillance.

Finally, in terms of **semantics and interoperability**, Kontopodis et al. (2022) introduce a domain ontology for land border surveillance, built upon the EUCISE-OWL maritime surveillance model. They model complex relationships between people, events, and locations, enabling semantic reasoning over visual sensor data. Their system infers high-level scenarios (e.g., person clusters) from low-level detections and integrates into a real-time alerting pipeline.

8.2.6. PASSPort

101004234

Operational Platform managing a fleet of semi-autonomous drones exploiting GNSS high Accuracy and Authentication to improve Security & Safety in port areas

SISTEMATICA SPA (IT)

€2,868,240

1/12/2020 – 30/11/2023

https://doi.org/10.3030/101004234

https://h2020-passport.eu

Programme
HORIZON 2020

Topic title
LC-SPACE-EGNSS-1-2019-2020 – EGNSS applications fostering green, safe and smart mobility

https://www.youtube.com/@passporteuspa5061

Objective

The purpose of the PASSport (Operational Platform managing a fleet of semi-autonomous drones exploiting GNSS high Accuracy and Authentication to improve Security & Safety in port areas) is to engineer and qualify a solution extending situational awareness based on aerial fixed/ rotary wing and underwater drones to improve safety and security in port areas. The need stems from the Directive 2005/65/CE asking to complement surveillance systems for the whole port area, in order to significantly improve security and safety for daily operations implanted in port area. This result also in saving citizen lives ensuring a high and equal level of safety and security for all European ports. Around one thousand European ports fall within the scope of the directive.

As a consequence, PASSport responds to the needs expressed by port authorities, harbour master and border control authorities which are active parties in the consortium and will be directly involved in the definition of the proposed solutions. The proposed solution is intended to complement already operational platforms by extending the surveillance perimeter using a fleet of drones to provide innovation and operational support to the recognition, management and analysis of safety and security aspects of daily operations with particular attention to: Pollution monitoring (safety), Support to e-navigation (safety), Critical buildings / Infrastructures protection (security), Protection against non-cooperative small craft approaching the port areas (security), Underwater threats monitoring (security).

The project is expected to activate commercialisation channels based on the novelty represented by the usage of a fleet of semi-automated drones integrating Galileo services (and other sensors) for a safe and efficient guidance, navigation and control even in a challenging environment in presence of obstacles - including buildings and other ground assets - and potentially unfavourable weather conditions.

8.2.7. NESTOR

101021851

aN Enhanced pre-frontier intelligence picture to Safeguard The EurOpean boRders

HELLENIC POLICE (EL)

€4,999,578

1/11/2021 – 30/4/2023

https://doi.org/10.3030/101021851

Programme
HORIZON 2020

Topic title
SU-BES03-2018-2019-2020 Demonstration of applied solutions to enhance border and external security

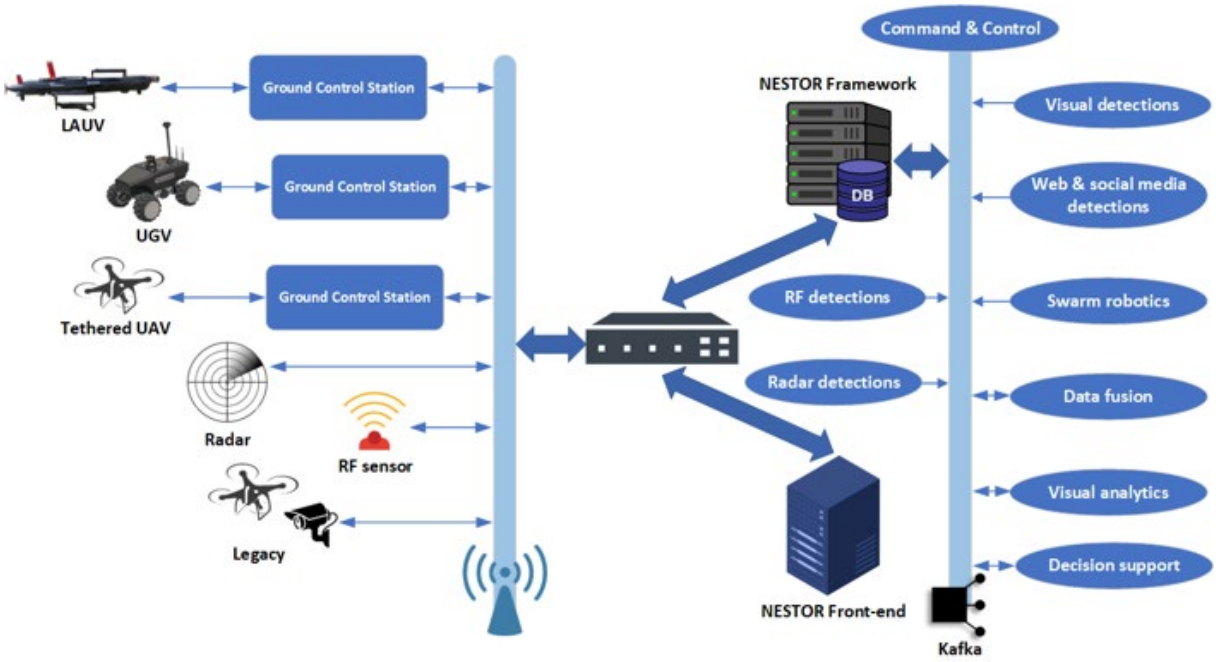
https://www.youtube.com/@NestorProjectH

Objective

For the past few years, Europe has experienced some major changes at its surrounding territories and in adjacent countries which provoked serious issues at different levels. The European Community faces a number of challenges both at a political and at a tactical level. Irregular migration flows exerting significant pressure to the relevant authorities and agencies that operate at border territories. Armed conflicts, climate pressure and unpredictable factors occurring at the EU external borders, have increased the number of the reported transnational crimes. Smuggling activity is a major concern for Eastern EU Borders particularly, as monitoring the routes used by smugglers is being hindered by mountainous, densely forested areas and rough lands aside with sea or river areas. Due to the severity and the abrupt emergence of events, the relevant authorities operate for a long-time interval, under harsh conditions, 24 hours a day.

NESTOR aims to demonstrate a fully functional next generation holistic border surveillance system providing pre-frontier situational awareness beyond maritime and land border areas following the concept of the European Integrated Border Management. NESTOR long-range and wide area surveillance capabilities for detection, recognition classification and tracking of moving targets (e.g. persons, vessels, vehicles, drones etc.) is based on optical, thermal imaging and Radio Frequency (RF) spectrum analysis technologies fed by an interoperable sensors network including stationary installations and mobile manned or unmanned vehicles (aerial, ground, water, underwater) capable of functioning both as standalone, tethered and in swarms. NESTOR BC3i system will fuse in real-time border surveillance data combined with web and social media information, creating and sharing a pre-frontier intelligent picture to local, regional and national command centers in AR environment being interoperable with CISE and EUROSUR.

Figure 40. NESTOR High Level Architecture.



Source: Cordis

Selected publications

Apostolidis et al. (2023) presents an enhanced approach to grid-based coverage path planning for unmanned aircraft, aiming to address key limitations that hinder their real-world adoption – particularly the inefficiencies introduced during grid representation of complex regions. Building on prior work that optimized grid alignment for better coverage, the authors introduce a refined method offering three adaptable modes: geo-fenced, better coverage, and complete coverage, each tailored for specific vehicle capabilities and mission needs. These modes factor in user-defined overlap settings essential for applications like orthomosaic and 3D mapping. Evaluated on 20 benchmark regions, the method proves effective even in irregular, obstacle-filled areas, achieving efficient, complete coverage with smooth trajectories. The results show that with proper parameterization, grid-based methods can overcome their traditional drawbacks and be viable for real-world unmanned aircraft operations.

Touska et al. (2023) present a survey of graph-based data association in multiple object tracking. It groups different graph-based tracking methods into three types based on how they connect object detections: linking detections to detections, detections to existing tracks, or tracks to tracks. It also compares traditional methods with newer deep learning approaches to show the latest trends and performance.

8.2.8. BorderForce

101168316

Flexible system extending automated border surveillance by increased situational awareness adaptable to uncertain times with unforseen events

AIT AUSTRIAN INSTITUTE OF TECHNOLOGY GMBH (AT)

€3,992,775

01/11/2024 – 30/04/2027

https://doi.org/10.3030/101168316

https://borderforce-horizon.eu/

Programme
HORIZON EUROPE

Topic title
HORIZON-CL3-2023-BM-01-01 – Capabilities for border surveillance and situational awareness

Objective

The EU has experienced a surge in fixed border surveillance solutions, including physical barriers, covering 13% of external land borders. The BorderForce project responds to evolving threats by enhancing real-time surveillance capabilities. It introduces a dynamic system, featuring self-sufficient, transportable Command and Control (C2) Stations with configurable and extendable capabilities. These stations incorporate versatile surveillance towers with anti-drone features, integrating data from autonomous monitoring sensors and UAV systems. Dedicated satellite resources, including CubeSat, strategically positioned at high-risk spots, ensure frequent revisits to critical areas. To enable early threat assessment, BorderForce leverages Open-Source Intelligence (OSINT), processing online data for border security threats.

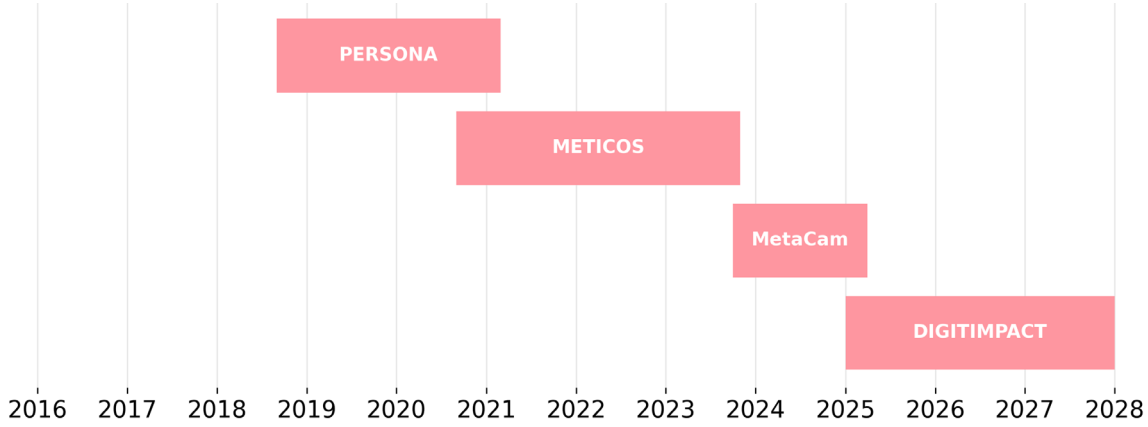
The project emphasizes ethical, legal, and social aspects, safeguarding fundamental rights in border surveillance capability development. BorderForce collaborates with EU and candidate country border authorities, customs, and Common Security and Defence Policy (CSDP) entities. It defines scenarios, gathers architecture feedback, and validates end user feedback. The project emphasizes data exchange, novel user interfaces (e.g., XR), and immersive training for collaborative threat assessment, promoting safety and security while upholding fundamental rights. This solution enhances resource sustainability by improving the coordination and deployment of reusable security measures. The BorderForce solution will be validated in 2 field trials to become TRL7. It ensures seamless operations in monitoring the flow of goods, people, and information. Addressing challenges like migration, smuggling, and geopolitical tensions, BorderForce contributes to regional stability, particularly in crises. Overall, it combines technological innovation and international collaboration to bolster border security, prioritizing fundamental rights and ethical considerations.

9. Societal and ethical dimensions

9.1. Introduction

Four projects focused on privacy, ethical regulatory and social aspects relevant to AI use in border management and customs controls. A timeline of these projects is shown below in Figure 41.

Figure 41. Timeline of projects on human rights and societal aspects, for AI in border management and customs controls.



Source: JRC

Other projects relevant to border management and customs controls and AI and included elsewhere in this report also considered societal and ethical aspects. For example, Sarlio-Siintola et al. (2019) developed a framework for (a) the identification of ethical, legal and societal aspects in technology innovation projects, and (b) the operationalisation of these aspects as concrete requirements. The proposed framework was applied to the outcomes of MARISA and RANGER maritime surveillance projects (see section 4 for information about the projects).

Within the framework of ITFLOWS (see section 6.3.2) researchers presented eight ethical requirements with their corresponding safeguards to be implemented at the design and development stages of AI predictive tools for humanitarian action, with the aim of operationalizing AI ethical principles and mitigating the inherent risks (Guillén & Teodoro, 2023). Also, Blasi Casagran (2021) looked at the challenges that have emerged from making EU databases and information systems 'interoperable' as well as the potential negative consequences that the Regulations may have for fundamental rights as enshrined in the EU Charter of Fundamental Rights.

9.2. Featured projects

9.2.1. PERSONA

787123

Privacy, ethical, regulatory and social no-gate crossing point solutions acceptance

VRIJE UNIVERSITEIT BRUSSEL (BE)

€2,984,700

01/09/2018 – 28/02/2021

https://doi.org/10.3030/787123

Programme
HORIZON 2020

Topic title
SEC-18-BES-2017 – Acceptance of no gate crossing point solutions

Objective
The increasing number of travellers crossing European borders is putting a mounting pressure on the everyday handling of border checks. Coping with limited resources, border control authorities have to perform a higher number of checks on travellers, while at the same time maintaining national security. Consequently, the experience of both European and third country travellers is deteriorating, due to the extra time they have to spend at the border crossing points. In that sense, flexible, automated and scalable no-gate crossing solutions may offer an effective support to border control authorities, to make border checks faster and more efficient. Nevertheless, the intensive use of these technologies bears the risk of infringing individuals' rights and freedoms, including privacy and personal data protection. Therefore, a thorough assessment of no-gate crossing point solutions, from asocial acceptance, ethical and legal perspective, is required prior to their implementation. PERSONA (Privacy, Ethical, Regulatory, and SOcial No-gate crossing point solutions Acceptance) project's main goal is to design and establish an integrated impact assessment method (IAM PERSONA) to appropriately evaluate the impacts of no-gate crossing point solutions, taking into account social acceptance, ethical and legal requirements, to ensure that these technologies meet the expectations of public authorities and travellers. The main outcome of the project will be a handbook, available in open access, that will provide guidance in order to drive the innovation and deployment of future no-gate crossing point solutions by industry and border authorities, while maintaining their societal acceptance

Reported results

PERSONA developed the *IAM PERSONA* impact assessment methodology to ensure no-gate border technologies respect privacy and data protection before deployment. Created an open-access handbook to guide ethical innovation, with a strong emphasis on rights, freedoms, and societal alignment. The project developed numerous models and solutions for measuring such metrics as technology anxiety and performance expectancy, amongst other variables. For example, through its technology acceptance models, researchers could study smart border crossing technologies from a user experience perspective. Researchers also developed a social sensing platform that allowed them to continuously monitor the social acceptance of a technology. To achieve this, the platform used a combination of modelling and simulation to identify different behavioural patterns exhibited when people pass through a smart EU border control system.

PERSONA highlighted the need to balance technological efficiency with individual rights and societal acceptance. Designed a big data analytics engine capable of harmonising data gathered from heterogenous sources. The engine uses machine and deep learning, along with statistical algorithms and data analytics, to uncover patterns regarding how a traveller or border control staff member accepts smart border control technologies.

Selected publications

Martins and Jumbert (2020) contributed to the understanding how and why security technologies came to be accepted as crucial for EU's borders and highlighted how the possession of technological expertise confers technology developers and other experts a discursive hegemony on border management and vests them with authority.

In the context of biometric technology for borders, Kloza et al. (2020) looked at the reformed data protection and border management legal frameworks and highlighted discrepancies between the two. The study provides an assessment to the new data protection reformed regime for biometric data.

9.2.2. METICOS

883075

A Platform for Monitoring and Prediction of Social Impact and Acceptability of Modern Border Control Technology

EUROPEAN UNIVERSITY - CYPRUS LTD (CY)

€4,997,481

01/09/2020 – 31/10/2023

https://doi.org/10.3030/883075

https://meticos-project.eu/

Programme
HORIZON 2020

Topic title
SU-BES01-2018-2019-2020 – Human factors, and social, societal, and organisational aspects of border and external security

Big Data Analytics, Real Time Analytics, Technology Acceptance methodologies, cross-sectorial, data correlations, No Gate Crossing Evaluation, Societal Acceptance Simulation Toolkit

https://www.youtube.com/@meticoseuproject449

Objective

METICOS aims to introduce Big Data Analysis of border control information systems, in order to provide a step-change towards more modern and efficient Smart Border management and towards gaining societal and political acceptance of modern control technologies of EU borders such as “no gate solutions”. It is therefore positioned as an initiative to create a real-time decision-support system with regard to different Smart Border control technologies that empowers two major stakeholder groups within the European border control sector (i.e., travellers and border control authorities & service providers) to ensure user acceptance, secure positive societal impact and maximize border control process efficiency.

To this end, the METICOS project will develop a platform that integrates information systems and networks of data sources in order to validate the efficiency and users acceptance of border control technologies. The proposed platform will provide metrics and KPIs to authorities and decision makers, based on a number of independent variables: performance expectancy, effort expectancy, facilitating conditions, physical privacy, accuracy, information privacy, ethical and societal perceptions.

To maximize impact of the project, METICOS will join the European Initiative for Smart Borders and contribute aforementioned cross-border and cross-cultural Big Data Analysis and decision support systems that feature harvesting of multi-lingual and cross-sectorial data from heterogeneous sources, different analytics approaches and risk assessment methods. METICOS will be demonstrated and validated under real operating conditions. This will involve two pilot implementations executed at five different countries involving real travellers and staff. Furthermore, the METICOS ecosystem aims to work closely with border control research initiatives to demonstrate its operational performance improvements by means of combined validation with these solutions.

Reported results

METICOS: Integrated privacy-related Key Performance Indicators (KPIs) to evaluate smart border systems from a human rights perspective. METICOS: Emphasized ethical decision-making and societal acceptance in real-time smart border management, grounded in participatory methods. Built systems that account for ethical, societal, and cultural variables to improve trust and acceptance of cross-border technologies. METICOS also initiated a Border External Security Cluster which currently counts 20 members: https://meticos-project.eu/related-projects/

Figure 42. The METICOS aimed at advancing the uptake and acceptance of no gate security solutions.



Source: Cordis, © Grispb/stock.adobe.com

Selected publications

Shaikh & Yayilgan (2023) study the use of agent-based approach to generate simulations of various scenarios for border crossing points and making predictions regarding technology acceptance. Shaikh et al. (2022) also explore the artificial intelligence (AI) techniques, available for analysing social media data. Along with brief information about the AI techniques, the paper also looks at studies which utilize the AI techniques for measuring user perceptions from the social media data. Gkioka et al. (2022) propose a data model which aims to capture the data needed for the assessment of border control technologies, their usability and acceptance. The data model is evaluated by domain experts.

9.2.3. MetaCam

101123231

Privacy-Preserving Security Cameras based on Metalenses

FONDAZIONE ISTITUTO ITALIANO DI TECNOLOGIA (IT)

€150,000

01/10/2023 – 31/03/2025

https://doi.org/10.3030/101123231

Programme
HORIZON EUROPE

Topic title
ERC-2023-POC – ERC PROOF OF CONCEPT GRANTS

Objective
Surveillance cameras are widely used by governments, businesses, and ordinary citizens to monitor private and public areas to deter crime and identify offenders. As of 2021, it was estimated that there are over 1 billion surveillance cameras worldwide. However, there are great concerns that continuous monitoring severely restricts people's privacy and freedom of movement. There have been reports of abuse of surveillance footage including by businesses and government institutions. Regulators, civil rights groups and citizens are thus rightly concerned about infringements on freedoms of speech and association, especially in this era of AI when monitoring, tracking, and profiling techniques have been made easier, cheaper and more accurate. An initial step in video scene analysis is pose estimation which enables monitoring the flux of people and detecting criminal actions or intent. Considering the privacy concerns of continuously recording citizens and automated image analysis, this project intends to develop a novel camera system that captures just enough detail required for pose estimation but excludes identifying information. Existing software methods of deidentifying videos require a clear video to be taken first, and so do not entirely solve this problem. Proposed experimental hardware methods that avoid this issue using lenses have bulky optics and a limited set of parameters. With MetaCam we will implement the optimized optical devices (meta lenses) that allow ultimate control of all the light degrees of freedom. In this project, we will design and fabricate the meta lenses and build an optical table prototype of the camera to demonstrate its secure privacy-preserving performance. Supported by business developers we will examine the business potential, and map and engage the stakeholders later required to get from a proof- of- concept to a market-ready product.

Reported results

- Design of a novel optical system that avoids capturing identifying features entirely, offering a hardware-level solution to privacy protection.
- Study of ethical issues in surveillance by ensuring that the technology cannot be used for intrusive identification or tracking.
- Promotion of privacy-by-design to align technological surveillance with public expectations and civil liberties.

9.2.4. DIGITIMPACT

101175991

The Impact of Digital Technologies on European Democracy and Society

UNIVERSITA DEGLI STUDI DI BARI ALDO MORO (IT)

€100,000

01/01/2025 – 31/12/2027

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43353764/101175991/ERASMUS2027

Programme
Erasmus+ (ERASMUS+)

Topic title
ERASMUS-JMO-2024-COE

Digital Technologies; European democracy; Disinformation; Artificial Intelligence; Online Platforms; Cybersecurity; Big Data

Objective

The Jean Monnet Centre of Excellence DIGITIMPACT (The Impact of Digital Technologies on European Democracy and Society) deals with researching and disseminating knowledge on the impact of digital technologies on European democracy and society. The topics covered will primarily but not exclusively concern the regulation of artificial intelligence, the protection of personal data online, the treatment of online content by digital platforms, the fight against hate speech and misinformation contents, the protection of minors online, the problem of "digital addiction", the market power of online platforms, recommendation algorithms for online content, the issue of microtargeting of political advertising, the use of big data, digital surveillance, cybersecurity rules, the impact of digital technologies on electoral processes, the EU's digital trade policy, and the digitisation of border controls.

The DIGITIMPACT Centre of Excellence's activities and deliverables aim at reaching the following target groups over the three-year period: university students (1 kick-off meeting, 2 annual conferences and 9 seminars in different Departments of the University of Bari), the national and international scientific community (2 annual conferences, 3 books), the general public (1 kick-off meeting, 3 general public literacy events, 3 small film festivals, 3 small book festivals, 3 digital literacy courses in politics and public administration, 10 in-depth TV pills, and 10 periodic newsletter), and specific targets (3 media literacy meetings for journalists, 6 high school literacy meetings, 3 "DIGITIMPACT Plate" award ceremonies to the benefit of high school students).

Fully open deliverables include 3 books (monographies and/or collective volumes) over the three-year period, in open-access format, and 10 periodic newsletter on the most recent European and Italian legislative and jurisprudential developments as well as past and future CoE initiatives.

10. Conclusions

This report describes how EU funded projects are advancing the use of artificial intelligence (AI) for border management and customs controls. We identified 75 projects funded between 2015 and 2024 – primarily under Horizon 2020 and Horizon Europe – representing a combined EU investment of approximately EUR 376 million. In chapters four to nine, we presented the projects and highlighted some of their most relevant results and scientific publications. The analyses are summarised below for each section.

Maritime and border surveillance

AI is used to fuse data from multiple sensor platforms (fixed and mobile) including UxVs (unmanned aerial, surface, and underwater vehicles) for improved situational awareness at sea. Techniques include multi-sensor data fusion, anomaly detection using machine learning, and integration of open-source intelligence. These approaches enable automated detection of suspicious behaviours, critical infrastructure threats, and environmental monitoring (e.g. oil spills, biodiversity). Projects such as *COMPASS2020*, *PROMENADE*, and *I-SEAMORE* exemplify the integration of fixed and mobile sensors with AI-enabled data fusion and anomaly detection. Others, like *SMAUG* and *VIGIMARE*, focus specifically on protecting underwater critical infrastructure or addressing environmental hazards, e.g. *AXOLOTL*. In the *AI-ARC* project, CNNs were used to estimate sea ice concentration in the Baltic Sea, whilst AIS data was used for detecting abnormal vessel behaviours and a selection of ML algorithms (K-means, DBSCAN, Affinity Propagation, Gaussian Mixtures). There is a large focus on maritime surveillance, but some projects also develop surveillance for land borders. For example, in the *FOLDOUT* project, AI-driven multi-sensor fusion techniques have significantly improved detection capabilities in complex terrains like forested land borders.

Goods inspection and detection technologies

AI is driving major advances in non-intrusive inspection (NII) systems for customs and border control. Projects such as *C-BORD* and *ENTRANCE* have used AI for interpreting X-ray and neutron imaging data to detect contraband in complex cargo environments. Complementary approaches like cosmic ray tomography, explored in *SilentBorder*, offer alternatives for detecting dense or shielded materials. In the *PARSEC* project, machine learning and spectral imaging techniques are used to automate the selection and secondary screening of high-risk parcels. Beyond goods inspection, projects such as *ENIGMA* and *ANCHISE* extend AI-driven image recognition, spectral analysis, and semantic data linking to the protection of cultural heritage, helping to trace and authenticate artefacts and combat illicit trafficking.

Risk analysis and predictive intelligence

Across a range of EU-funded initiatives, AI is being used to anticipate threats, detect emerging risks, and disrupt organised criminal networks through data integration, pattern recognition, and predictive modelling. *TRESSPASS* applied AI-based simulations and red-teaming to develop risk-based border management systems that assign dynamic threat scores to passengers and cargo, improving the efficiency and selectivity of border checks. Meanwhile, *ITFLOWS* used deep learning models to predict migration flows and assess potential social tensions by analysing digital traces from social media and conflict data.

Projects such as *ATLANTIS* and *AHEAD* extend predictive intelligence to broader civil security domains, using explainable AI (XAI) and foresight modelling to anticipate risks to critical infrastructure and

inform law enforcement strategies. In disrupting criminal networks, AI supports large-scale data fusion and intelligence analysis. The *PREVISION* platform integrates heterogeneous data streams – from financial records to darknet activity – using knowledge graphs and machine learning to map and predict illicit network behaviour. Similarly, *CEASEFIRE*, *PERIVALLON*, and *VANGUARD* apply AI techniques such as natural language processing, computer vision, and multimodal analytics to detect and investigate crimes including firearms trafficking, environmental offences, and human trafficking. Finally, in operational research, projects such as *BA-AC* demonstrate how AI-driven optimisation supports planning and coordination at complex border and transport hubs.

Identity, biometrics, and document security

Artificial intelligence and machine learning have the potential to advance biometric identification and document verification for border management. Early initiatives such as *FLYSEC* applied AI-based anomaly detection to passenger movement data, using agent-based simulation to optimise security checks without disrupting flow. *iBorderCtrl* explored privacy-preserving solutions through cancellable biometrics, a method that transforms biometric data using encryption keys to allow secure authentication while preventing misuse of raw data—a precursor to privacy-by-design principles in large-scale biometric systems. In *PROTECT*, convolutional neural networks (CNNs) were used to enhance iris recognition from low-resolution or mobile-captured images, demonstrating how deep learning can improve identification accuracy in challenging operational environments.

Similarly, *eBORDER* compared fingerprint and finger-vein recognition systems, applying AI-based image processing and feature extraction techniques to achieve higher precision and resistance to spoofing. The *D4FLY* project pushed biometric innovation further by introducing somatotype—a full-body shape model—as a "soft biometric" for recognising individuals at a distance or while on the move. Deep learning architectures were used to extract stable body-shape features even when appearance changed. Finally, *iMARS* addressed the growing threat of morphing attacks on face-recognition systems by evaluating both human and AI detection capabilities. The project developed benchmark datasets and compared the performance of trained border personnel with automated morph detection algorithms, underscoring the need for AI-assisted tools and training to mitigate this vulnerability.

Unmanned vehicles and autonomous systems

AI is increasingly central to the operation and coordination of unmanned aerial, surface, and underwater vehicles (UxVs) used in maritime and land border surveillance. Projects such as *SafeShore* and *ALFA* demonstrated early advances in AI-based object detection and classification, addressing one of the most pressing challenges in border surveillance: distinguishing legitimate aerial activity from threats. Using deep convolutional networks (ResNet) and probabilistic reasoning methods such as Dempster–Shafer theory, these projects achieved automated differentiation between drones, aircraft, and birds under real-world conditions. Through SafeShore, the drone-vs-bird detection challenge highlighted the growing role of AI competitions benchmarking and accelerating innovation.

ROBORDER developed a comprehensive multi-domain surveillance architecture that integrates heterogeneous unmanned platforms – aerial, surface, and underwater – equipped with visual, radar, and radio-frequency sensors. AI algorithms perform multi-sensor data fusion, semantic interpretation, and real-time anomaly detection to improve situational awareness. Machine learning models were also applied to specialised tasks such as oil spill detection from satellite imagery and efficient on-board object detection using lightweight neural networks for resource-constrained drones. Projects such as *ARESIBO* and *NESTOR* further advanced swarm intelligence and cooperative mission planning.

Using AI-driven path optimisation and coordinating multiple drones for applications ranging from hazardous gas detection to large-area border monitoring. AI is also applied to ensuring security and trust in these autonomous networks. For example, *ARESIBO* introduced a trust-based mechanism for detecting malicious nodes in mobile *ad hoc* networks, while *ROBORDER* and related studies explored intrusion detection in cyber-physical systems for unmanned vehicles.

These projects illustrate a technological shift toward autonomous, intelligent, and interoperable unmanned systems capable of operating collaboratively across domains. By combining machine learning, optimisation, and data fusion, they aim to provide a new generation of border and maritime surveillance technologies that can detect, analyse, and respond to emerging threats with minimal human intervention.

Societal and ethical dimensions

As AI becomes increasingly integrated into border management, questions of privacy, accountability, and societal acceptance have come to the forefront. The *PERSONA* project developed an impact assessment methodology to evaluate no-gate border crossing technologies before deployment. Combining big data analytics with machine and deep learning, the project analysed behavioural patterns, technology acceptance, and user perceptions among travellers and border personnel. Similarly, *METICOS* built a data-driven platform to evaluate the social and ethical acceptability of smart border systems in real time. Using AI techniques such as social media analytics and agent-based modelling, *METICOS* assessed public sentiment, human factors, and cultural influences affecting trust in automated border management. The *MetaCam* project explored privacy-preserving surveillance from a technical standpoint, developing optical systems that prevent the capture of identifying features altogether. This hardware-based approach advances the principle of privacy-by-design by limiting the potential for intrusive identification or tracking. At a broader societal level, *DIGITIMPACT* examines how digital technologies including AI affect democracy, data protection, and online discourse. Projects mentioned previously, like *ITFLOWS* and *MARISA* also incorporate ethical design principles and propose frameworks for identifying and mitigating risks related to AI use. Together, these initiatives illustrate the growing recognition that technological advancement in border security must be accompanied by robust ethical governance, transparent evaluation frameworks, and inclusive dialogue.

Future work

This report is based exclusively on publicly available project information (e.g. CORDIS summaries, project websites, and open deliverables). Classified or commercially sensitive reports were not accessed. To provide additional value and a deeper layer of insight, a systematic assessment of project impact – for example in terms of technological maturity, patents, commercialisation pathways, or operational uptake – would be a logical next step. Conducting such an assessment would require a dedicated methodology, structured engagement with project partners, and significant time and resources, which were beyond the scope of this study. It is hoped that the present report will serve as a solid baseline to inform and support such future work.

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List of abbreviations and definitions

acronym	definition
ABC	automated border control
AI	artificial intelligence
AIS	automatic identification system
API	application programme interface
API	advanced passenger information
APIS	Advance Passenger Information Systems
AUV	autonomous underwater vehicle
CCEI	Customs Control Equipment Instrument
CISE	Common Information Sharing Environment
CNN	Convolutional Neural Network
CPH	cyber-physical-human
CRT	cosmic ray tomography
CSDP	Common Security and Defence Policy
CSS	Copernicus Security Services
DAP	Drugs Action Plan
DARPA	Defence Advanced Research Projects Agency
DCNN	deep convolutional neural network
DL	deep learning
DLT	distributed ledger technologies
DMA	differential mobility analyser
D-MAD	differential morphing attack detection
DoA	direction-of-arrival
EES	Entry/Exit System
EFE	European Firearms Experts
EMT	EUMigraTool
ERC	European Research Council
ETIAS	European Travel Information and Authorisation System
EU	European Union
EUAA	European Union Agency for Asylum
EUMSS	EU's maritime security strategy
EWPS	early warning and preparedness system
FET	future and emerging technologies

acronym	definition
FIND	Fixed INTERPOL Network Database
FRS	face-recognition system
FRT	facial recognition technology
FVR	finger vein recognition
GCS	ground control station
GDELT	Global Database of Events, Language, and Tone
GDPR	General Data Protection Regulation
GeoAI	Geospatial Artificial Intelligence
GIS	geographic information system
GTD	Global Terrorist Database
HE	Horizon Europe
IBMS	Integrated Border Management System
IR	infrared
ISF	Internal Security Fund
ISFB	Internal Security Fund - Borders and Visa
ISFP	Internal Security Fund - Police
ISR	intelligence, surveillance, and reconnaissance
ITC	illicit trafficking combat
JRC	Joint Research Centre
KPI	key performance indicator
LADAR	laser detection and ranging
LEA	law enforcement agency
LIDAR	light detection and ranging
LTA	lighter-than-air
MAD	morphing attack detection
MC	member countries
MCC	Matthews correlation coefficient
ML	machine learning
MOC	maritime operation centre
MS	mission system
MSCA	Marie Skłodowska-Curie Actions
NFP	national focal point
NII	non-intrusive inspection
NIIS	non-intrusive inspection systems

acronym	definition
OPV	offshore patrol vessel
OSINT	open-source intelligence
PCAO	parametrized cognitive-based adaptive optimization
R&I	research and innovation
RF	radio frequency
ROV	remote operation vehicle
RPAS	remotely piloted aircraft systems
RRTNIS	rapidly relocatable tagged neutron inspection system
SAR	synthetic aperture radar
SDR	software defined radio
SEDIA	Single Electronic Data Interchange Area
SIS	Schengen Information System
S-MAD	single image morphing attack detection
SME	small and medium-sized enterprise
SNA	social network analysis
SNM	special nuclear materials
SSD	Single Shot Detector
SWIR/LWIR	shortwave/longwave infrared
TCN	third country nationals
THB	trafficking in human beings
TNIS	tagged neutron inspection system
UA	unmanned aircraft
UAS	unmanned aircraft system
UAV	unmanned aerial vehicle[10]
USV	unmanned surface vehicle
UUV	unmanned underwater vehicle
UXO	unexploded ordnance
UxV	unmanned vehicle (i.e. UAV or USV or UUV)
VIS	Visa Information System

[10] The preferred term used in EU legislation (Regulation EU 2019/945) is *unmanned aircraft* (UA).

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